Analytics Pipeline for Data from Recipe Oriented Portals

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Abstract

Data regarding user interactions from recipe-focused platforms offers significant insights into culinary preferences, engagement motivations, and decision-making processes within digital food environments. Nevertheless, this information is frequently unstructured, intricate, and sensitive to context. A structured analytics pipeline is essential to derive valuable business intelligence and enhance user experience.

This initiative develops a modular analytics pipeline. It gathers, organizes, and interprets raw user interaction logs from recipe platforms to facilitate metric-driven evaluations of engagement and conversion trends. The pipeline encompasses data preprocessing, feature extraction, exploratory data analysis (EDA), and visualization workflows utilizing Python libraries such as pandas and matplotlib, in conjunction with Tableau.

Crucial behavioral metrics, including the success rate of widget interactions, frequency of recipe re-engagement, patterns in session duration, trends in dietary preferences, and conversion ratios, are systematically computed to evaluate platform usability and user intent. The project investigates user-level interaction patterns and provides both actionable metrics and strategic insights aimed at improving content personalization, refining the user interface, and maintaining user engagement.

Keywords:

User behavior analytics, Recipe portals, Data pipeline, Interaction metrics, Engagement analytics, Conversion analysis, Tableau visualization, Python data processing

Declaration

We, the undersigned students, hereby declare that this project report entitled **"Analytics Pipeline for Data from Recipe Oriented Portals"** is the result of our own work carried out during the internship period from 25th August 2025 to 31st October 2025 under the guidance of **Atanu Roy Chowdhury.**

We further declare that this report is an original piece of work and has not been submitted previously, in part or in full, for the award of any degree, diploma, or certificate at any other institution or university.

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Chapter 1

Introduction

1.1 The Rise of Data-Driven Culinary Platforms

Recipe platforms have grown to become sophisticated online platforms that offer user engagement and the delivery of personalized culinary experience and culinary discovery. Users engage with multiple food choices by exploring recipe platforms in addition to the platforms capturing engagement data by monitoring search activities, recipe saves, shopping cart adds, as well as user feedback systems. The collected behavioral and preference data enables platforms to optimize their operations and create better experiences for users.

The increasing usage of these platforms does not match the development of unified analytics systems which can handle user interaction data effectively. The current data processing methods for cleaning and engineering and reporting metrics operate independently which hinders the discovery of meaningful platform insights. The current analytical shortcomings prevent platform administrators from tracking user preference changes and detecting successful content and assessing new feature performance.

A well-defined analytics framework addresses existing issues by providing a standardized approach to handle data and monitor metrics and decision support based on evidence. This kind of infrastructure is critical for platforms to ensure their competitiveness because it allows them to monetize their data to deliver better selection of content and improve the user interface.

1.2 Limitations of Traditional Analytical Approaches

The traditional analytical methodologies used in food discovery platforms tend to rely on static reporting, or ad-hoc SQL queries, that examine user behavior. While these methodologies provide insights at the snapshot level, they lack scalability and flexibility. The lack of automated data preprocessing, standardized metric definitions, and integrated visualization hinders the ability to conduct longitudinal trend analysis.

Many forms of user engagement such as "views," "saves," "adding to cart," or "dietary preferences" are often not connected, meaning that even if behavior can be tracked, there is not an overall understanding of engagement pathways or conversion funnels. Without a dedicated analytics pipeline it becomes difficult for teams to monitor and track behavioral metrics, such as user retention, conversions, and patterns of re-visiting recipes over time, and across demographic or contextual segments.

1.3 Opportunities and Challenges in Building Analytical Pipelines

The fast growth of structured and semi-structured interaction data on recipe portals creates both positive and negative aspects. The data provides valuable insights about user behavior and helps identify system obstacles and enhance content suggestion systems. The diverse nature of this data which includes timestamps and user preferences and text logs and multi-step interaction paths needs an efficient automated cleaning and aggregation and transformation system.

The current state of machine learning and exploratory data analysis tools enables developers to create adaptable pipelines which monitor user behavior and reveal relationships and display patterns between users and time intervals. The development of these systems requires achieving a proper equilibrium between analytical precision and system understanding and deployment simplicity.

1.4 Proposed System: Analytics Pipeline for Recipe-Oriented Portals

This initiative plans to develop an end-to-end analytics pipeline for recipe-based platforms. This proposed pipeline will take raw interaction data, process it, and deliver easy-to-use datasets for visualization purposes for Tableau and Power BI.
The framework emphasizes three core objectives:

1. **Data Standardization** – Ensuring consistent formats for timestamps, categorical attributes, and user interactions.

2. **Metric Computation** – Automating the derivation of key behavioral indicators such as *Widget Success Rate, Save-to-View Ratio, Average Tab Changes per Device, and User Retention Patterns.*

3. **Visualization Enablement** – Exporting clean, structured data to BI tools for intuitive dashboards and storytelling-driven analytics.

Through this pipeline, the study aims to empower data teams to transition from descriptive to diagnostic and predictive analytics, ultimately supporting better decision-making in recipe recommendations, UI design, and engagement strategy.

1.5 Data Sources and Analytical Framework

The dataset utilized for this project was derived from simulated user interactions on a recipe discovery portal. It includes logs detailing session timings, recipe views, cart actions, device types, widget usage, dietary preferences, and other contextual variables.
Preprocessing was performed using **Python (Pandas, NumPy, Matplotlib)**, followed by metric formulation and export to **Tableau** for visualization. A structured set of **Exploratory Data Analysis (EDA) and Behavioral Metrics** was computed—ranging from *conversion rate by time of day* to *recipes with high views but low saves.*
Each metric was implemented as a modular component of the pipeline, ensuring reusability and clarity of logic. This framework enables future integration with live APIs or production databases for real-time reporting.

1.6 Technical Preparation and Tools

Before the pipeline implementation, extensive preparatory work was conducted to establish a foundation in data processing, visualization, and machine learning concepts. The training included the following core areas:

1. Python for Data Analysis (Pandas, NumPy)
2. Exploratory Data Analysis and Feature Engineering
3. Data Visualization with Tableau and Matplotlib
4. Metric Definition and Behavioral Analytics
5. Fundamentals of Data Storytelling and Insight Communication

This preparatory training played a crucial role in ensuring systematic pipeline design, metric validation, and visualization accuracy across the analytics workflow.

1.7 Scope and Future Directions

Although the present project is centered on static batch analytics, its modular architecture permits future enhancements into real-time data monitoring, tracking of recommendation performance, and frameworks for A/B testing. The potential integration with generative AI-based summarization tools or LLM-driven recommendation explainers also presents a promising avenue.

Ultimately, this pipeline acts as a foundational step towards the development of scalable, interpretable, and user-focused analytics systems within the realm of digital food platforms.

Chapter 2

Literature Review

2.1 Introduction

Users now discover and evaluate and interact with online culinary content through recipe-oriented platforms which have become widespread. The massive daily production of recipes and user interactions requires data analytics to understand user behavior and enhance recommendation systems and platform engagement. The current research benefits from three essential areas of development which include data analytics in online food platforms and machine learning methods for personalization and engagement modeling and pipeline automation for data-driven insights.

The existing body of research investigates recommendation systems and user engagement prediction but lacks studies that demonstrate the complete analytics process from user interaction data to business intelligence-ready visualizations. The following sections review existing research to establish the value of this project which develops a complete modular analytics system for recipe data platforms.

2.2 Data Analytics in Recipe-Oriented Platforms

Data analytics in online culinary ecosystems typically focuses on understanding user interaction, content popularity, and behavioral segmentation. Studies in this area emphasize the role of structured interaction logs (e.g., views, saves, and cart events) as indicators of user intent and engagement.

1. **Behavioral Analysis:** Platforms such as *AllRecipes*, *Tasty*, and *Yummly* employ descriptive analytics to identify recipe trends, ingredient popularity, and seasonal variations in user activity.

2. **Engagement Metrics:** Research highlights metrics such as “save-to-view ratio,” “widget conversion rate,” and “session duration” as proxies for user satisfaction and decision efficiency (Lee et al., 2020).

3. **User Segmentation:** Analytical clustering approaches are used to distinguish casual browsers from high-intent users (Gupta & Sharma, 2021), improving targeted content delivery.

However, most studies rely on static reporting or isolated models rather than integrated, automated workflows. The lack of unified data cleaning, transformation, and visualization frameworks often limits scalability and reproducibility. This gap motivates the development of an **analytics pipeline** capable of seamlessly connecting raw data preprocessing, feature extraction, and interactive dashboard visualization.

2.3 Machine Learning for Recipe Recommendation and User Engagement

Machine learning has become central to modern recipe portals, particularly for personalization and engagement prediction. The literature identifies three major categories of techniques:

1. **Collaborative Filtering and Content-Based Models:** Early systems focused on matching users to recipes based on historical behavior or ingredient similarity (Liu et al., 2019).

2. **Hybrid and Context-Aware Models:** Recent works incorporate contextual factors such as dietary preferences, device type, or session time to refine recommendations (Zhao et al., 2022).

3. **Behavioral Prediction Models:** Supervised learning methods—including Random Forests, Gradient Boosting, and Neural Networks—are used to predict conversion events such as saves or add-to-cart actions.

Despite strong predictive capabilities, these models often function as black boxes, offering limited interpretability for decision-makers. Moreover, the deployment of ML models in production requires continuous monitoring, retraining, and metric tracking — components rarely integrated into analytical studies. This reinforces the necessity of a **pipeline-based approach**, ensuring transparency, repeatability, and maintainability across the ML lifecycle.

2.4 Data Pipeline Automation and Visualization Frameworks

In recent years, the emphasis in data analytics research has shifted toward **pipeline automation**, emphasizing modularity and scalability.

1. **ETL and Workflow Automation:** Tools such as Apache Airflow, Prefect, and Kubeflow have enabled structured Extract-Transform-Load (ETL) pipelines, reducing manual preprocessing overhead.

2. **Visualization and BI Integration:** Tableau, Power BI, and Metabase are frequently used to transform processed datasets into actionable insights through metrics like user retention, conversion funnels, and engagement heatmaps.

3. **Metrics-Driven Dashboards:** Prior work (Patel & Verma, 2023) emphasizes the importance of defining *standardized engagement metrics* — such as “average tab changes per session” or “recipes revisited per user” — as consistent analytical touchpoints across the data pipeline.

However, most implementations remain domain-agnostic. In recipe-based ecosystems, the heterogeneity of user behavior, interaction patterns, and dietary diversity poses unique challenges to pipeline design. Therefore, an analytics pipeline tailored to the **culinary domain** must integrate behavioral, temporal, and contextual features to deliver nuanced insights.

2.5 Gaps in Existing Literature

While considerable progress has been made in data analytics and machine learning applications for digital platforms, the literature reveals several notable gaps:

- **Lack of unified pipelines** connecting preprocessing, metric computation, and visualization stages.
- **Limited interpretability** of behavioral metrics beyond raw engagement counts.
- **Absence of domain-specific frameworks** tailored for recipe-oriented or nutrition-based user data.
- **Underrepresentation of actionable analytics**, where results translate directly into platform design or recommendation strategies.

These limitations motivate the development of an **end-to-end analytics pipeline** that not only computes behavioral and conversion metrics but also presents them through interpretable, modular dashboards for continuous monitoring.

Chapter 3 — Methodology

3.1 Overview

This chapter presents the methodological framework designed for the development of an **Analytics Pipeline for Data from Recipe-Oriented Portals**.
The goal of this project is to design a scalable analytical system capable of processing, cleaning, enriching, and visualizing user interaction data from online recipe recommendation platforms.

Unlike conventional studies relying on static data, this project focuses on creating a **dynamic, end-to-end analytics workflow** — spanning from synthetic data generation to metric computation and dashboard visualization in **Tableau** and **Python**.
The framework emphasizes behavioral insight extraction rather than predictive modeling, helping understand user engagement, conversion trends, and interaction behavior within recipe-based environments.

The pipeline consists of five major components:

1. **Data Acquisition and Simulation**,
2. **Data Preprocessing and Feature Engineering**,
3. **Metric Definition and Analytical Computation**,
4. **Visualization and Dashboard Development**,
5. **Insight Extraction and Interpretation.**

3.2 Data Acquisition

The dataset used in this project was **synthetically generated** through a controlled simulation environment that replicates user interactions within recipe discovery platforms.
Since real-world proprietary datasets were unavailable, synthetic generation ensured full control over event frequency, interaction variability, and schema design.

The data simulates user journeys across key activities — browsing, searching, saving, carting, and tab switching — each corresponding to a potential engagement signal.

Key data fields include:

1. **UserID** — unique identifier per user session,
2. **SessionStart / SessionEnd** — timestamps for user activity duration,

3. **RecipeID** — unique recipe identifier,
4. **ViewedRecipes, RecipeSaved, AddToCart** — core behavioral metrics,
5. **Device, UserAgent, and Location** — contextual engagement attributes,
6. **WidgetDuration, ChatInteraction, TabChange** — session-level activity descriptors,
7. **DietaryPreferences and Allergy** — user-level personalization parameters.

This dataset was exported in **CSV format** and later adapted into **Excel sheets** to facilitate Tableau integration.

3.3 Data Preprocessing and Feature Engineering

Data preprocessing formed the foundation of the analytical pipeline. Several operations were applied to make the dataset Tableau-ready:

- **Timestamp Parsing:** SessionStart and SessionEnd columns were converted into datetime format to enable duration calculations.
- **Boolean Normalization:** Interaction columns (e.g., RecipeSaved, AddToCart) were standardized as integer flags (0/1) for consistent computation.
- **Derived Features:**
  - **WidgetSuccessFlag:** Sessions where a widget led to save/cart actions.
  - **TabChangeCount:** Extracted from `ActivityLog` using regex patterns.
  - **ReturnWithin24h:** Flag for users returning within one day.
  - **DietPrefCount:** Number of dietary preferences per session.
- **Aggregation Logic:** For higher-order metrics (e.g., per user or per week), group-by operations were performed using Pandas and validated in Tableau.

The resulting dataset, `eda_ready.csv`, served as the unified source for both **notebook-based analysis** and **Tableau dashboards**.

3.4 Analytical Metric Development

Instead of predictive model training, this project emphasized the **development of domain-relevant engagement metrics**.
Each metric was framed, computed, and visualized through Python and Tableau. Examples include:

1. **Widget Interaction Success Rate** — % of widget sessions leading to a save or cart.
2. **Conversion Rate by Device and Browser** — cross-sectional user engagement behavior.
3. **Average Tab Changes per Session** — multitasking and attention measure.
4. **User Retention Rate** — returning users within 24 hours.
5. **Recipes with High Views but Low Saves** — identifying underperforming content.
6. **Average Time Between Recipe Views** — engagement flow measure.
7. **Cart Conversion by Dietary Restriction** — behavioral segmentation.
8. **User-Level Non-Conversion Pattern** — identifying passive users.

Each metric was expressed mathematically, verified using Pandas-based transformations, and visualized through Tableau bar charts, scatter plots, or line graphs.

3.5 Visualization and Dashboard Development

Visual exploration and storytelling were conducted using **Tableau Public**, chosen for its ability to integrate seamlessly with preprocessed CSV/Excel outputs.

The process involved:

- Creating **calculated fields** for dynamic expressions such as:
  - `IF [RecipeSaved] = TRUE OR [AddToCart] = TRUE THEN 1 ELSE 0 END`
  - `DATEDIFF('minute', [SessionStart], [SessionEnd])`
- Generating **interactive dashboards** combining multiple plots (success rates, conversions, retention patterns).

- Implementing filters for **Device, DayType, Region**, and **DietaryPreference** to allow granular exploration.

This visualization layer translates numerical patterns into **actionable insights** about user flow, conversion, and preference behavior.

3.6 Insight Generation

Insights were derived through comparative and temporal analysis.
For instance:

- A higher widget success rate indicated intuitive user interface design.
- Tab changes varied significantly by device — indicating context-driven engagement.
- Users with specific dietary preferences (e.g., vegan or gluten-free) exhibited distinct cart behaviors.

Patterns were qualitatively evaluated to identify optimization areas — for example, aligning widget timing with user browsing intent or refining recommendation algorithms.

3.7 Technical Implementation

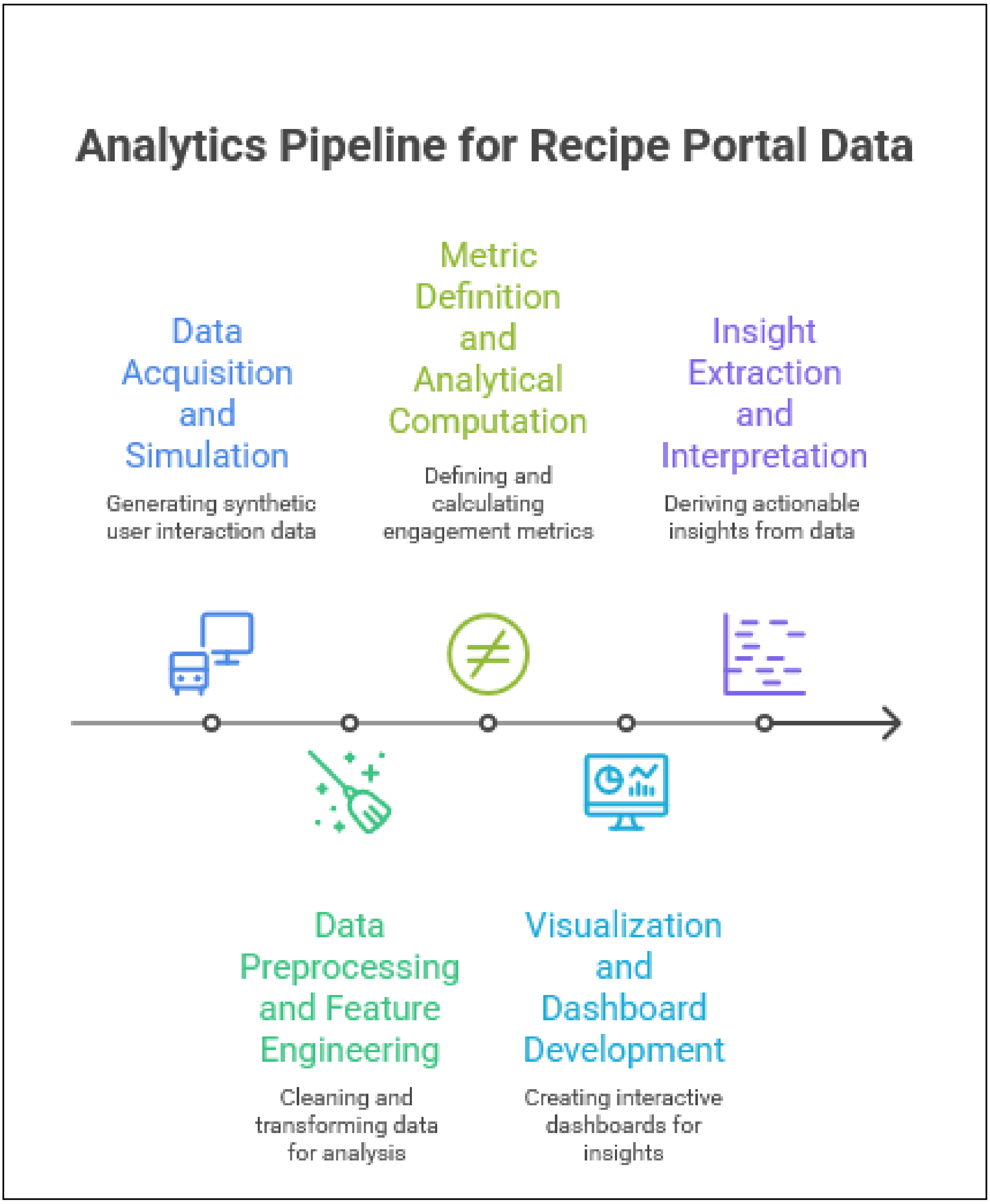
The analytical pipeline was developed entirely in **Python and Tableau**, leveraging the following stack:

- **Python (Pandas, NumPy):** Data preprocessing, cleaning, and metric computation.
- **Matplotlib / Seaborn:** Exploratory visualization for validation and comparison.
- **Tableau:** Interactive dashboards for executive reporting.
- **Google Colab:** Cloud-based execution environment for reproducibility and sharing.
- **Excel/CSV:** Data interchange between computation and visualization modules.

All transformations were documented as reproducible notebooks, ensuring traceability from raw data to insights.

3.8 Tools and Frameworks

1. **Python** – Core programming environment for EDA and metric computation.
2. **Pandas** – Tabular data manipulation and group-based aggregation.
3. **Matplotlib** – Visual validation and trend analysis.
4. **Tableau Public** – High-quality visual analytics and storytelling.
5. **Google Colab/ / Jupyter Notebook** – Interactive notebook execution environment.
6. **Excel** – Lightweight dataset conversion layer for Tableau compatibility.



3.9 Summary

This chapter outlined the **end-to-end analytical methodology** implemented to explore engagement behavior within recipe recommendation platforms.

The complete implementation, including preprocessing scripts, metric computation notebooks, and Tableau visualization workflows, is available in the public GitHub repository for reproducibility and further exploration:

🔗 https://github.com/ankitghosal/ISI_Internship_Autumn25

Chapter 4: Data Analysis and Results

This chapter presents the outcomes of the analytical pipeline designed and implemented for data originating from recipe-oriented portals. Unlike traditional predictive modeling approaches, this project focuses on **data preparation, metric formulation, and visualization-based insight generation**. The aim was to transform raw, unstructured user interaction data into a structured, interpretable analytics system capable of measuring behavioral, engagement, and conversion trends across the platform.

4.1 Overview of Analytical Process

Users now find and assess and engage with online culinary content through recipe-based platforms which have spread across the internet. The large amount of daily recipe production and user interactions needs data analytics to analyze user behavior and improve recommendation systems and platform engagement. The current research benefits from three fundamental areas of advancement which combine data analytics for online food platforms with machine learning approaches for personalization and engagement modeling and pipeline automation for data-driven insights.

The current body of research studies recommendation systems and user engagement prediction but fails to show the complete analytics process which transforms user interaction data into business intelligence-ready visualizations. The following sections evaluate existing research to validate the worth of this project which creates a complete modular analytics system for recipe data platforms.

4.2 Data Cleaning and Preparation

The preprocessing pipeline was designed to transform raw, semi-structured interaction data into an analysis-ready format. It systematically addressed data quality, consistency, and interpretability issues across multiple dimensions.

Key operations included:

- **Standardization of session data:** Combined *Date* and *SessionStart* fields to generate uniform datetime indices, ensuring temporal consistency across sessions.

- **Extraction of interaction features:** Parsed *ActivityLog* entries to capture user actions such as tab changes, recipe views, saves, and cart additions.

- **Categorical normalization:** Standardized categorical fields such as *Device*, *UserAgent*, and *DietaryPreferences* to ensure consistency in grouping and analysis.

- **Derived engagement metrics:** Computed columns like *WidgetSuccessFlag*, *tab_change_count*, and *ScrollTime* were created as proxies for user engagement and interaction depth.

- **Session-level aggregation:** Grouped user activity by *UserID* and *RecipeID* to facilitate both per-session and per-recipe analytics.

Beyond these, numerous additional derived metrics and composite indicators were generated to capture a wider spectrum of behavioral, operational, and content-level dynamics. These included measures of user retention, interaction density, recipe performance, navigation flow, and engagement intensity, among others.

The cleaned and enriched dataset subsequently served as the analytical foundation for all metric computations, visualizations, and exploratory analyses conducted through both Python scripts and Tableau dashboards.

4.3 Metric Computation and Interpretation

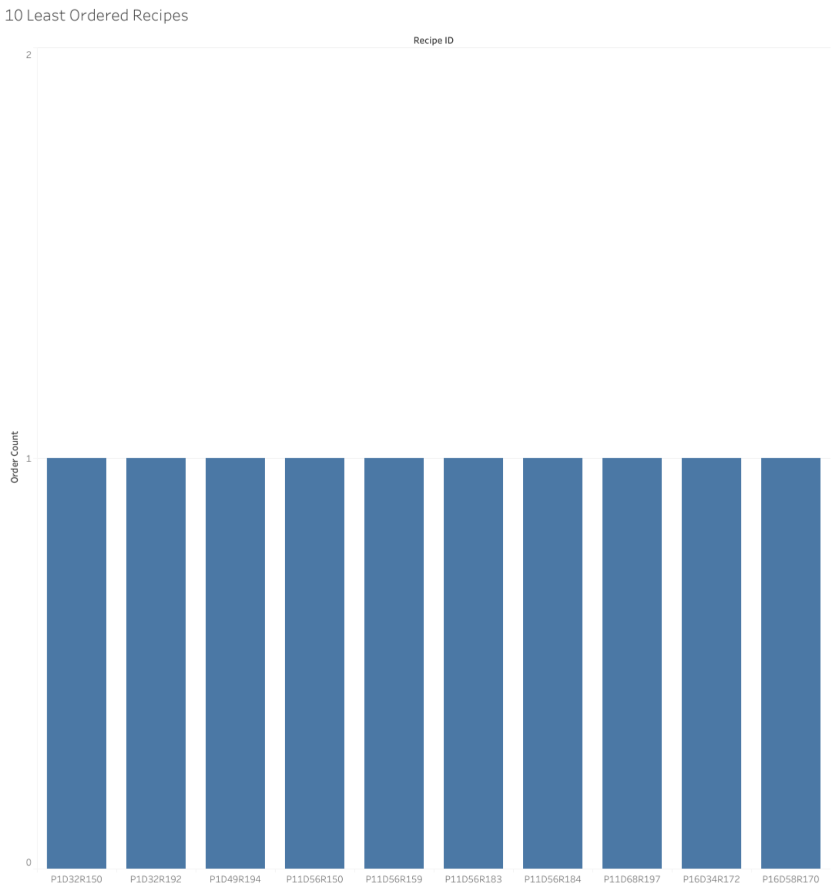
Each metric was expressed as a logical or mathematical transformation designed to quantify platform behavior.
For instance:

- **Widget Interaction Success Rate:** Ratio of widget-triggered sessions that resulted in saves or carts.
- **Average Tab Changes per Device:** Indicates multitasking tendencies and focus levels by device type.
- **User Retention Rate:** Proportion of users returning within a 24-hour window after their last session.
- **Save-to-View Ratio:** Evaluates recipe attractiveness and conversion potential.
- **Average Session Duration:** Measures engagement depth during browsing.
- **Cart Conversion by Dietary Restriction:** Explores how dietary filters influence purchase intent.
- **Users Trying the Same Recipe Multiple Times:** Detects recipe loyalty and repeat behavior patterns.

The metrics were calculated with clear formula definitions, supporting code implementations, and consistent naming conventions to ensure reproducibility.

1. 10 Least Ordered Recipes

This metric counts how many times each recipe was added to cart, then selects the ten recipes with the lowest non-zero counts (i.e., least chosen among those that were chosen at least once). It helps isolate recipes with very low user purchase intent.



Relevance

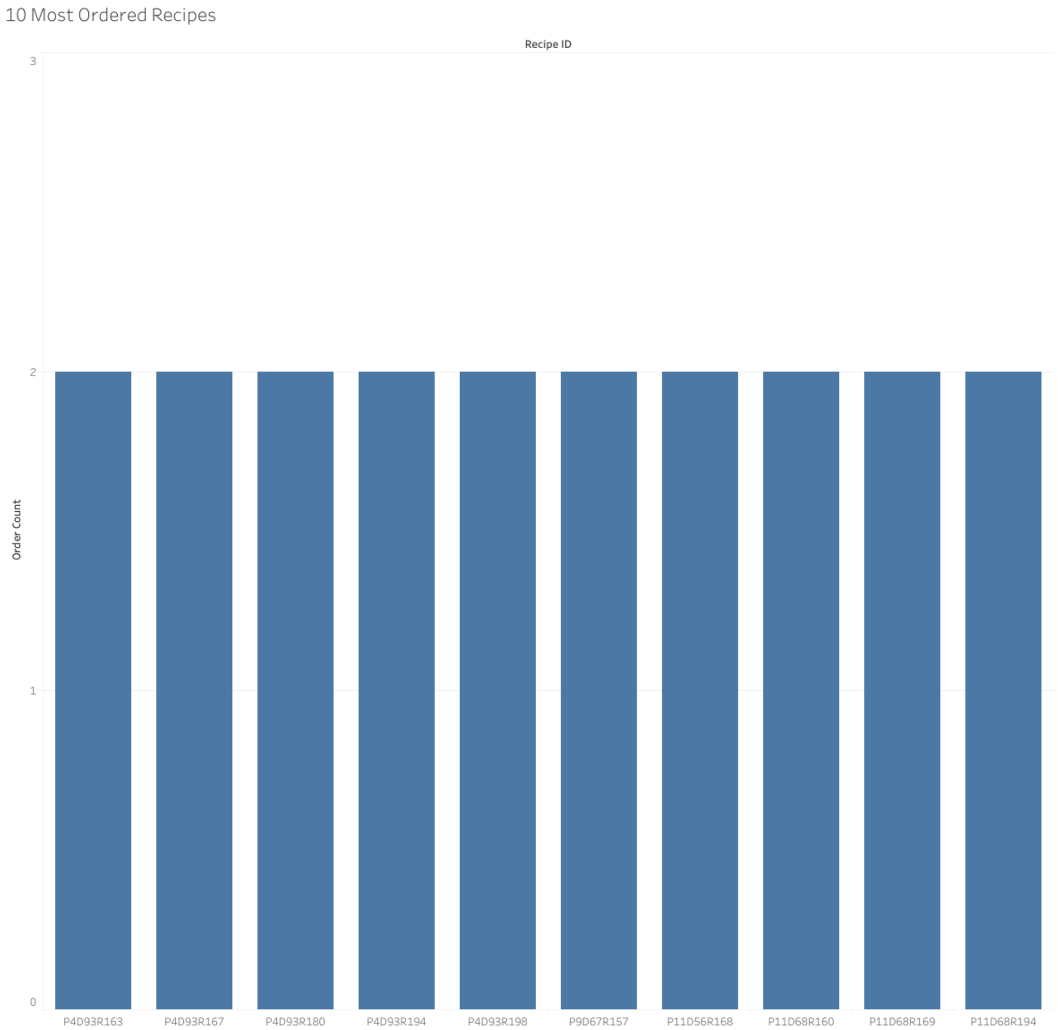
Low cart-add frequency can indicate weak attractiveness, discoverability problems, or friction in presentation. These items are the highest-risk recipes in terms of user engagement and can drag overall platform conversion.

Possible Insights

These low-performing recipes are candidates for improvement through better placement, image/title changes, simplification of ingredients, or different recommendation logic. Cross-checking them with impressions or page views later can help distinguish unpopular recipes vs. poorly surfaced ones.

2. 10 Most Ordered Recipes

This metric ranks all recipes by the number of times they were added to cart and selects the top ten with the highest counts. It reflects recipes with the strongest purchase intent or user preference.



Relevance

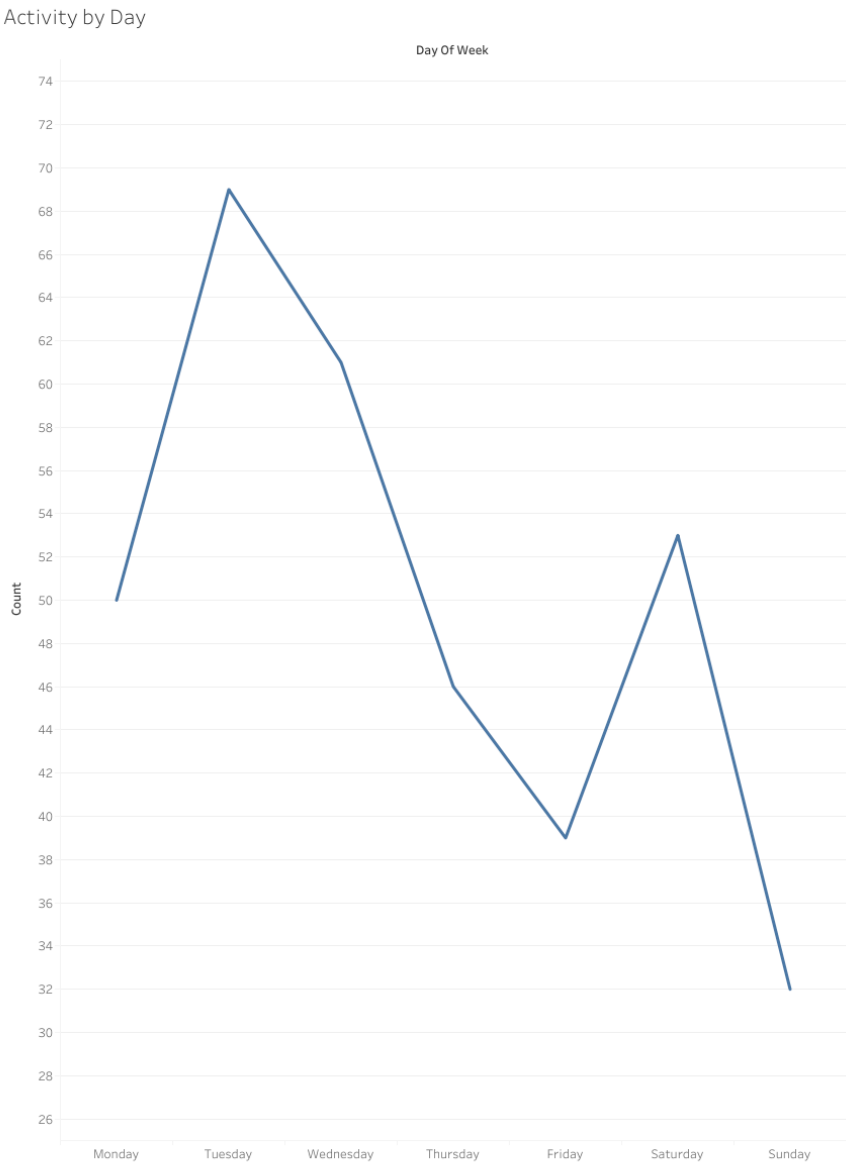
These are the highest-performing recipes in terms of actionable engagement. They serve as a benchmark for what users find appealing, both in content style and presentation.

Possible Insights

High performers can guide recommendation logic, homepage placement, and content replication (e.g., similar cuisine/style). They also serve as a reference for A/B testing — changes that negatively impact these would signal UX or preference misalignment.

3. Activity By Day

Counts the number of user sessions grouped by day of the week to determine when platform engagement is highest or lowest.



Relevance

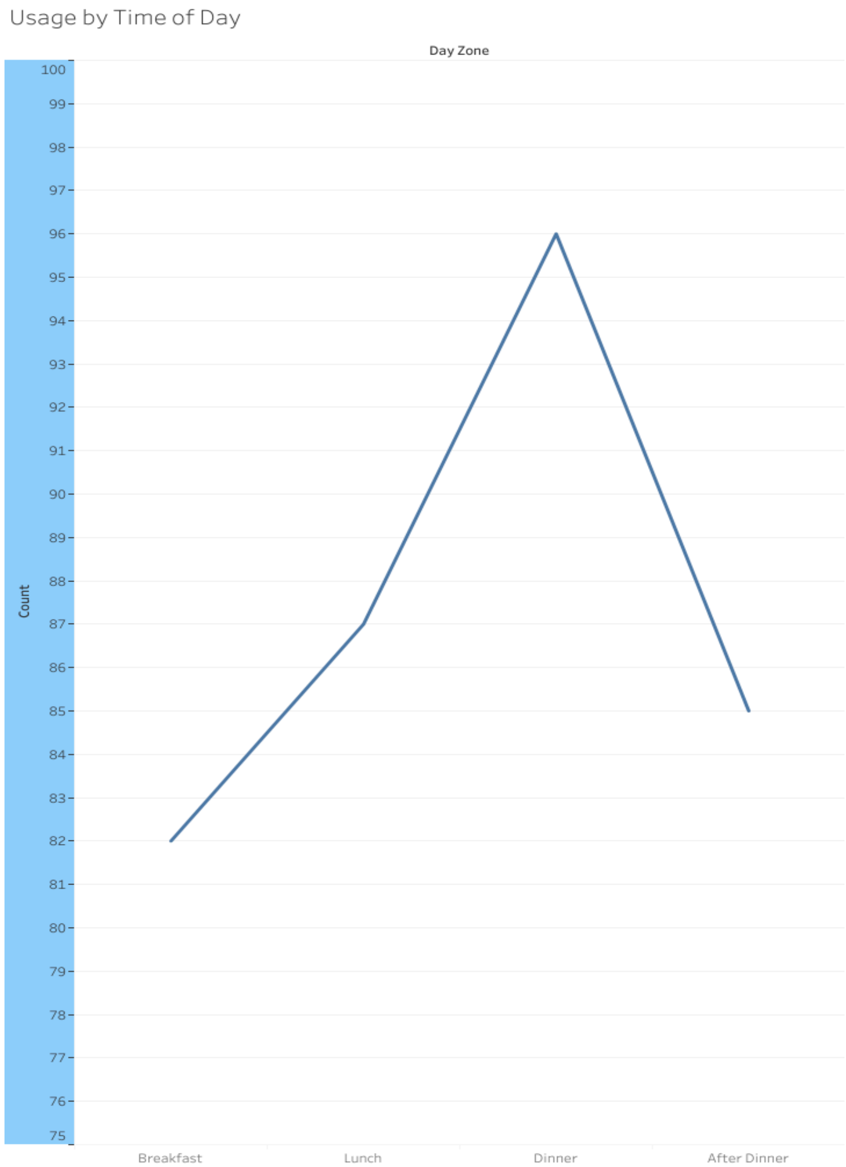
Helps identify peak traffic periods which are useful for scheduling promotions, caching strategies, deployment windows, and notification timing.

Possible Insights

Peak days can indicate when users are most willing to explore or order recipes, while low-activity days can be targeted for retention strategies or UI experiments with lower risk. Patterns may also correlate with weekend vs weekday cooking behavior.

4. Usage by Time of Day

Counts how many user sessions occur during each meal/time category (Breakfast, Lunch, Dinner, After Dinner), based on the DayZone column.



Relevance

Shows when users are most active during the day, which helps infer when they are most likely planning or browsing for food-related decisions.

Possible Insights

Time-based engagement can guide when to surface recommendations, push notifications, or dynamic promotions. It can also inform meal-specific personalization (e.g., quick breakfast recipes vs elaborate dinner recipes).

5. Users by Region

Counts the number of unique users segmented by geographic region to understand where the platform has the highest and lowest user presence.

Relevance

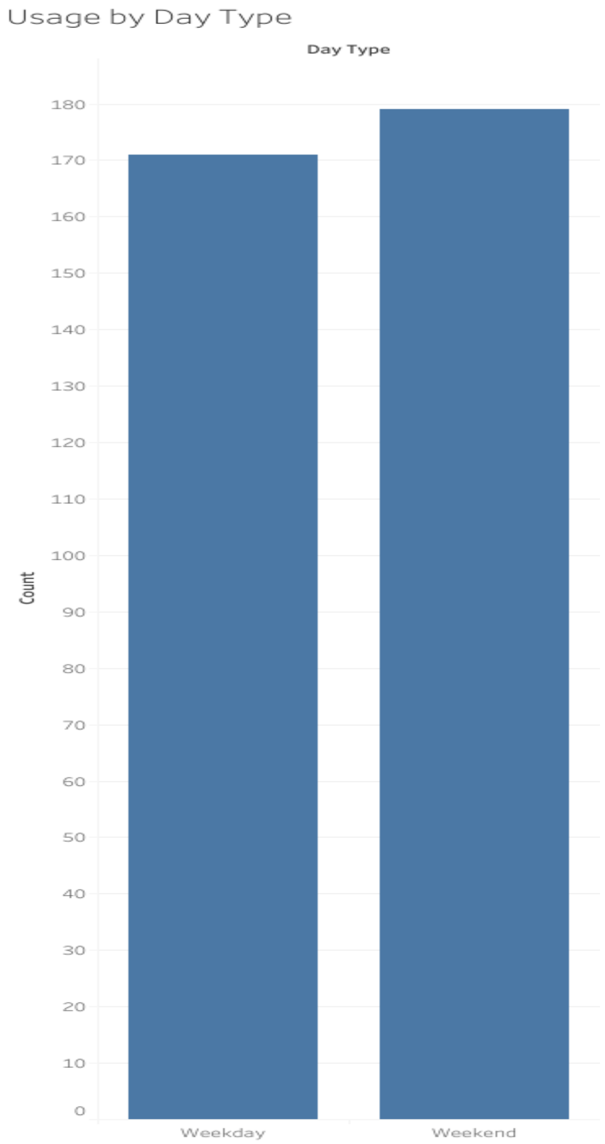
Regional distribution helps assess market reach and adoption across different locations, making it useful for targeting and platform growth decisions.

Possible Insights

High-user regions can be used for localized content or regional cuisine tailoring, while low-user regions highlight potential expansion or marketing opportunities. Can also be paired later with cart/add-to-cart metrics to detect geographic preference patterns.

6. Usage By Day Type

Counts user sessions grouped into weekday vs weekend activity to compare behavior across working and non-working days.



Relevance

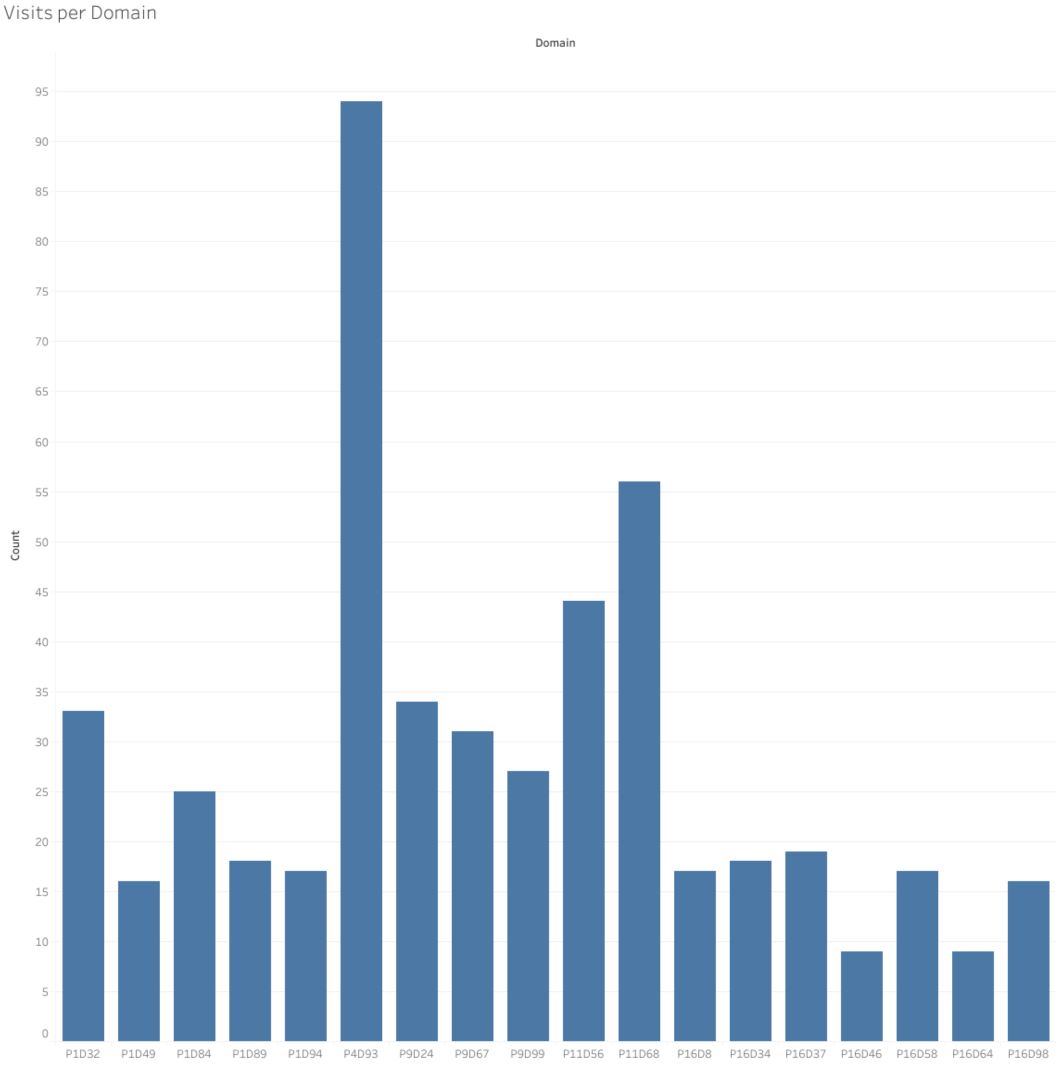
Helps identify whether engagement is more routine-based (weekday usage) or leisure-driven (weekend usage).

Possible Insights

If weekends dominate, users treat the platform as an exploratory or planning tool; if weekdays dominate, it may be tied to meal-prep convenience during workdays. Can guide timing of promotions or feature rollouts.

7. Visits Per Domain

Counts how many sessions originated from each domain or section of the platform (e.g., homepage, recipe page, chat, widget pages).



Relevance

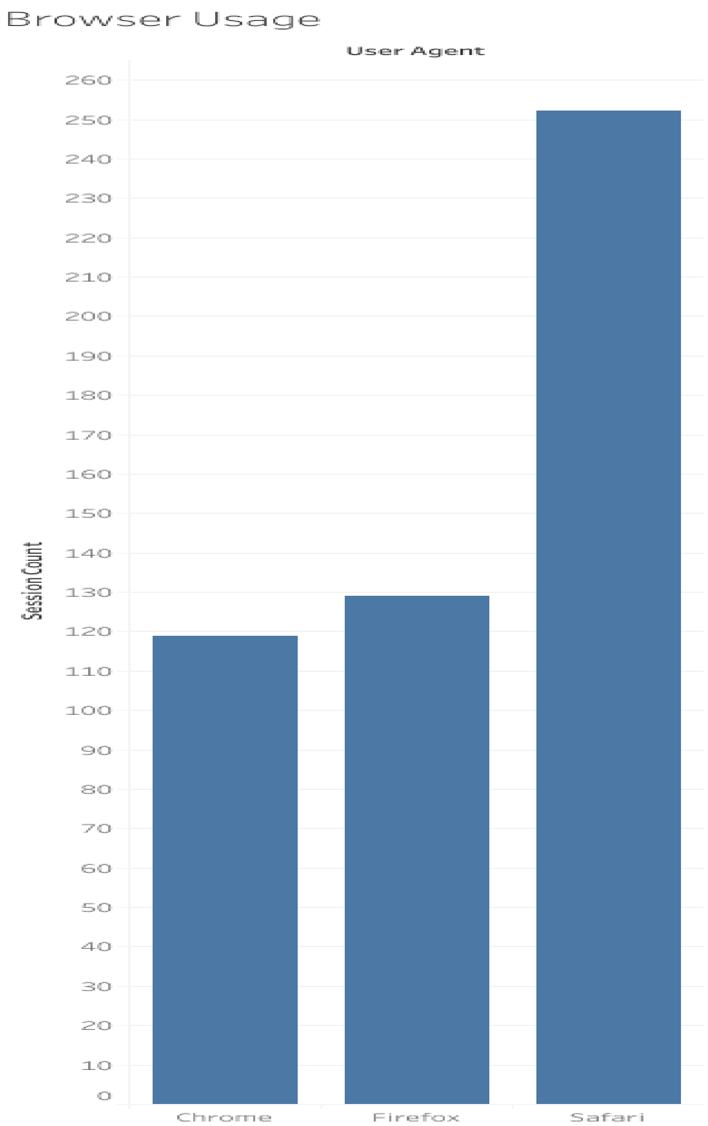
Shows which parts of the platform attract the most traffic and which features users actively navigate to.

Possible Insights

High-traffic domains indicate strong user interest or good entry points, while lower-traffic ones may need restructuring or better linking. Can later be paired with conversion metrics to identify high-view but low-action domains.

8. Browser Usage

Extracts the browser type from each session's user-agent string and counts the number of sessions per browser.



Relevance

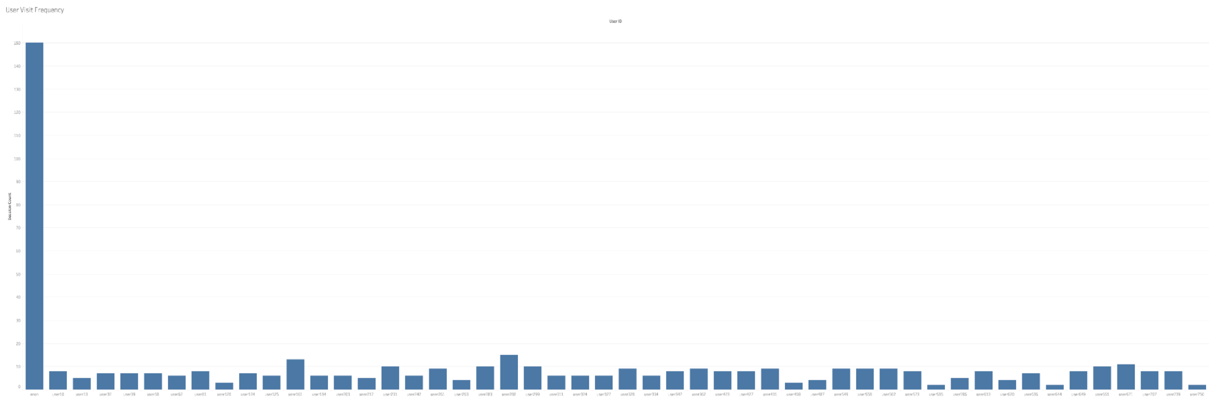
Useful for understanding the dominant browsing environments so the platform can be optimized for the devices and engines most of the audience actually uses.

Possible Insights

Helps prioritize browser-specific testing and performance tuning. Also highlights legacy or low-usage browsers that may not justify heavy optimization effort.

9. User Visit Frequency

Counts how many sessions each user generated, and plots the distribution to show how often users return to the platform.



Relevance

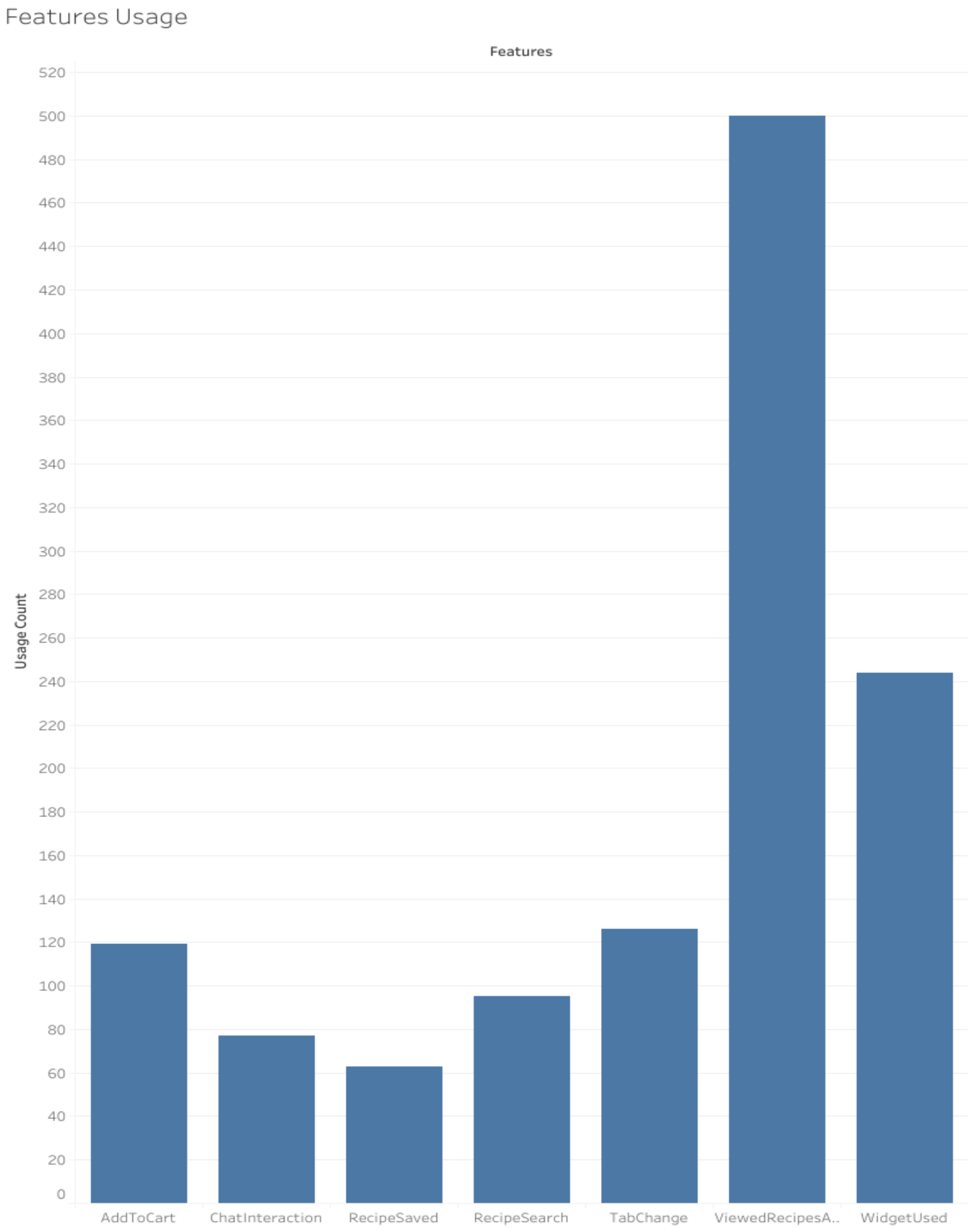
Measures repeat usage and early signs of retention. A higher share of repeat visitors usually indicates stronger engagement or habitual use.

Possible Insights

Can help separate casual one-time users from loyal or high-intent users. Later, this metric can feed into cohort or churn analysis and identify which features correlate with repeat visits.

10.Features Usage

Counts how many sessions involved each major platform feature (chat, search, add-to-cart, saves, tab changes, recipe views, widget interactions) to determine which features are most actively used.



Relevance

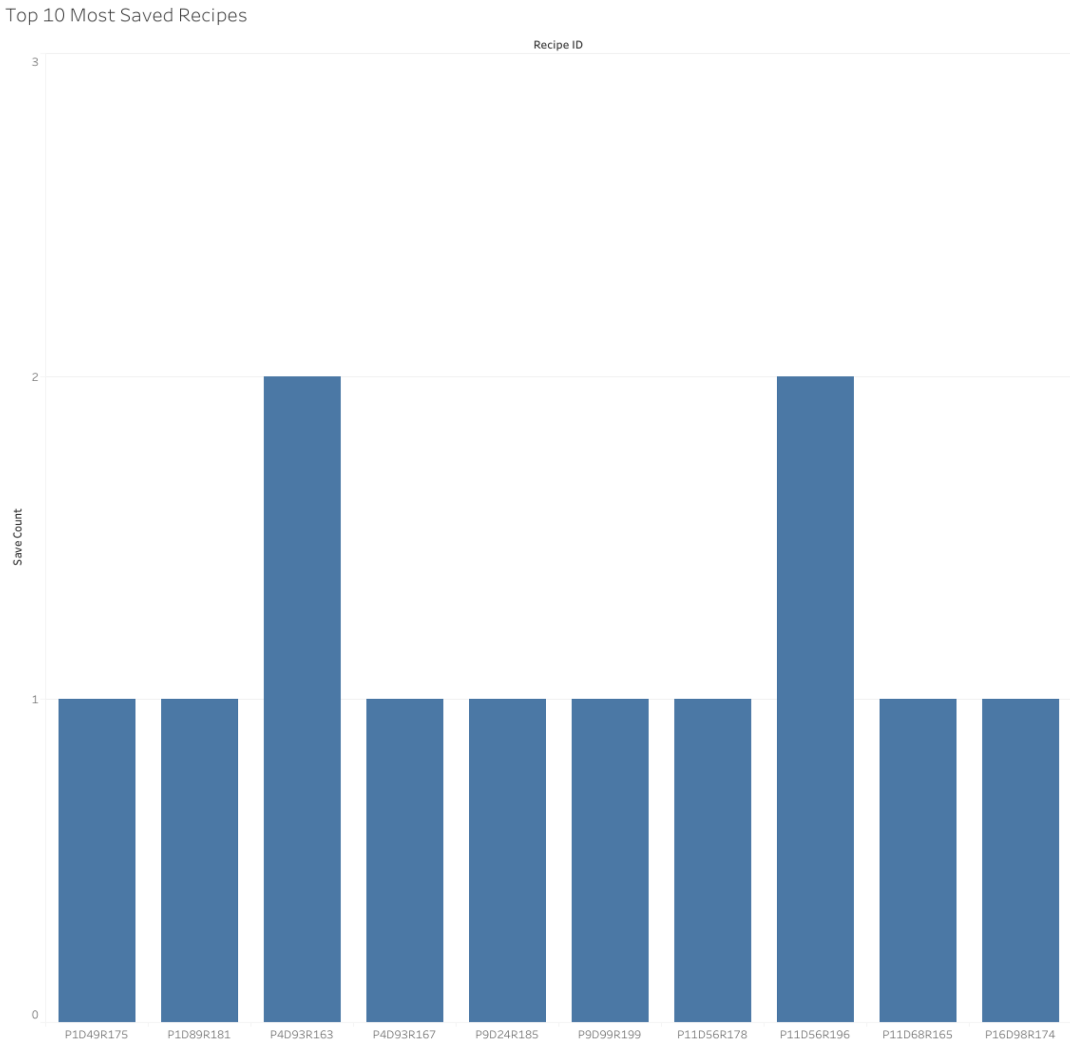
Shows which interactions drive actual engagement and which features have low utilization or may need UX improvements.

Possible Insights

High-usage features can be prioritized for optimization or scaling, while low-usage ones may require redesign or better surfacing. Can also be paired later with conversion metrics to see which features correlate with actual orders.

11.Top 10 Most Saved Recipes

Counts how many times each recipe was saved to favorites/collections and displays the top ten recipes with the highest save frequency.



Relevance

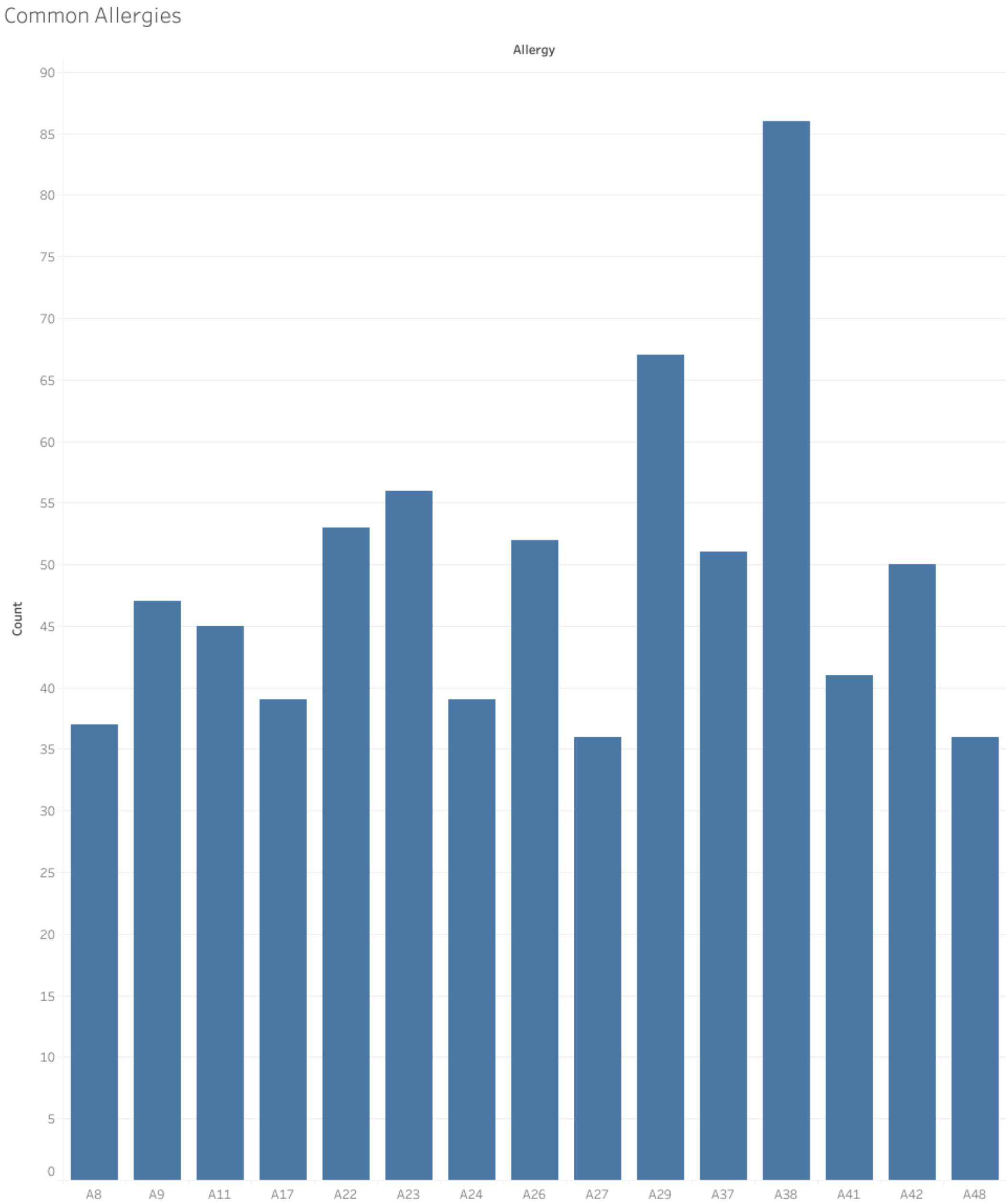
Saving a recipe reflects long-term interest or intent to revisit later, which is a stronger engagement signal than just viewing.

Possible Insights

Highly saved recipes can indicate content users consider valuable or aspirational, even if they don't immediately convert to cart. Later analysis can compare "saved vs ordered" to detect recipes that are liked conceptually but not acted on (conversion gap).

12.Common Allergies

Extracts and counts the frequency of allergy codes reported by users to identify which dietary restrictions are most common.



Relevance

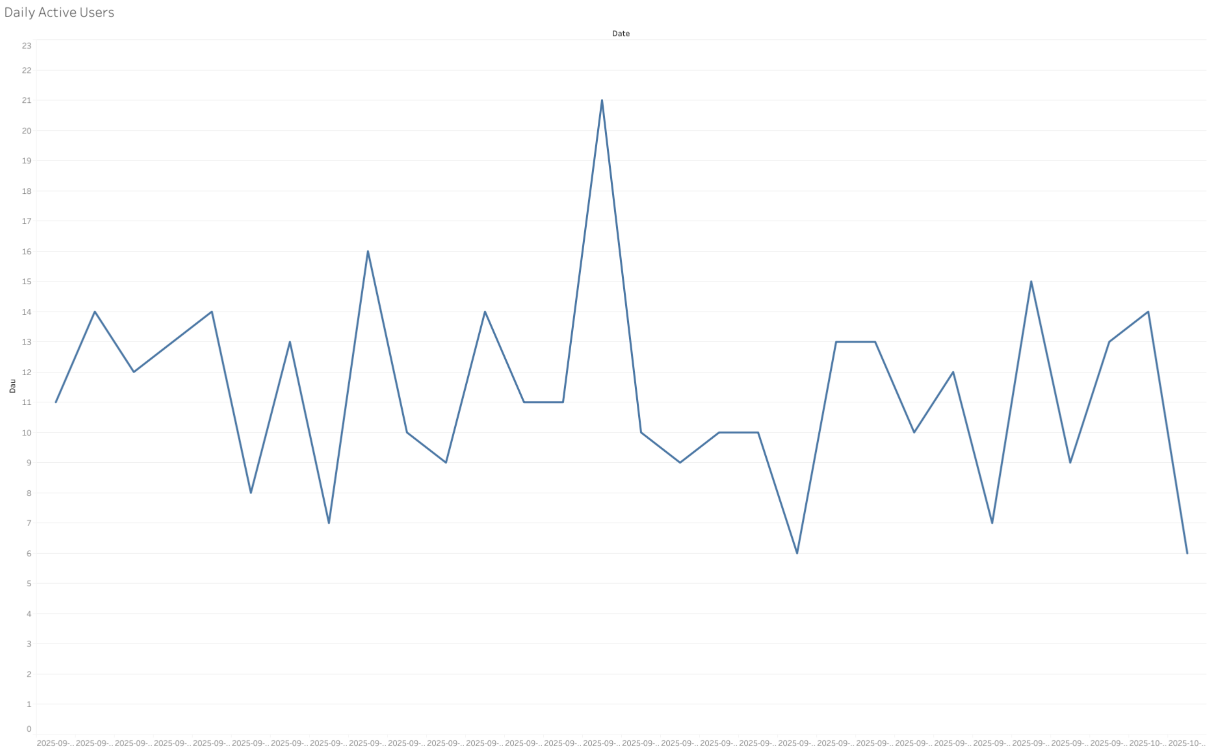
Understanding allergy prevalence helps tailor recipe recommendations, filtering, and ingredient substitutions for accessibility and safety.

Possible Insights

High-frequency allergies (like nuts, dairy, gluten etc.) can guide recipe tagging, UI prompts, or auto-filter options. Later this can be cross-checked with saved/ordered recipes to see whether allergy-friendly recipes influence engagement.

13.Daily Active Users

Counts the number of unique users active on each day, giving a day-level view of recurring platform usage.



Relevance

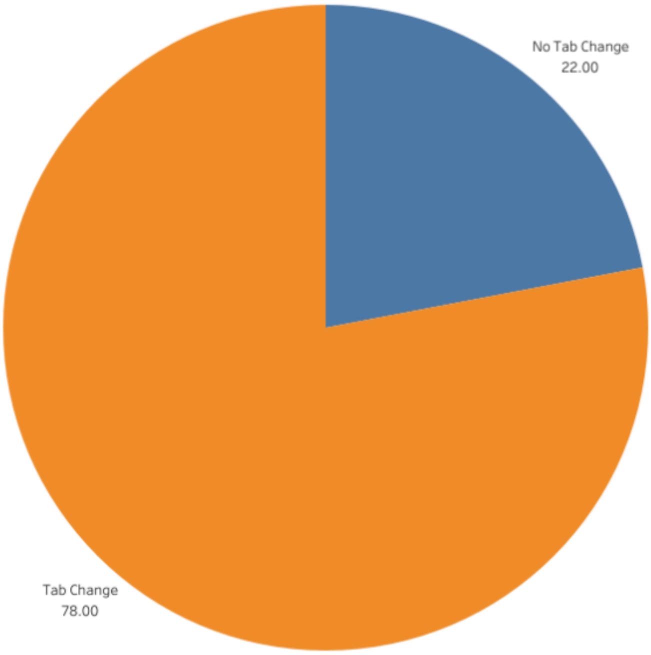
DAU is a core engagement indicator that reflects how consistently users return to the platform on a daily basis.

Possible Insights

Can show spikes linked to events, promotions, or meal planning behavior. Later this metric can be paired with time-of-day or weekday/weekend usage to detect repeating engagement cycles.

14.Tab Change Percentage

Calculates the percentage of unique users who switched between tabs at least once during their session.



Relevance

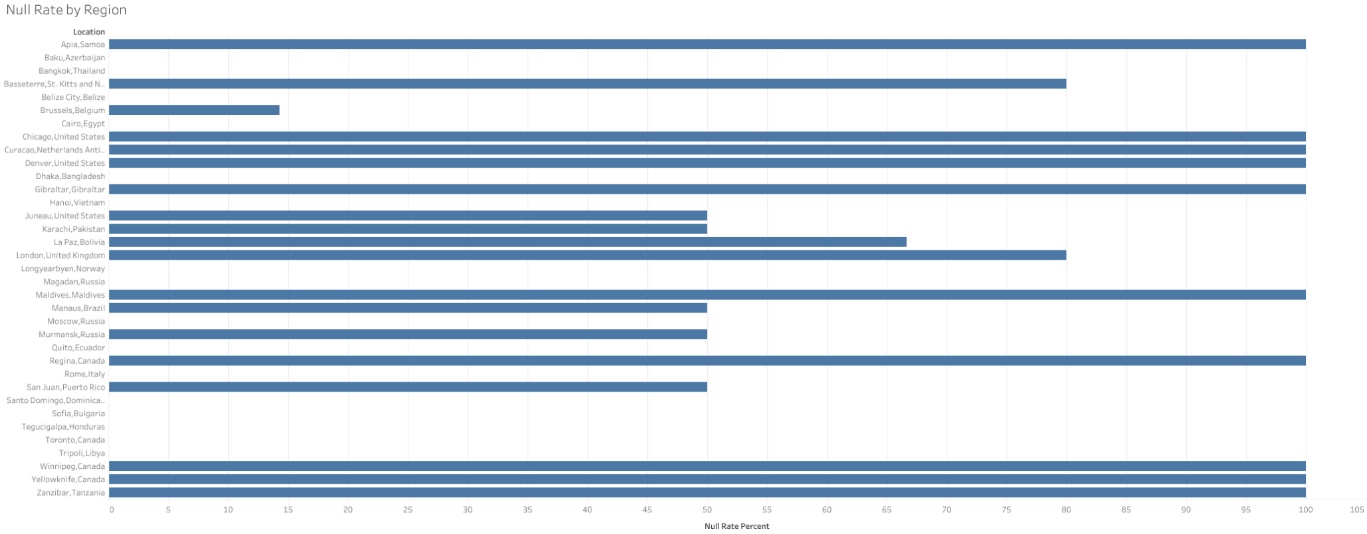
Tab switching is a proxy for deeper navigation and multi-section exploration, indicating higher interaction depth than simple page viewing.

Possible Insights

A higher percentage suggests users actively explore the platform beyond a single screen, while a low percentage may indicate friction or poor discovery of available sections. Later this can be compared with engagement metrics like saves/orders to see if tab navigation correlates with conversions.

15.Null Rate by Region

Measures the percentage of search sessions in each region where the user did not follow up with any meaningful action (no recipe viewed and no add-to-cart).



Relevance

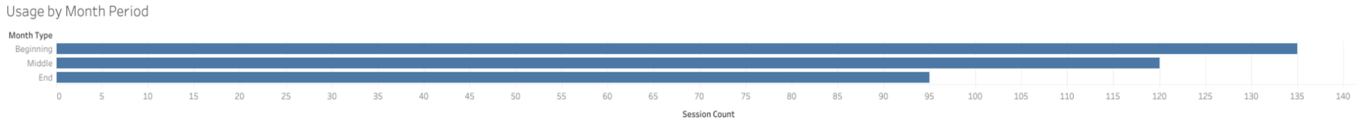
This is an early signal of search inefficiency or mismatch between what users expect vs what the platform returns. High null rate = users search but find nothing worth engaging with.

Possible Insights

Regions with high null rate may need localized search tuning, better ranking logic, or region-specific surfacing of cuisines/ingredients. Later this can be paired with allergy or preference data to see whether irrelevance is cultural, dietary, or UI-driven.

16.Usage by Month Period

Groups user sessions into beginning, middle, and end of the month to analyze when engagement tends to peak within the monthly cycle.



Relevance

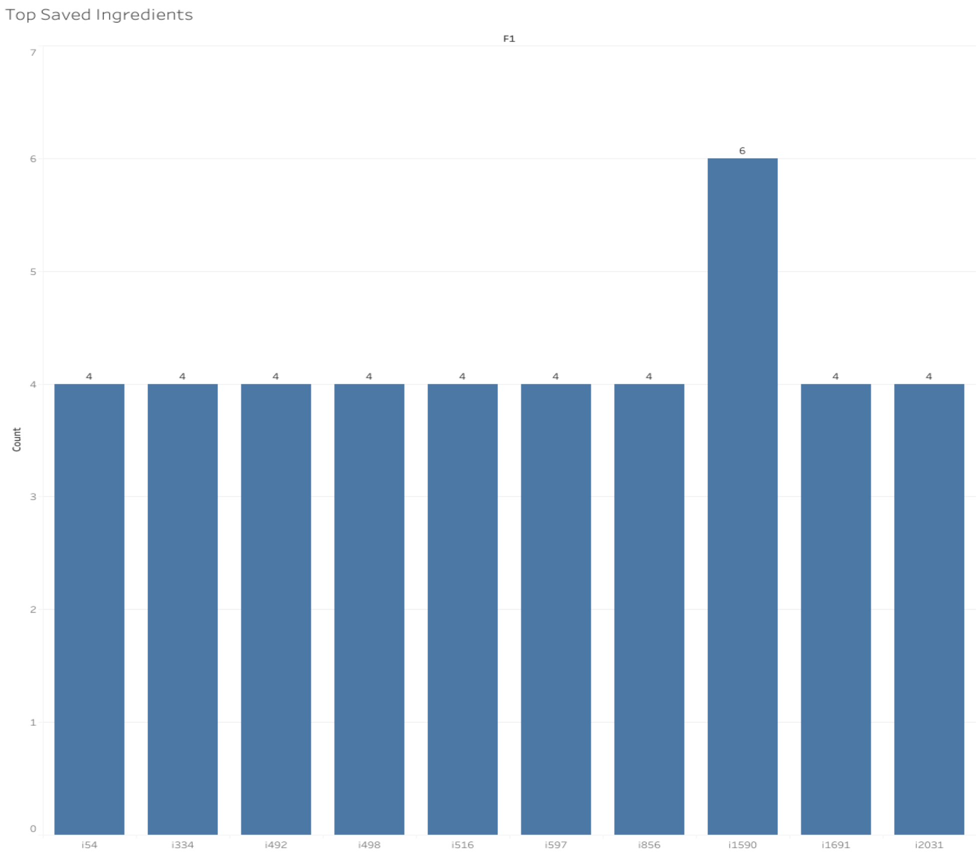
Month-period behavior can reflect budgeting patterns, grocery restock cycles, or routine meal planning timing.

Possible Insights

If beginning-of-month usage is higher, users may be planning fresh meals after grocery purchase; if end-of-month dominates, it might indicate late-cycle planning or deal-seeking. This can later inform scheduling of campaigns tied to spending cycles.

17.Top Saved Ingredients

Looks at the ingredients used in the top 10 most saved recipes, counts their frequency, and identifies which ingredients appear most often in highly saved content.



Relevance

Ingredients act as a proxy for user taste preference. If certain ingredients repeatedly show up in saved recipes, they are likely driving desirability.

Possible Insights

These common ingredients can guide recommendation logic, personalization, and inventory/availability hints. Later this can be cross-referenced with add-to-cart data to see which ingredients move users from “interest” to “intent.” If some ingredients appear often in saved but rarely in ordered recipes, that signals aspiration vs purchase friction.

18.Ingredient Synergy Pairs

Counts how frequently pairs of ingredients co-occur within the top saved recipes, revealing common ingredient combinations rather than just individual popularity.

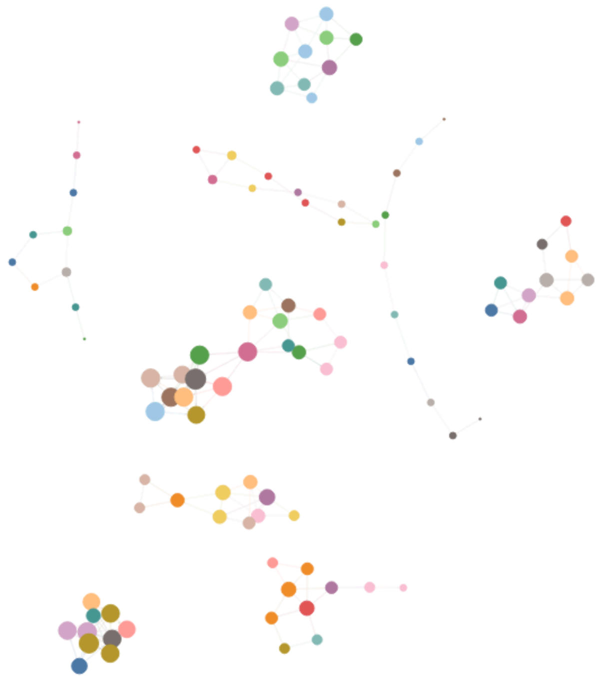


Fig. 18 a.

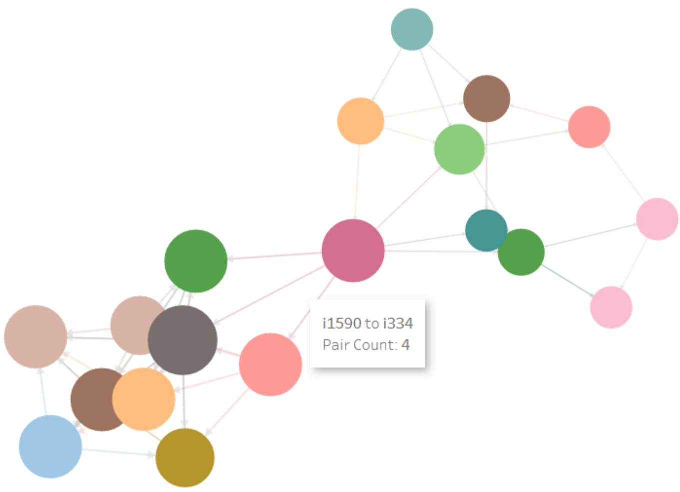


Fig 18 b.

Fig. 18 a. is a view of the overall network, where each node signifies an ingredient and connected nodes represent ingredient pairings. The pair count as shown in Fig 18 b. Is the number of recipes in which the two ingredients are found together.

Relevance

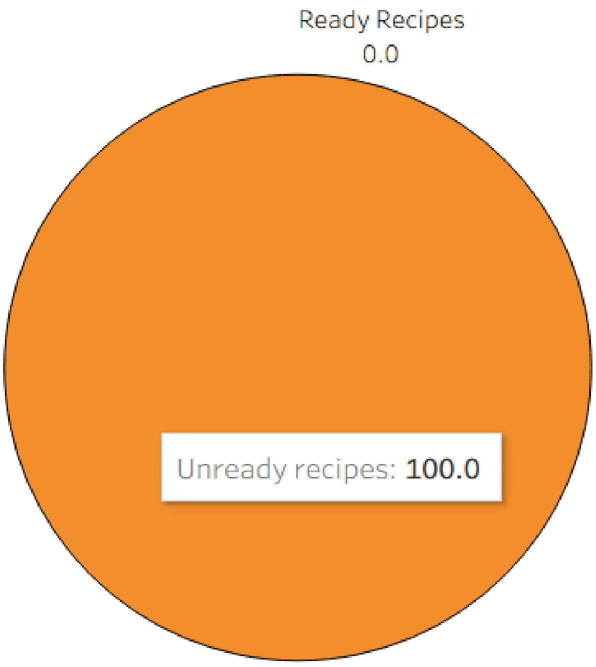
Identifying which ingredients are most often used together gives insight into flavor pairing preferences and typical recipe composition patterns favored by users.

Possible Insights

These ingredient pairs can be used to power smarter recommendation systems (e.g., "if a user likes X, suggest recipes containing X+Y"), and can also inform recipe suggestions, bundling logic, or ingredient substitution models. They can later be compared against cart data to see if popular pairings also convert into purchases.

19. Users with Ready Recipes

Identifies users who already have ≥50% of the required ingredients for at least one recipe they viewed or saved, based on their pantry list.



Relevance

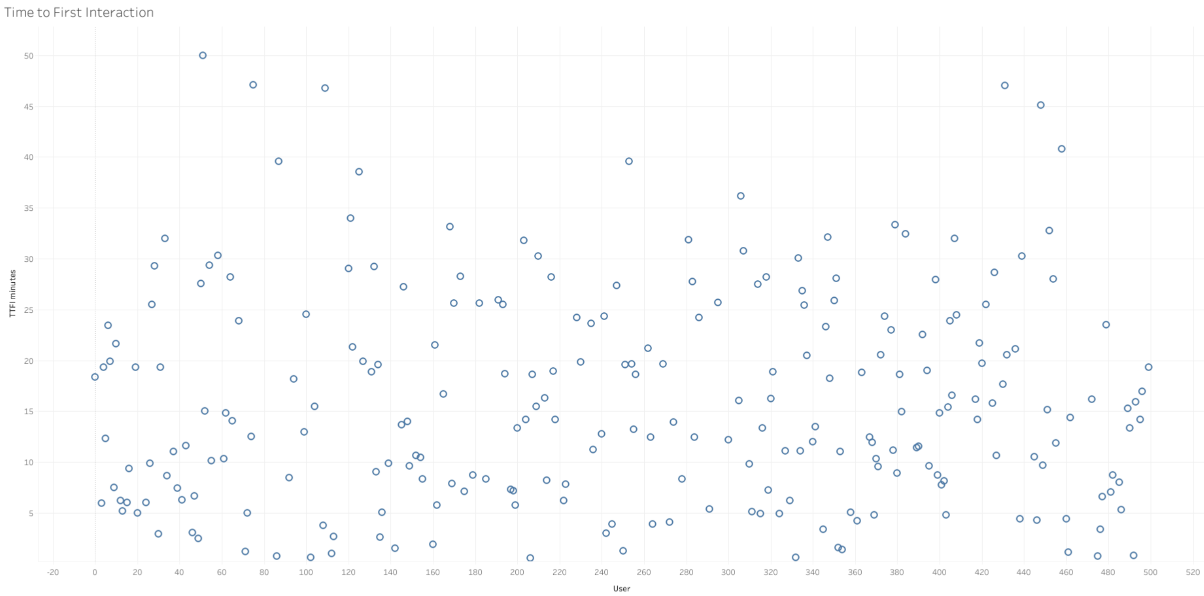
This measures how many users are practically able to cook what they are interested in, not just browsing. It links intent to feasibility.

Possible Insights

A high ready-match rate indicates strong conversion potential (users can actually cook what they like), while a low rate signals the need for substitutes, alternative ingredient suggestions, or Pantry-aware recommendations. Later this can support "one-click cookable" filtering or nudging users toward recipes they can immediately make.

20.Time to First Interaction

Measures the time delay (in minutes) between session start and the user's first meaningful interaction (widget open or chat start).



Relevance

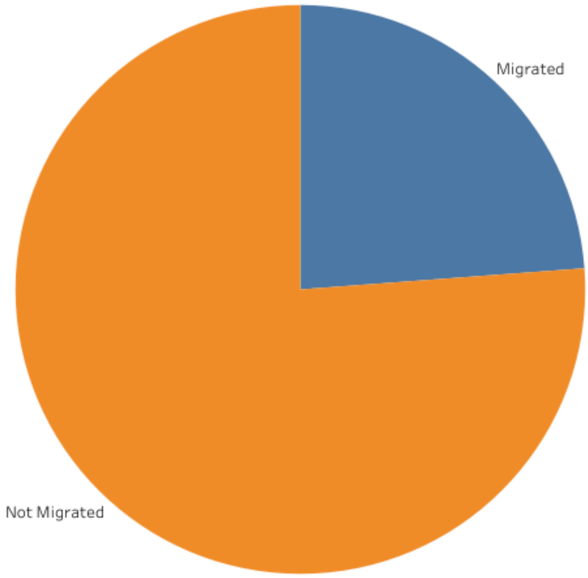
TTFI reflects how quickly users engage beyond passive browsing. Lower values indicate faster engagement and a smoother onboarding/entry flow.

Possible Insights

High TTFI suggests hesitation or confusion before interaction, hinting at UI friction or weak initial hooks. This metric can later be tracked against UX changes or landing-page tweaks to evaluate improvement in early engagement.

21.Save to Cart Migration

Measures the percentage of user–recipe pairs that move from a save event to an add-to-cart event, indicating whether saved interest eventually turns into purchase intent.



Relevance

This is a commitment-progression metric: it shows how many users actually act on recipes they save instead of treating saving as a passive bookmark.

Possible Insights

Low migration means users like the idea of the recipe but aren't ready to cook it (missing ingredients, complexity, effort, etc.). A higher rate means saving is a strong leading indicator of conversion. Later this metric can help identify which recipe traits encourage actual follow-through.

22.Widget Reopen

Calculates the percentage of sessions where the widget was opened more than once, indicating that the user closed it and later returned to it within the same session.

Widget Reopen

Widget Reopen	
Widget One-time Open Rate	100
Widget Reopen Rate	0

Relevance

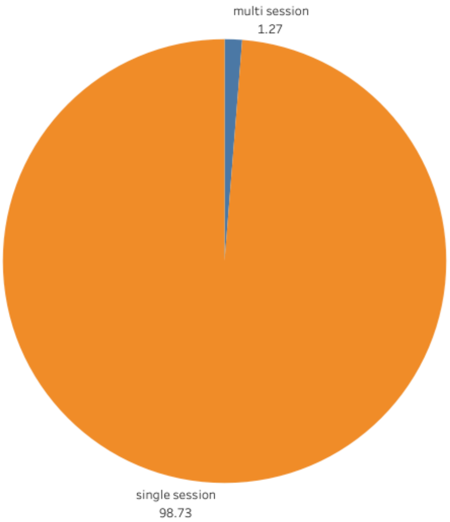
Widget reopening is a proxy for persistent curiosity or utility — if users come back to it, the widget is serving an ongoing decision-making role rather than a one-off interaction.

Possible Insights

A high reopen rate suggests that users treat the widget as a reference tool while browsing other features. A low rate may indicate that once opened, it doesn’t provide enough value to return to, or users complete their task in one go. This metric can later be compared to conversion behavior from widget-assisted sessions.

23.Multi Session Commitment

Measures how many user–recipe pairs move from a view in one session to a save or add-to-cart in a later session, indicating deliberate consideration over time rather than impulsive action.



Relevance

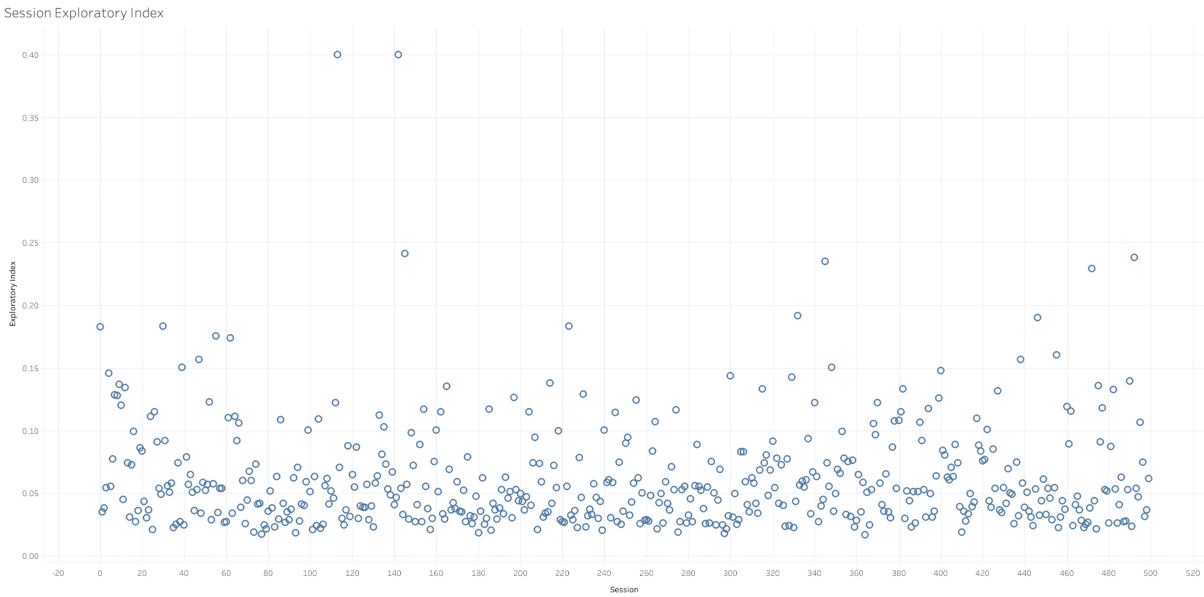
Captures long-term intent and planning behavior instead of single-session curiosity. This metric reflects thoughtful commitment and recipe "stickiness."

Possible Insights

A higher commitment rate means recipes are compelling enough for users to come back to them after initial browsing. A lower rate may indicate weak recall value or lack of follow-through. Later it can help isolate recipes that require multi-touch nurturing vs instant-appeal recipes.

24.Session Exploratory Index

Defines how exploratory a session is by dividing the total number of unique recipes + unique domains viewed by the session duration (in minutes). It captures how actively a user browses per unit time.



Relevance

This goes beyond raw engagement counts by controlling for time. It helps distinguish slow, passive browsing from fast, high-curiosity exploration.

Possible Insights

Higher values indicate active discovery behavior (good for recommendation models), while low values may reflect focused intent or low engagement. Over time this metric can help segment users into explorers vs direct-intent users.

25.Conversion Rate for Users with Allergies

Compares the conversion rate (save + add-to-cart) of highly allergy-constrained users (3 or more allergies) against the overall platform average.

Conversion Rate for Users with Allergies

Metric Name	
All Users Conversion Rate	33.40000000
Allergy Users' Conversion Rate	33.45070423
Allergy Users' Success Ratio	1.00151809

Relevance

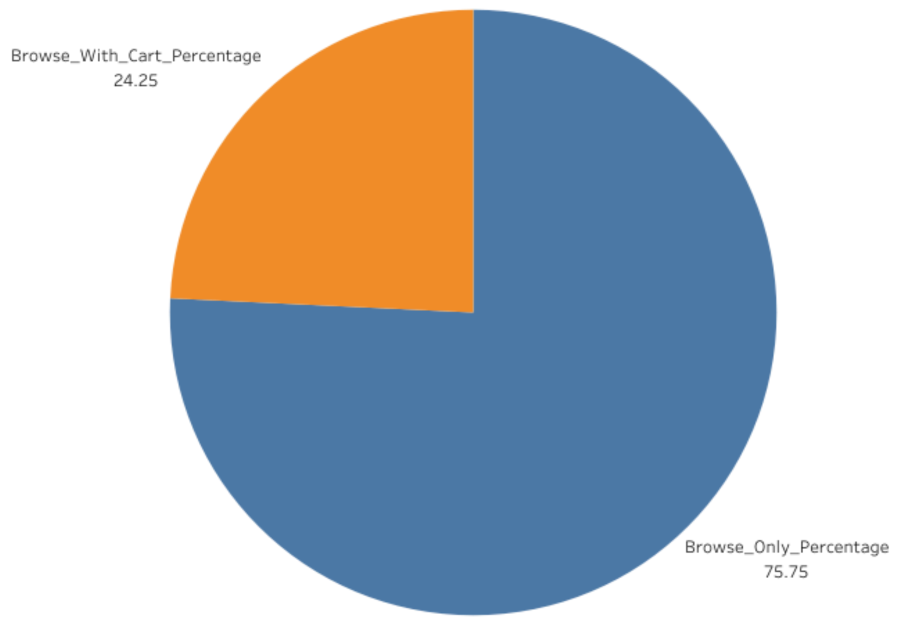
Shows how well the platform serves users with dietary constraints. If their conversion is lower than average, recipe accessibility or filtering may be failing them.

Possible Insights

A lower rate signals users struggle to find "safe" recipes, implying need for better allergy-aware recommendations or substitutes. A higher rate means filtering/availability is working well. This metric can later drive accessibility-focused personalization.

26.Browsing & Added to Cart

Measures the percentage of sessions where users viewed or searched for recipes but did not proceed to add anything to the cart.



Relevance

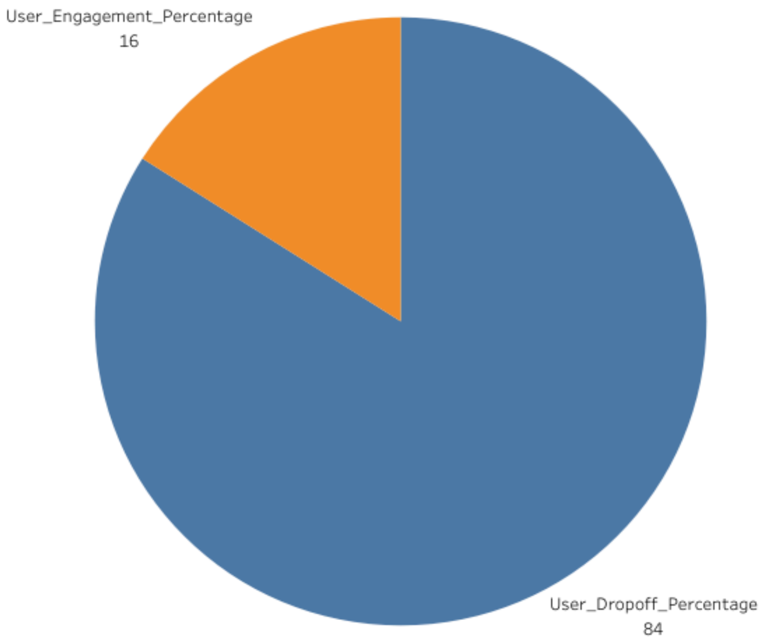
This represents pure exploration without purchase intent, highlighting the gap between discovery and action.

Possible Insights

A high browsing-only rate may indicate friction after discovery (ingredients, complexity, dietary mismatch) or low motivation to commit. Tracking this over time helps evaluate how effectively the platform nudges users from interest to intent.

27.User Drop Off

Percentage of users who start a session but perform no meaningful interaction at all (no view, no search, no save, no cart, no widget use, no chat).



Relevance

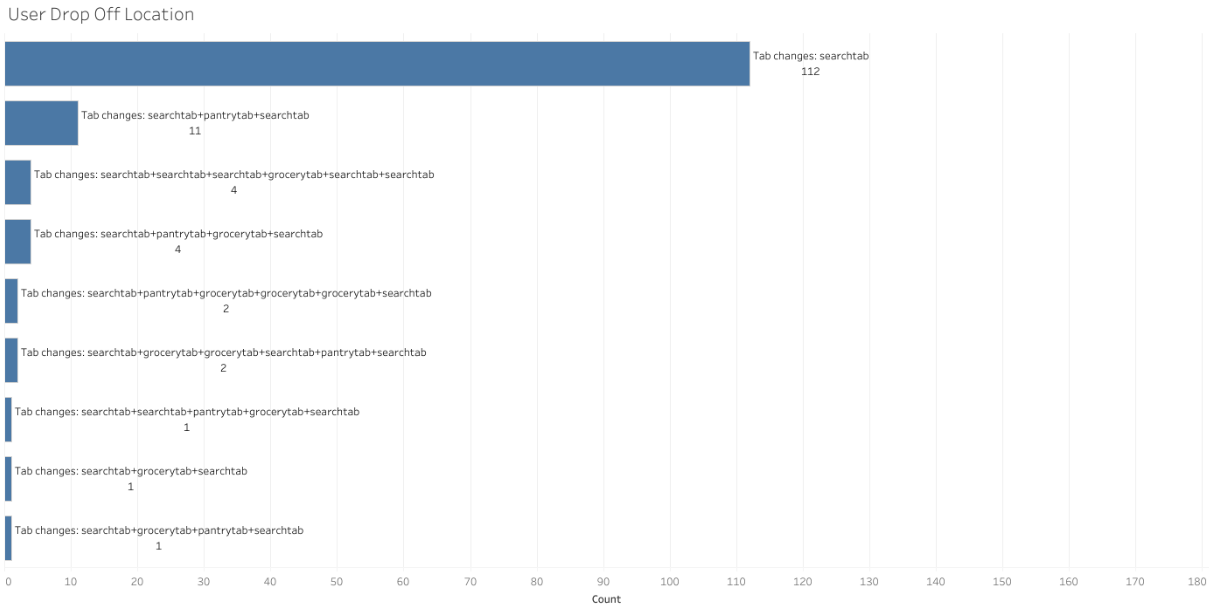
This captures users who bounce immediately, indicating low initial engagement or poor first-touch experience.

Possible Insights

A high drop-off rate suggests onboarding or discovery friction, meaning users don't find value fast enough. Tracking this over time helps evaluate whether UI improvements or better entry-point recommendations reduce abandonment.

28.User Drop Off Location

Identifies the last recorded event or page before a user abandoned the session without performing any meaningful engagement.



Relevance

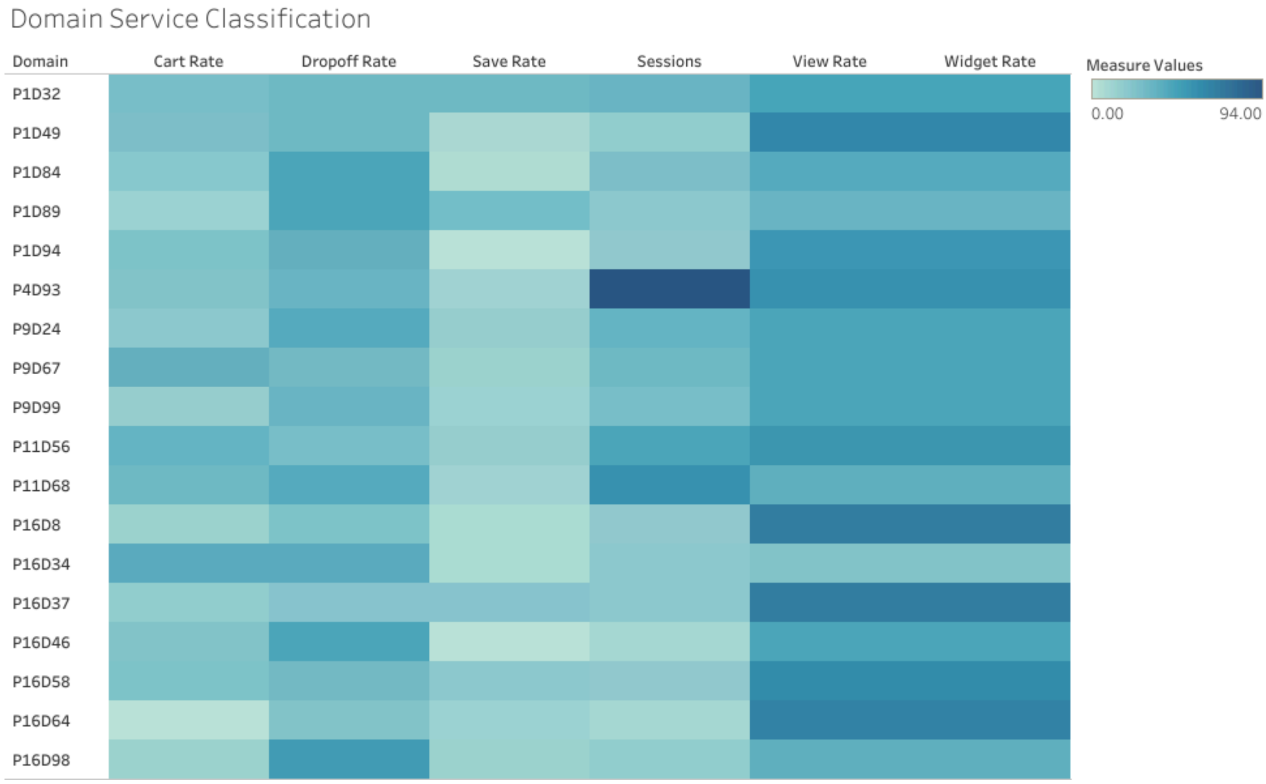
This helps pinpoint where users lose interest in the flow, not just how many drop off.

Possible Insights

If drop-offs cluster around certain pages (e.g., landing page, list view, search entry), it likely indicates friction or poor clarity at that step. This metric guides where UX fixes or onboarding nudges are most needed.

29.Domain Service Classification

Aggregates engagement behavior per domain (view, save, cart, widget usage, or drop-off) and classifies each domain by its dominant user behavior.



Relevance

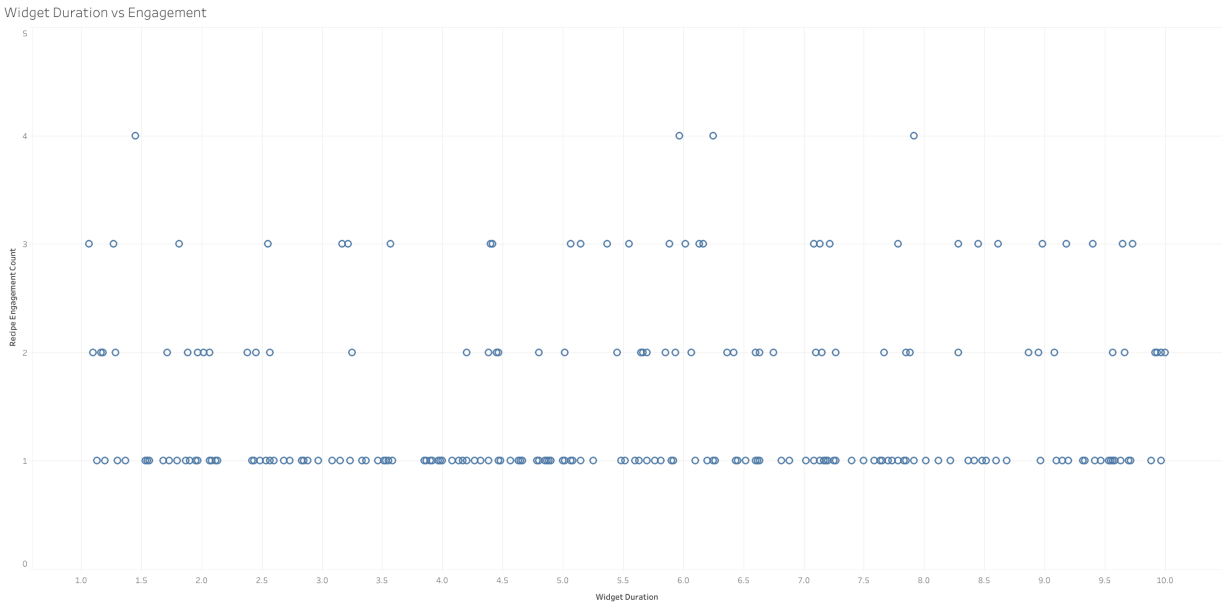
This shows what role each domain is actually serving for users — browsing-oriented, intent-focused, commerce-heavy, or assistive — instead of assuming its design purpose matches its usage.

Possible Insights

Helps detect mismatches between intended design and real user behavior. For example, a supposed commerce domain behaving as a browsing domain indicates weak conversion flow. This classification also supports partner evaluation and UI restructuring by service-type function.

30.Widget Duration vs Engagement

Measures the relationship between how long users keep the widget open and how many recipes they subsequently view or save in the same session.



Relevance

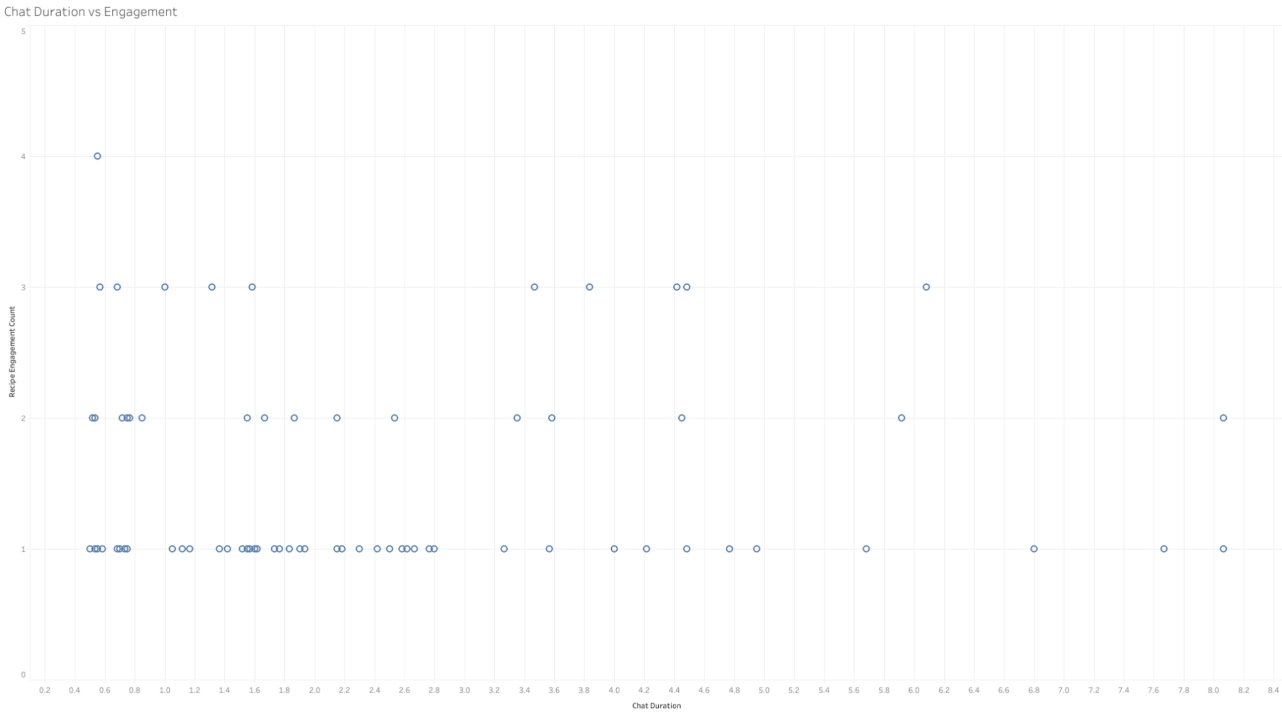
Widget duration acts as a proxy for information-seeking depth. If longer widget use correlates with more engagement, it validates the widget's role in discovery and recipe decision-making.

Possible Insights

A positive correlation suggests the widget actively drives exploration. Weak or negative correlation would imply the widget isn't helping users progress and may need UI/content improvements. This metric can later support tuning widget design for stronger downstream engagement.

31.Chat Duration vs Engagement

Measures how chat session length correlates with downstream engagement, i.e., how many recipes are viewed or saved after or during chat usage.



Relevance

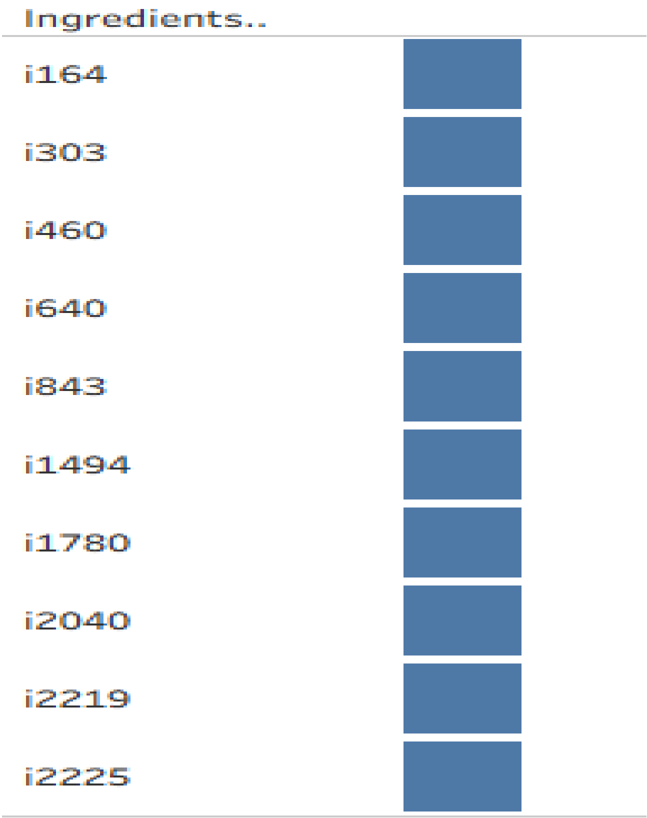
This captures whether the chat actually assists decision-making or discovery, rather than functioning as a passive help feature.

Possible Insights

A strong positive correlation implies chat guidance drives curiosity or intent. A weak correlation means users talk but don't act afterward, suggesting the chat's recommendations or usefulness may need refinement. Later this can inform improvements to conversational nudges or hint systems.

32. Rare Ingredients (Bottom 5%)

Rare Ingredients (Bottom 5%) represent the least frequently used ingredients across all recipes. Determined through frequency analysis, this metric identifies ingredients falling in the bottom 5% of occurrence, helping uncover uncommon items, niche preferences, or potential areas for expanding recipe diversity and ingredient variety.



Relevance

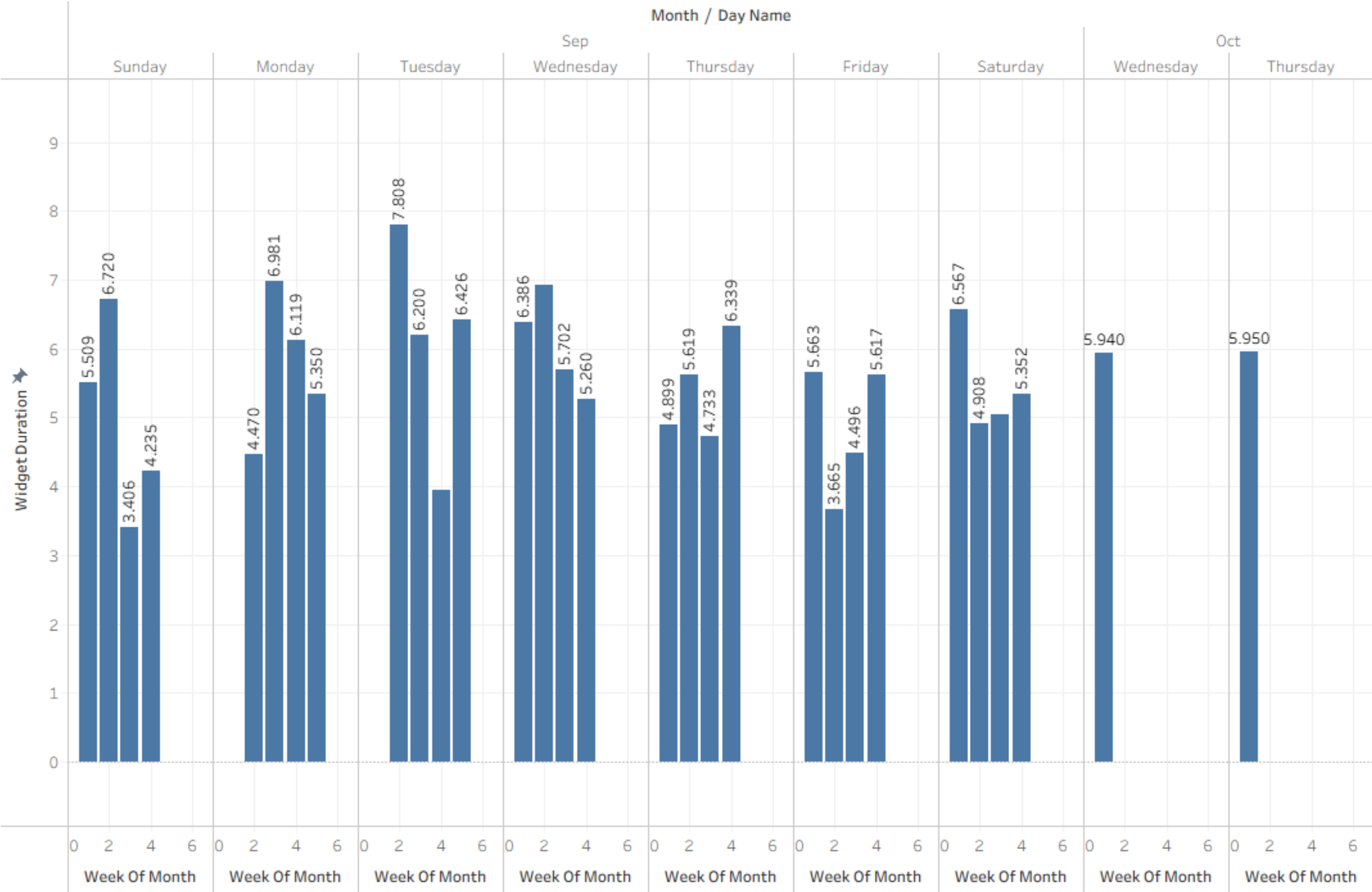
This metric identifies the **least frequently used ingredients**, helping uncover niche, unique, or underutilized items that may represent special recipes, seasonal trends, or potential supply inefficiencies.

Possible Insights

- Each listed ingredient (e.g., **i2040, i2219**) appears only **once**, indicating very low usage.
- Could reflect **rare or specialty ingredients** catering to limited preferences.
- Useful for **inventory optimization**—reducing rarely used stock.
- May highlight **innovation opportunities** for introducing new recipe variations.

33. User Engagement Trends

User Engagement Trends analyze how long different user segments interact with widgets by comparing **WidgetDuration** values. This reveals variations in engagement between groups like new and returning users, helping identify retention patterns, content effectiveness, and opportunities for improving personalized experiences and overall platform engagement strategies.



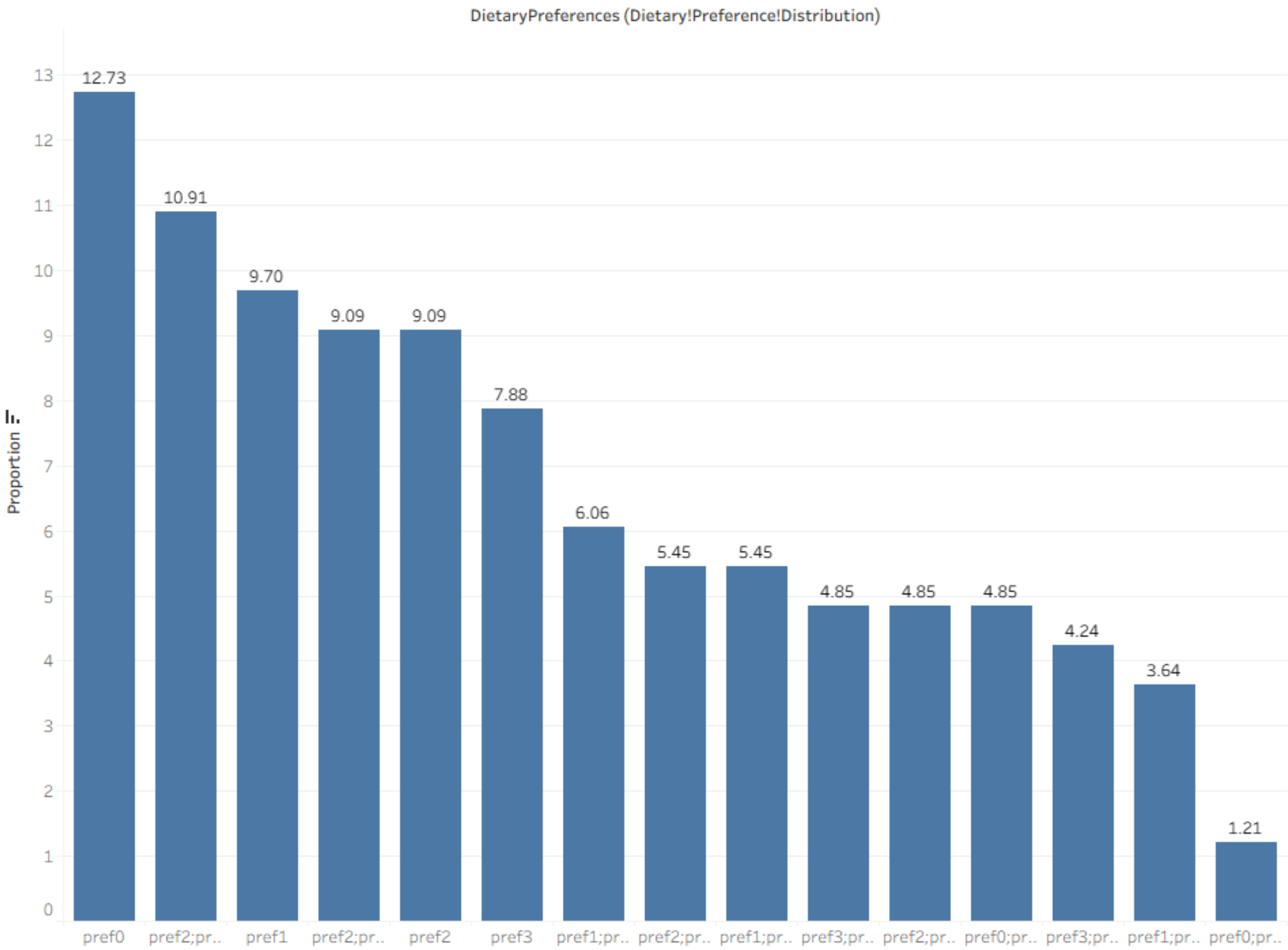
Relevance
This metric tracks **daily variations in widget engagement duration**, revealing when users are most interactive and how engagement patterns shift over time or across user segments (e.g., weekdays vs weekends, new vs returning users).

Possible Insights

- Engagement peaks on **Tuesdays (7.8s)** and **Wednesdays (6.9s)** — users are most active midweek.
- Slight dips on **Fridays (3.6s)** and Mondays suggest lower focus or reduced exploratory behavior.
- Consistent engagement across Week 36–37 indicates stable user interaction patterns.
- Insights can guide timing for promotions, feature rollouts, or content refreshes to align with high-engagement days.

34. Dietary Preference Distribution

Dietary Preference Distribution shows how different dietary preferences are represented in the dataset. It's calculated by counting how many times each unique preference appears, helping identify dominant eating habits, user diversity, and potential focus areas for personalized recommendations or nutrition-based content strategies.



Relevance

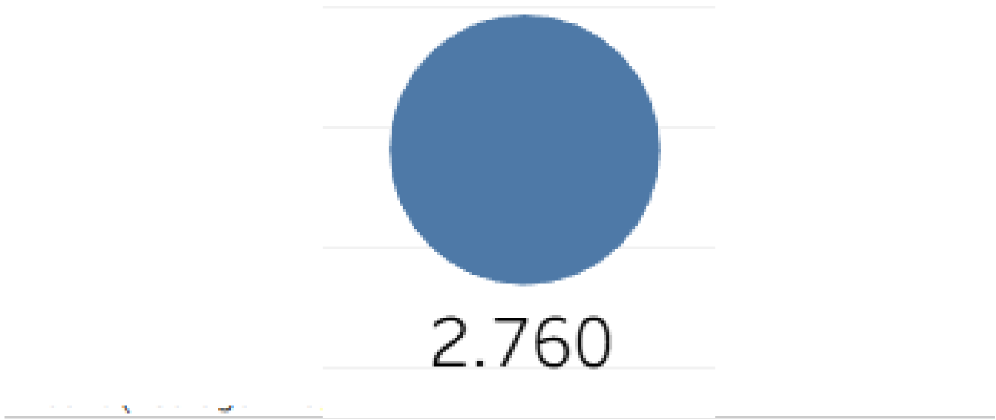
This metric highlights the **distribution of dietary preferences** among users, helping understand audience composition, dietary diversity, and opportunities for personalized recipe or feature recommendations.

Possible Insights

- **Pref0 (12.7%)** is the most common single preference, indicating a dominant dietary trend.
- Mixed combinations like **pref2;pref1;pref3 (10.9%)** suggest many users have flexible or overlapping dietary habits.
- Moderate shares of **pref1 (9.7%)** and **pref2 (9.1%)** indicate secondary but notable preference clusters.
- Insights can guide **personalized content**, targeted recipe filters, or menu recommendations aligning with dominant diet types.

35. Average Widget Load Time

Average Widget Load Time measures the mean duration taken for widgets to fully load. It's calculated by summing all **WidgetLoadTime** values and dividing by the total number of widgets, helping assess system performance, responsiveness, and user experience efficiency during widget initialization across sessions.



Relevance

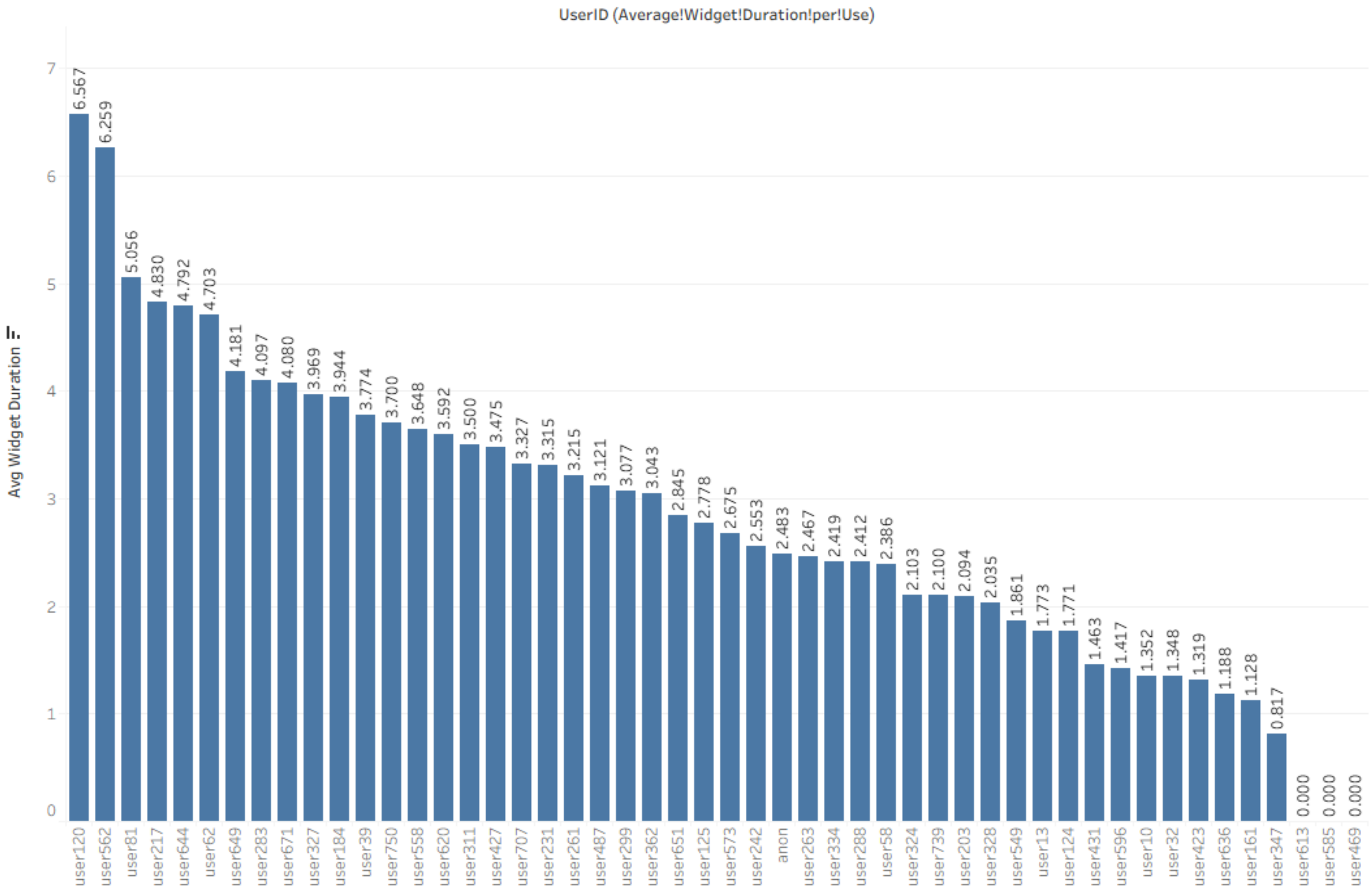
This metric measures the **average time taken for the widget to become interactive after being triggered**, directly impacting user experience, satisfaction, and engagement rates.

Possible Insights

- An average load time of **2.76 seconds** is acceptable but may feel slightly slow on mobile or low-speed networks.
- Consistent delays can reduce user retention or interrupt interaction flow.
- Optimization of backend queries or frontend rendering can improve responsiveness.
- Monitoring trends over time helps detect performance regressions after updates.

36. Average Widget Duration per UserID

Average Widget Duration measures the mean time users spend interacting with widgets. It helps identify which widgets maintain user attention the longest, revealing engagement effectiveness, usability, and overall content appeal within the interface or application experience across different sessions.



Relevance

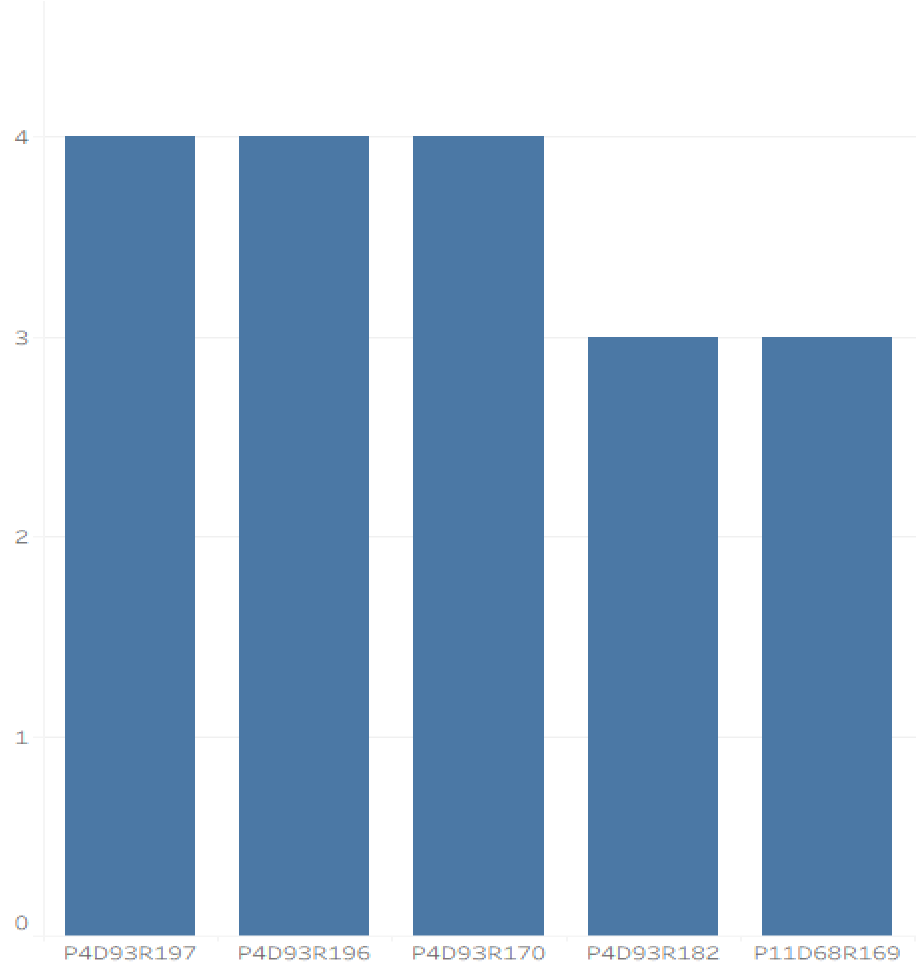
This metric captures how long users actively engage with a widget on average, indicating the **level of user interaction and attention** each widget retains before disengagement.

Possible Insights

- Average durations vary widely — **user120 (6.56s)** shows notably higher engagement than others.
- Longer durations may reflect interactive or content-rich widgets.
- Shorter times (1–2s) could mean quick actions or low engagement.
- Helps identify which widget types or designs most effectively sustain user interest.

37. Top 5 Most Viewed Recipes

This metric represents the **most common number of recipes viewed** by users within the dataset. It's determined by identifying which view count value occurs most frequently, helping reveal typical user behavior patterns and engagement levels with recipe content across the platform or application.



Relevance

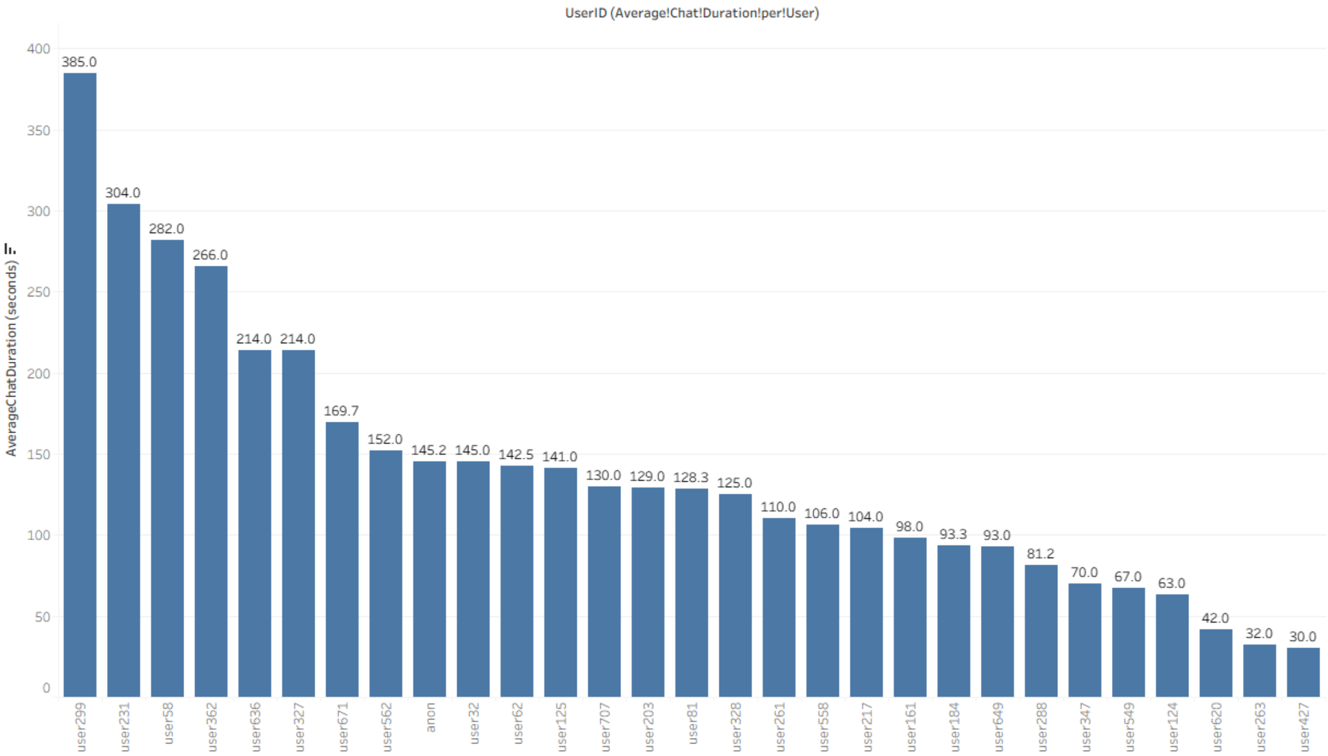
This metric highlights which recipes attract the most user attention, helping identify popular content themes, recipe types, or presentation styles that drive engagement.

Possible Insights

- Top recipes (e.g., **P4D93R197, P4D93R196, P4D93R170**) each viewed **4 times** show consistent audience interest.
- Recipes with **3–4 views** indicate emerging favorites or high discoverability.
- Can inform future recommendations or highlight trending recipe categories.

38. Average Chat Duration per User

Average Chat Duration measures the mean time users spend in chat sessions. It's calculated by subtracting **ChatStart** from **ChatEnd** for each session and averaging the results. This metric helps evaluate user engagement, chat efficiency, and the overall effectiveness of customer interactions or chatbot performance.



Relevance

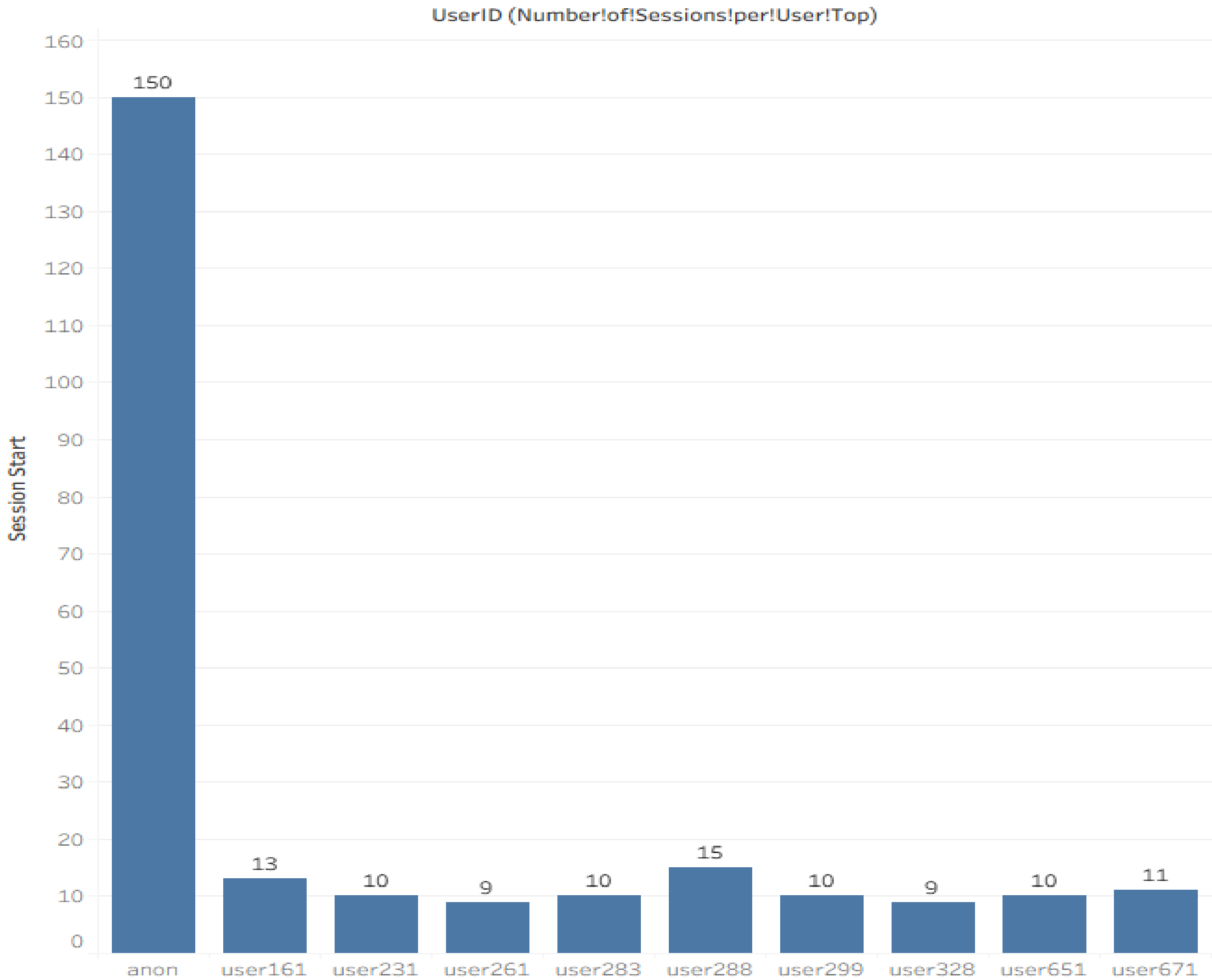
This metric measures the **average time users spend in chat interactions**, reflecting engagement depth and the usefulness of conversational support. Longer durations often indicate detailed assistance or strong user curiosity, while shorter ones may suggest quick resolutions or low engagement.

Possible Insights

- Average chat duration of **~2.43 minutes** suggests moderate user interaction and engagement.
- Users like **user299 (385s)** and **user231 (304s)** show high engagement — potential power users or complex query cases.
- Shorter sessions may indicate either efficiency or disengagement; needs cross-check with satisfaction or outcome data.
- Can inform chatbot design—balancing quick resolutions with richer, value-adding conversations.

39. Number of Sessions per User (Top 10)

Number of Sessions per User measures how frequently each user engages with the system. It's calculated by counting all **SessionStart** entries grouped by **UserID**, helping analyze user activity levels, consistency, and overall engagement frequency across the platform or application.



Relevance

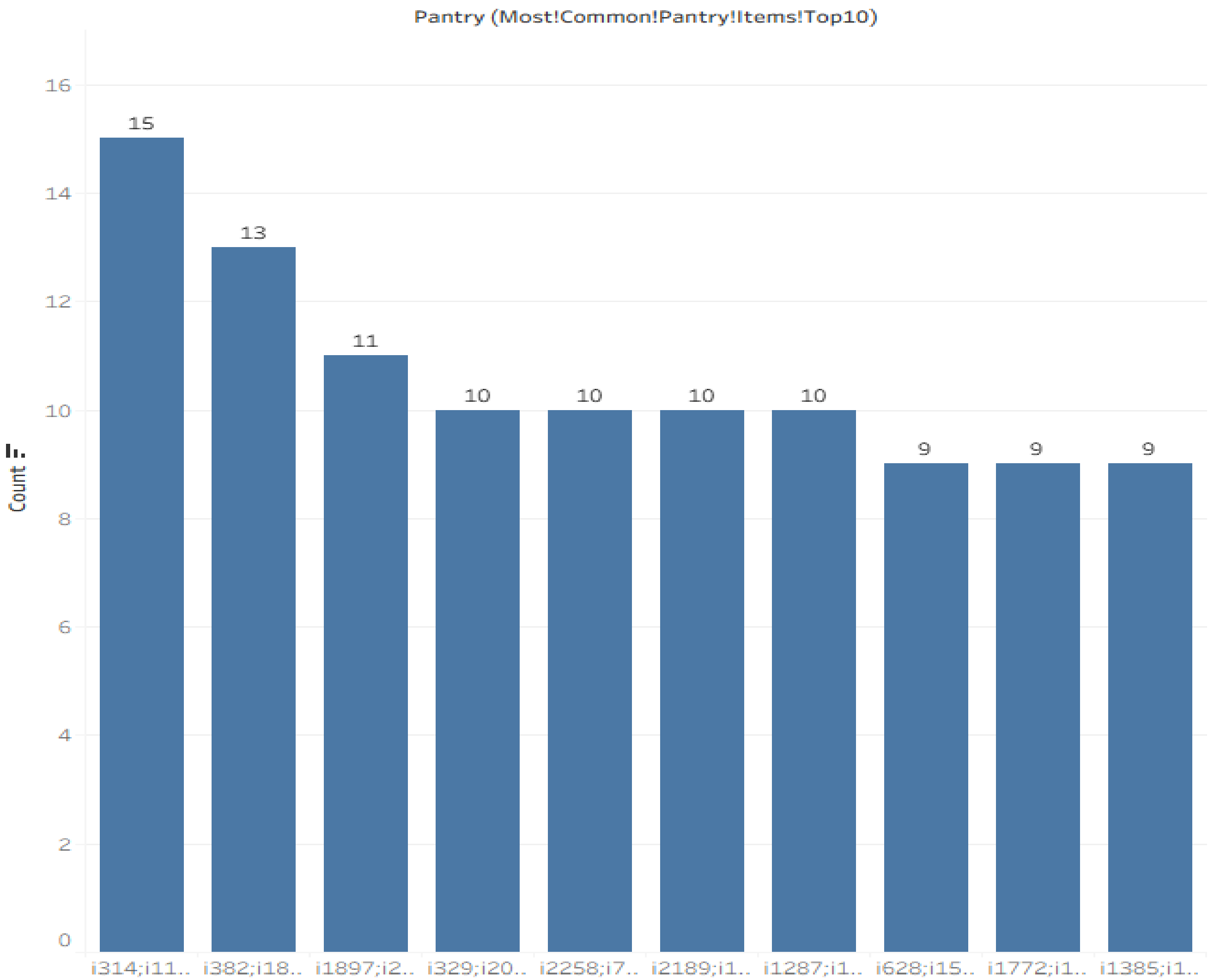
This metric measures **how frequently each user interacts with the system**, reflecting user retention, engagement intensity, and overall platform stickiness over time.

Possible Insights

- **anon (150 sessions)** shows very high anonymous or guest activity—potentially unregistered or exploratory users.
- Registered users like **user288 (15)** and **user161 (13)** demonstrate strong recurring engagement.
- Users with **9–10 sessions** represent consistent usage, valuable for long-term retention tracking.
- Can inform **user segmentation**, targeting strategies, or onboarding improvements to convert frequent anonymous users into registered ones.

40. Most Common Pantry Items (Top 10)

Most Common Pantry Item identifies the pantry ingredient that appears most frequently in the dataset. It's determined by counting occurrences of each item and selecting the one with the highest frequency, revealing popular or essential ingredients preferred by users across their stored or saved pantry lists.



Relevance

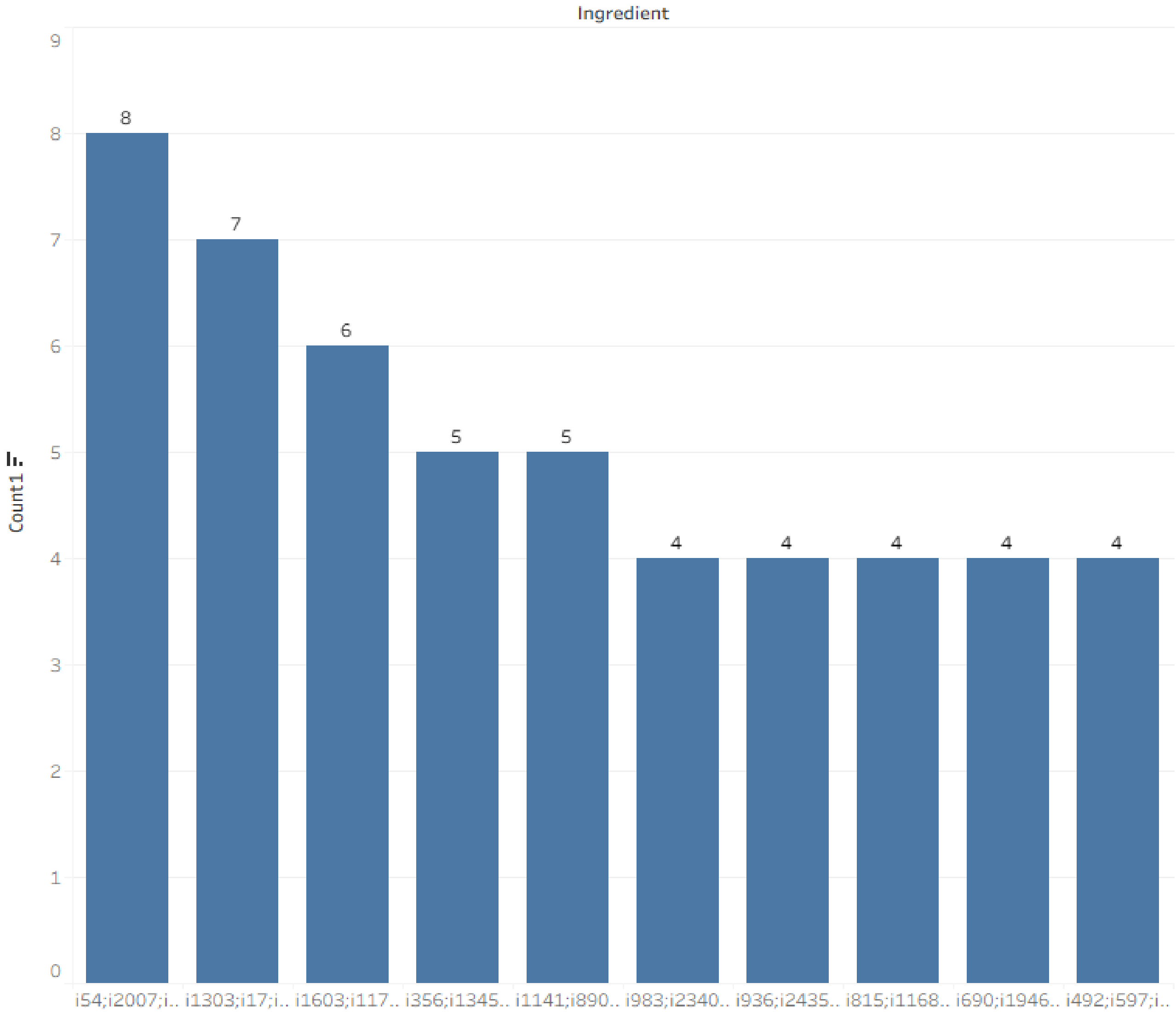
This metric identifies the **most frequently stocked pantry items**, revealing user preferences, cooking habits, and ingredient demand trends that can guide recommendation and inventory optimization.

Possible Insights

- The top pantry combination (**i314;i1172;i2362;...**) appears **15 times**, showing a strong recurring ingredient pattern.
- Frequent overlaps suggest popular ingredient sets or meal preparation routines.
- Common groupings can inform **personalized recipe suggestions** or grocery partnerships.
- Tracking shifts in top items helps predict **seasonal or trend-based demand changes**.

41. Most Common Ingredient Overall (Top 10)

Most Common Ingredient Overall highlights the ingredient appearing most frequently across all recipes. It's calculated by counting each ingredient's occurrences and identifying the one with the highest frequency, helping reveal widely used or preferred ingredients and overall trends in recipe composition and user preferences.



Relevance

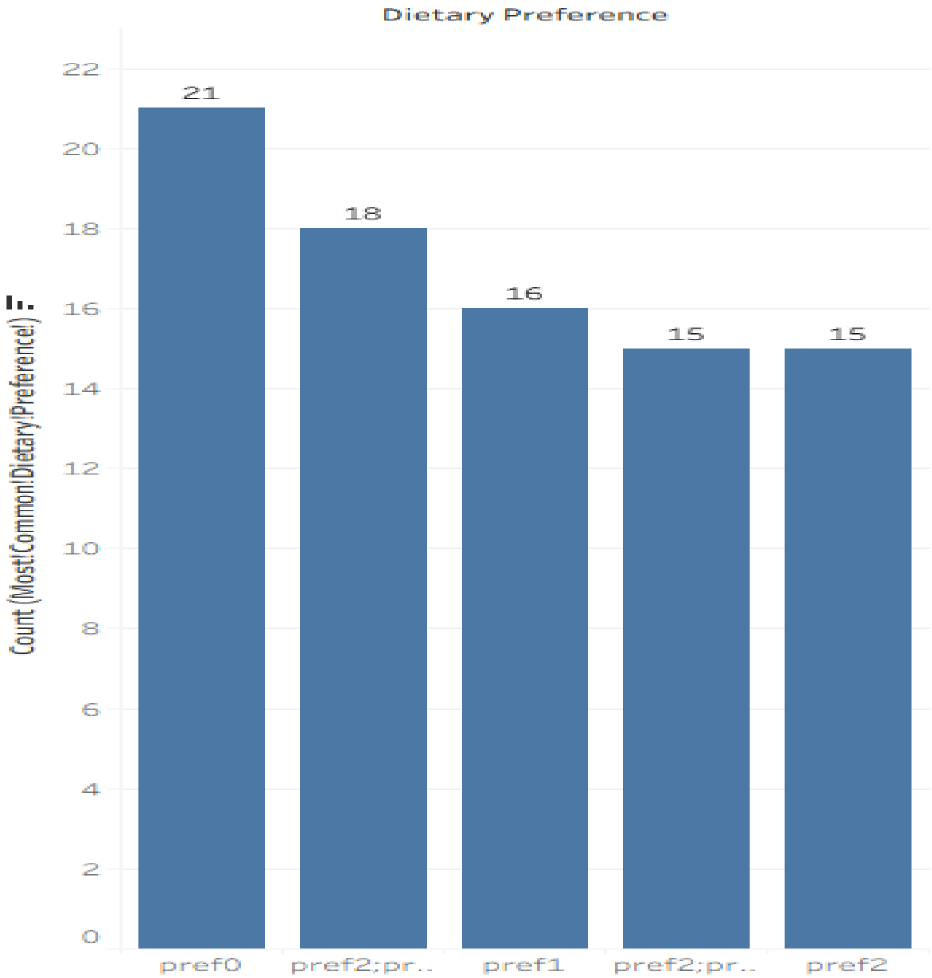
This metric pinpoints the **most frequently used ingredients across all recipes**, helping understand ingredient popularity, core components of meals, and potential focus areas for sourcing or recommendations.

Possible Insights

- The leading ingredient set (**i54;i2007;...;i1080**) appears **8 times**, showing strong recipe centrality.
- High recurrence implies staple or universally appealing ingredients.
- Patterns reveal **ingredient clustering** useful for cross-selling or combo suggestions.
- Can guide recipe development or highlight ingredients driving consistent engagement.

42. Most Common Dietary Preference (Top 5)

Most Common Dietary Preference identifies the dietary preference most frequently chosen by users. It's determined by finding the value with the highest count in the **DietaryPreferences** column, helping understand dominant eating habits and guiding content personalization or targeted recipe recommendations.



Relevance

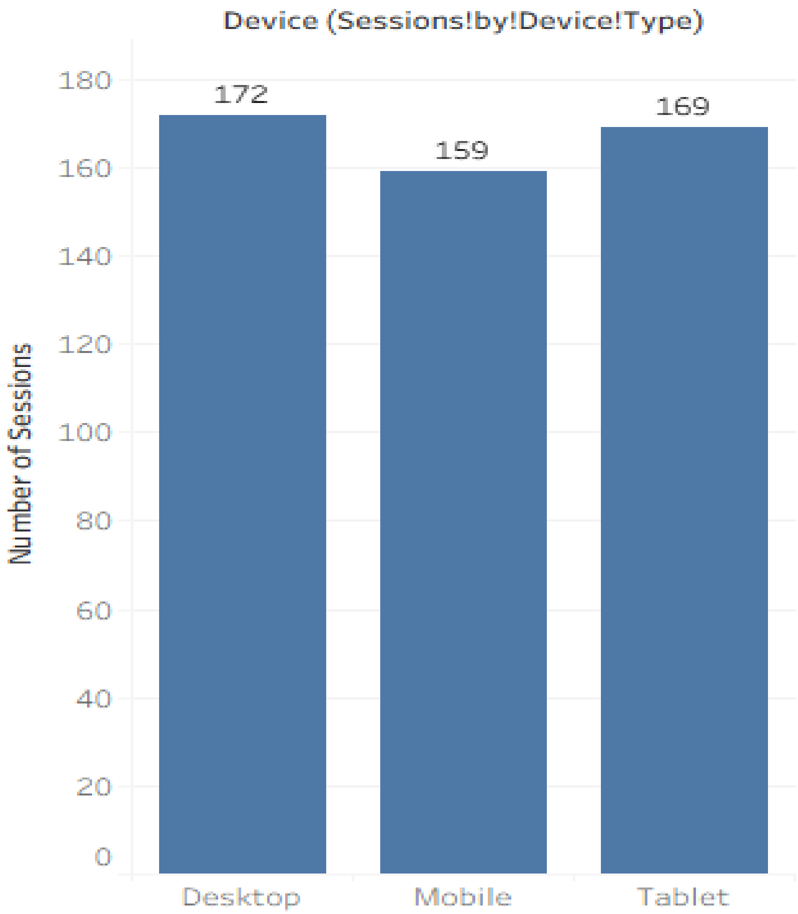
This metric identifies the **most common dietary preference** among users, offering insights into the dominant eating habits and guiding recipe personalization, content prioritization, or targeted marketing.

Possible Insights

- **pref0 (21 counts)** leads as the most frequent preference, marking it as the primary dietary trend.
-
- **Mixed preferences (pref2;pref1;pref3, 18 counts)** show diverse user needs and flexible diets.
- Balance between single and multi-preference users indicates a **broad dietary spectrum**.
- Can inform recipe tagging, dietary filter design, and tailored recommendation strategies.

43. Sessions by Device Type

Sessions by Device Type measures how often users access the platform from different devices. It's calculated by counting occurrences of each device type in the **Device** column, helping analyze platform accessibility, user behavior patterns, and device-based engagement preferences across sessions.



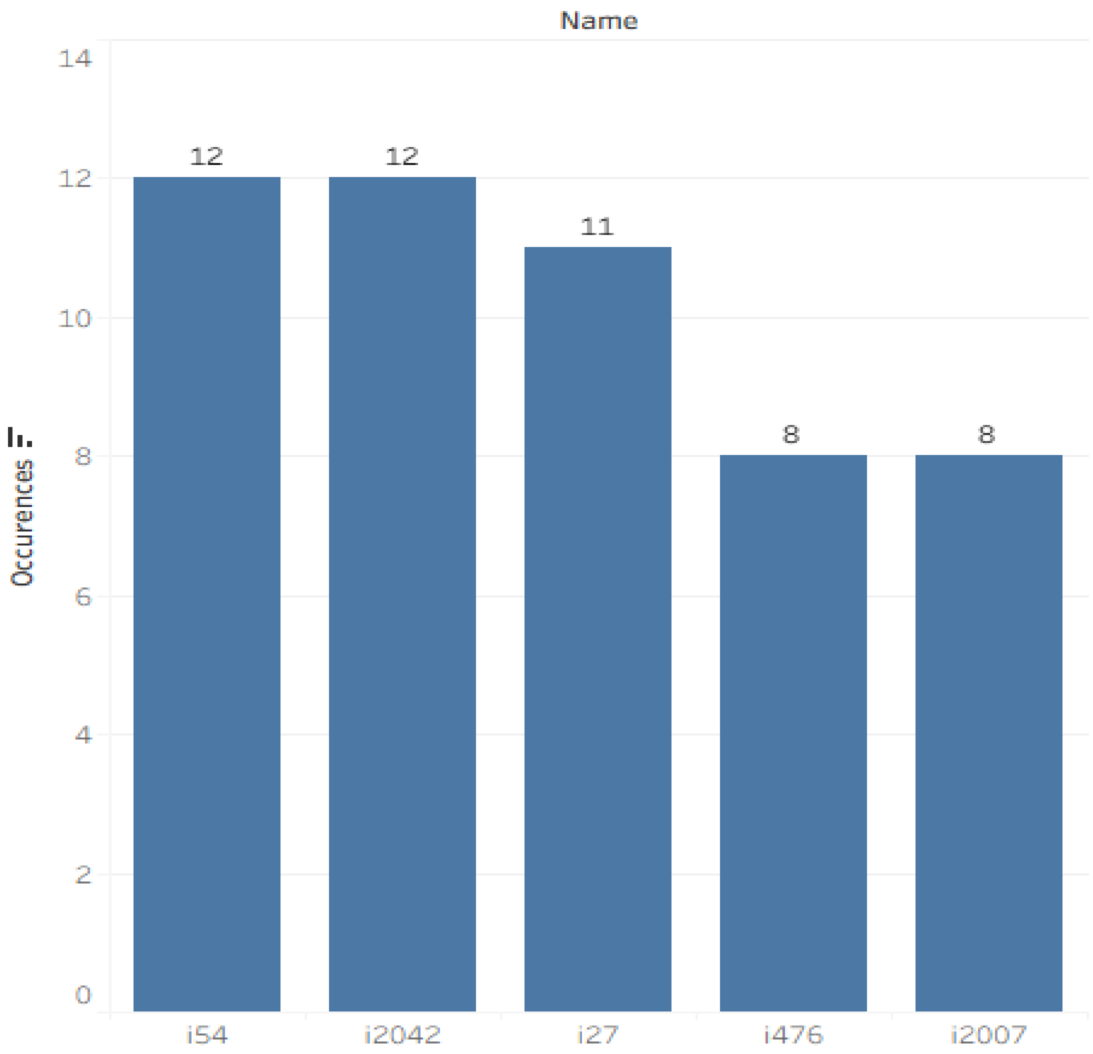
Relevance
This metric measures the **distribution of user sessions across device types**, helping understand user access behavior and optimize the interface for the most-used platforms.

Possible Insights

- **Desktop (172 sessions)** slightly leads, showing a preference for larger-screen interactions.
- **Tablet (169)** and **Mobile (159)** counts indicate balanced multi-device usage.
- Suggests strong **cross-device accessibility** and user flexibility.
- Insights can guide **UI/UX optimization** and adaptive design focus for high-traffic devices.

44. Average Ingredient Count per Recipe

Average Ingredient Count per Recipe measures the mean number of ingredients used in recipes. It's calculated by splitting each recipe's ingredient list, counting the items, and averaging those counts across all recipes to understand recipe complexity and typical ingredient variety.



Relevance

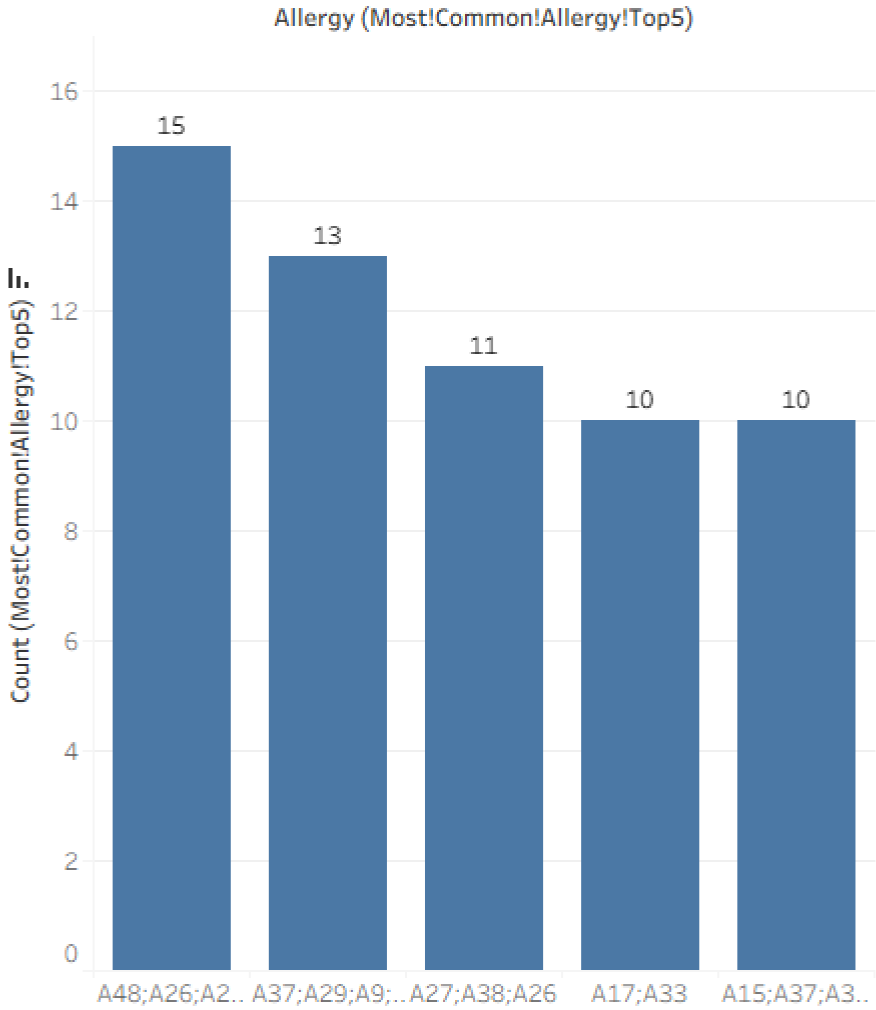
This metric reflects the **average complexity of recipes** based on ingredient count, offering insights into preparation effort, ingredient diversity, and potential appeal to different user segments.

Possible Insights

- An average of **10 ingredients per recipe** indicates moderately detailed meal preparation.
- Common items like **i2042** and **i54** (12 uses each) are likely staple components.
- High overlap in top ingredients suggests recurring flavor bases or core recipe themes.
- Useful for tailoring recipe recommendations to user skill levels or ingredient availability.

45. Most Common Allergy (Top 5)

Most Common Allergy identifies the allergy most frequently reported by users. It's determined by finding the value with the highest count in the **Allergy** column, helping understand prevalent dietary restrictions and supporting better recipe recommendations or allergen-aware content personalization.



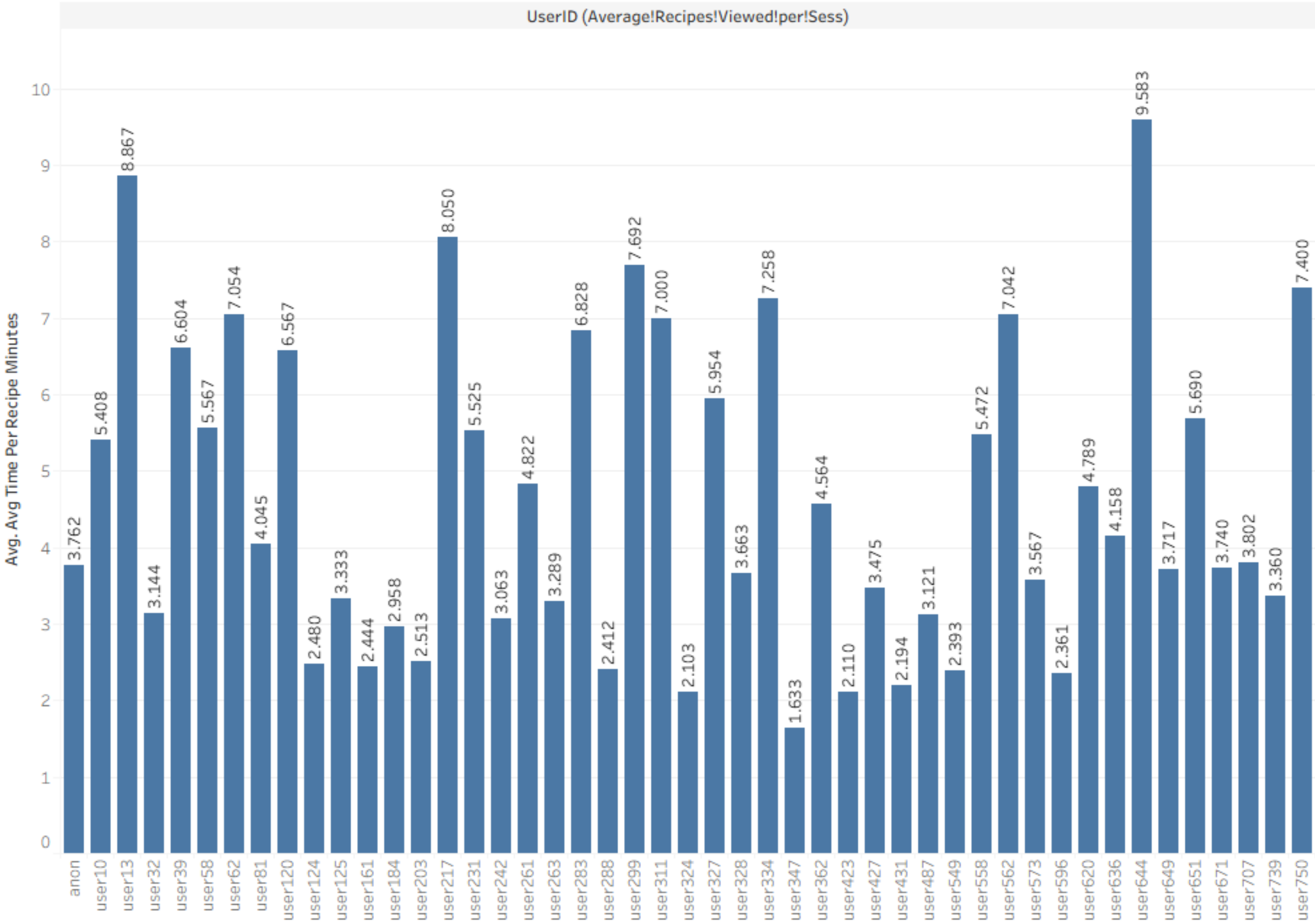
Relevance
This metric identifies the **most common allergy combinations** among users, offering crucial insights for personalization, safety alerts, and allergen-aware recipe recommendations.

Possible Insights

- **A48;A26;A23;A6;A41 (15 counts)** is the dominant allergy group—must be prioritized in filtering or warnings.
- Frequent overlaps suggest users often have **multiple concurrent allergies**.
- Tracking common allergy sets helps improve **safe recipe suggestions**.
- Supports better **user trust and compliance** through allergen-sensitive design.

46. Average Recipes Viewed per Session

Average Recipes Viewed per Session measures how many recipes users typically view during a single session. It's calculated by dividing the total number of recipes viewed by the total number of sessions, helping assess user curiosity, browsing behavior, and engagement depth across the platform.



Relevance

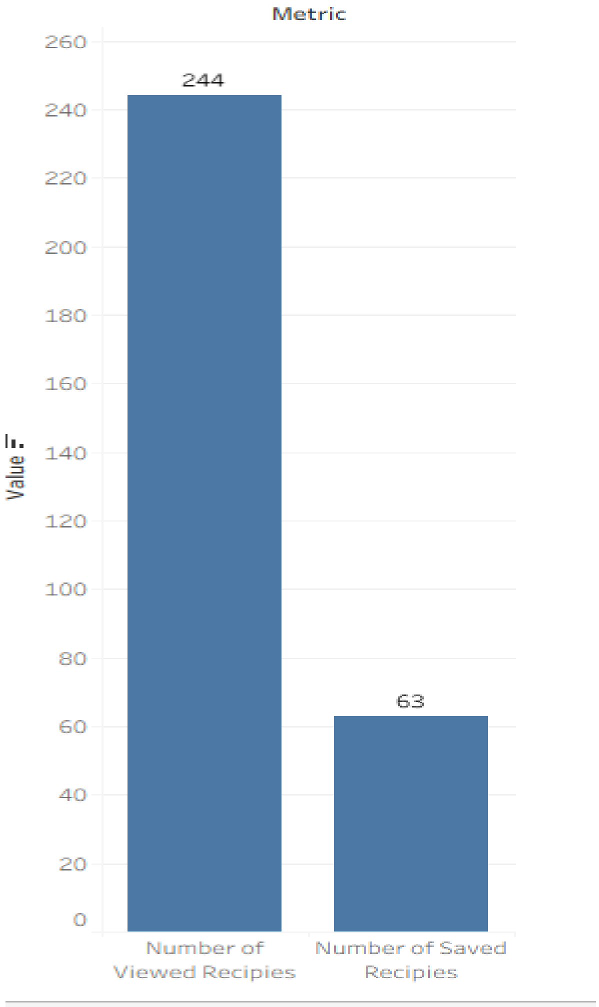
This metric measures the **average number of recipes viewed per session**, reflecting browsing depth, exploration behavior, and user curiosity toward available recipes.

Possible Insights

- **user120 (6.56 min/recipe)** shows high engagement and deliberate exploration.
- **anon (3.76)** and **user125 (3.33)** indicate moderate browsing habits.
- Shorter averages (e.g., **user124: 2.48**) may reflect quick scanning or indecision.
- Helps refine **content layout or recommendation flow** to sustain longer exploration.

47. Save-to-View Ratio

This expression calculates the proportion of viewed recipes that were actually saved by users, essentially representing the conversion rate from viewing to saving a recipe.



Relevance

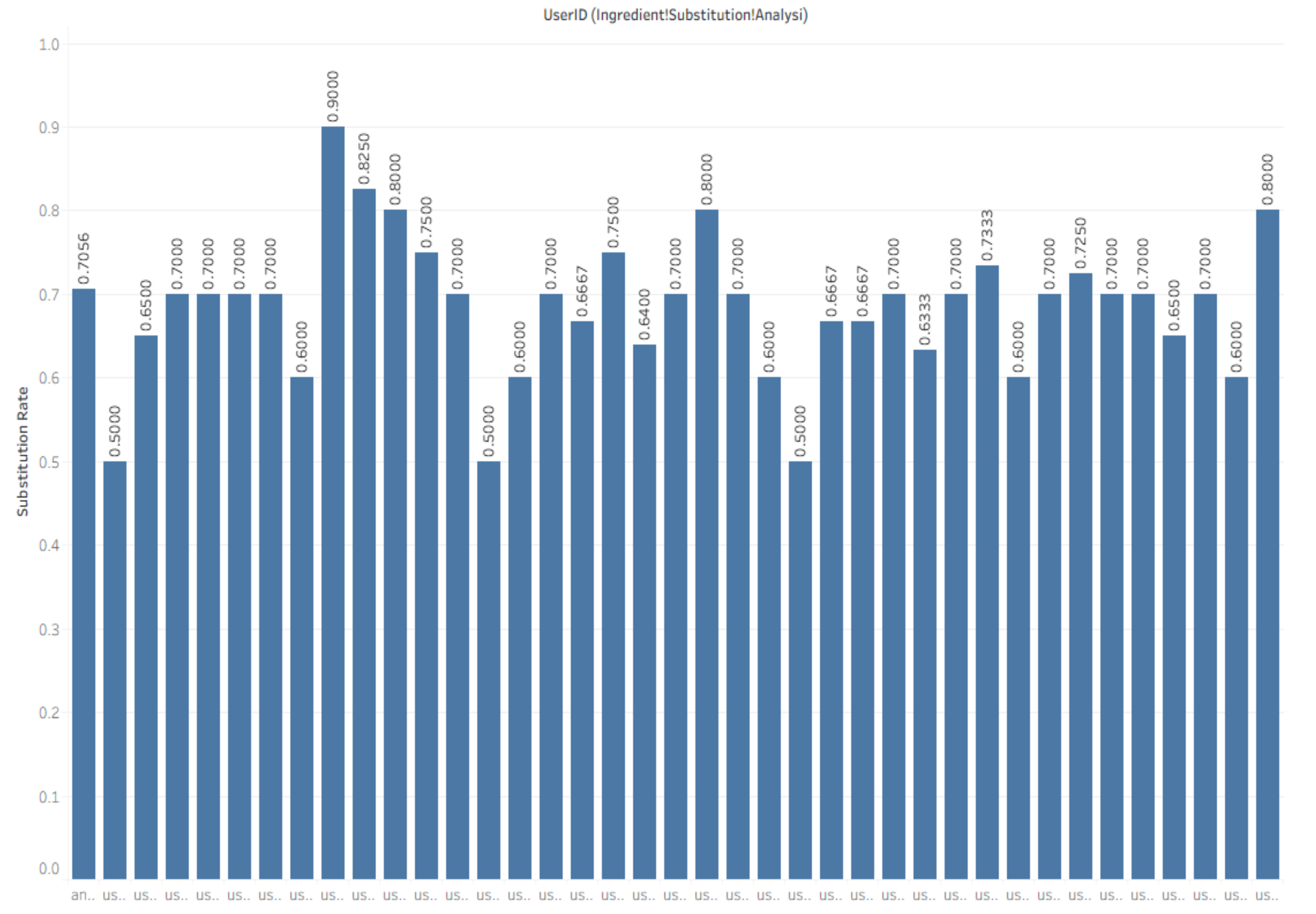
This metric indicates how effectively viewed recipes convert into saved ones, showing user interest strength and the impact of recipe content or recommendations on engagement.

Possible Insights

- A save-to-view ratio of **≈25.8% (63/244)** suggests moderate engagement and content appeal.
- High ratio → users find recipes valuable enough to revisit or try later.
- Low ratio → users browse but lack motivation to save, hinting at weak personalization or visual appeal.
- Can guide improvements in recommendation quality, recipe presentation, or relevance of search results.

48. Ingredient Substitution Analysis (Element-wise Comparison)

Ingredient Substitution identifies when users replace suggested ingredients with alternatives. By comparing **Ingredients** and **AddedToCartItems**, this analysis detects substitutions, revealing user preferences, dietary adjustments, and shopping behavior patterns that can improve recipe recommendations and personalization strategies.



Relevance

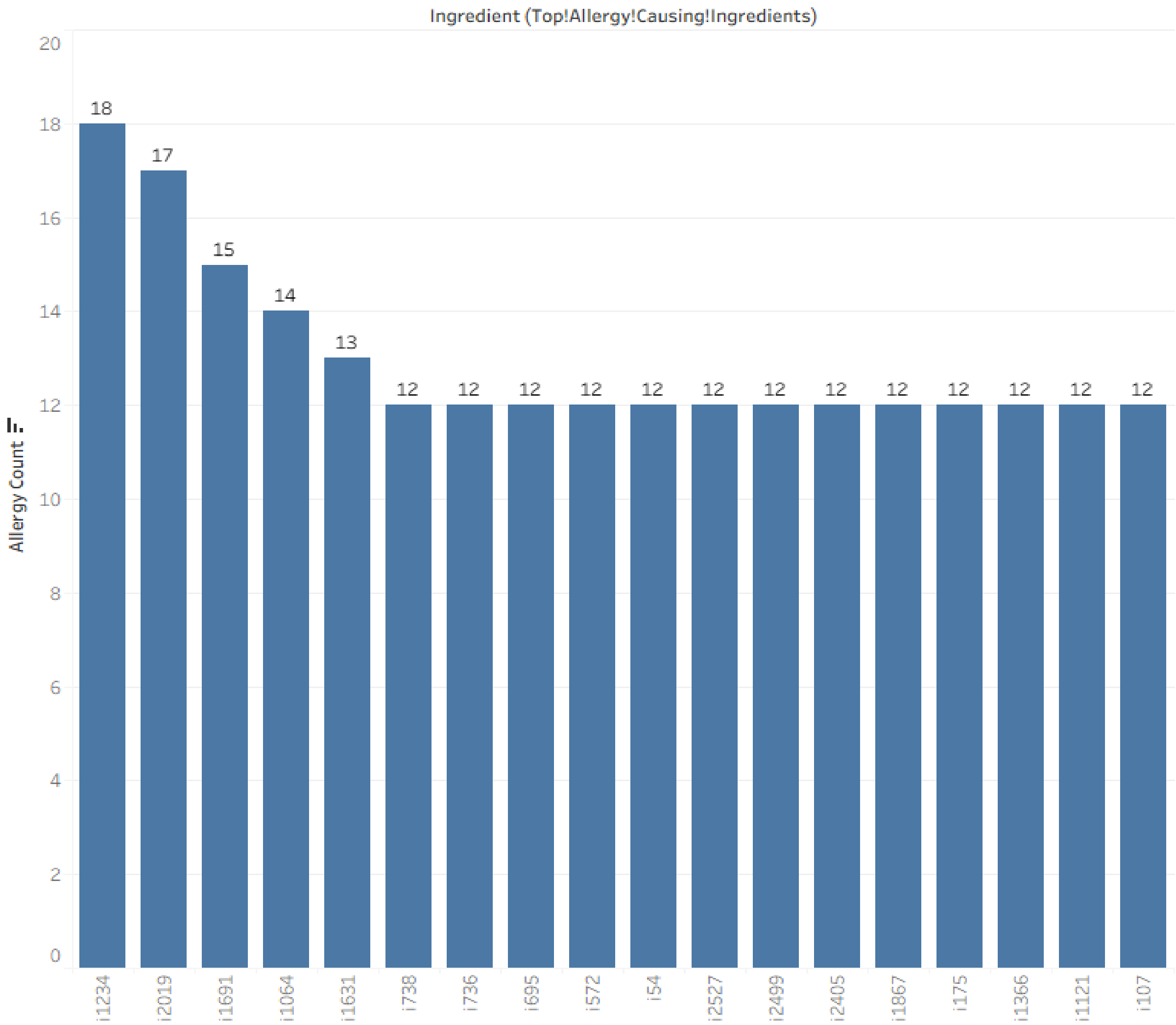
This metric measures how often users **substitute ingredients** when adding items to their cart, revealing flexibility in recipe adaptation, availability constraints, or personalization in cooking behavior.

Possible Insights

- **anon (70.5%)** shows strong substitution activity, suggesting experimentation or limited pantry overlap.
- **user120 (90%)** demonstrates near-complete match—indicating consistent recipe adherence.
- High substitution rates reflect **creative or resource-driven users**.
- Helps refine **ingredient recommendation systems** and substitution suggestions for better user support.

49. Top Allergy-Causing Ingredients

Top Allergy-Causing Ingredients highlight ingredients most linked to user allergies. By mapping **allergy codes** to their corresponding ingredients, this analysis identifies common triggers, helping enhance safety alerts, refine recipe filters, and support allergen-aware personalization for a safer user experience.



Relevance

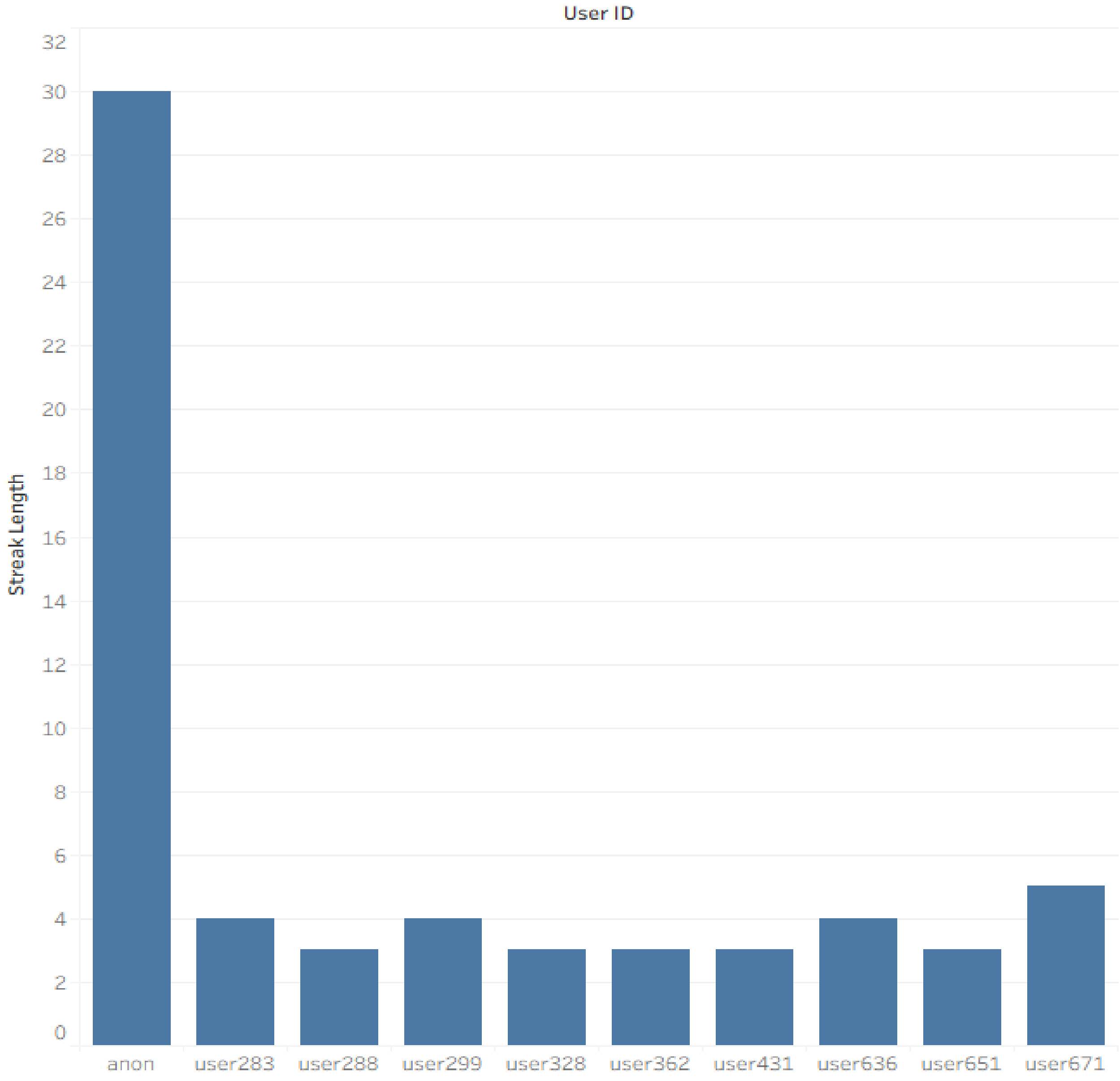
This metric identifies **ingredients most frequently linked to user allergies**, essential for improving recipe safety, allergen alerts, and user trust in personalized recommendations.

Possible Insights

- **i1234 (18 counts)** and **i2019 (17)** are the top allergy triggers—must be prioritized in safety labeling.
- Clustered allergens (e.g., **i1691, i1064**) indicate common intolerance patterns.
- Useful for **risk mitigation** by excluding or flagging high-risk ingredients.
- Can guide development of **allergen-free recipe alternatives** or substitute suggestions.

50. Active User Streaks

Active User Streaks track how many consecutive days each user remains active by recording sessions per **UserID**. This metric highlights user retention, consistency, and engagement habits, helping identify loyal users and measure the effectiveness of features in maintaining continuous platform activity.



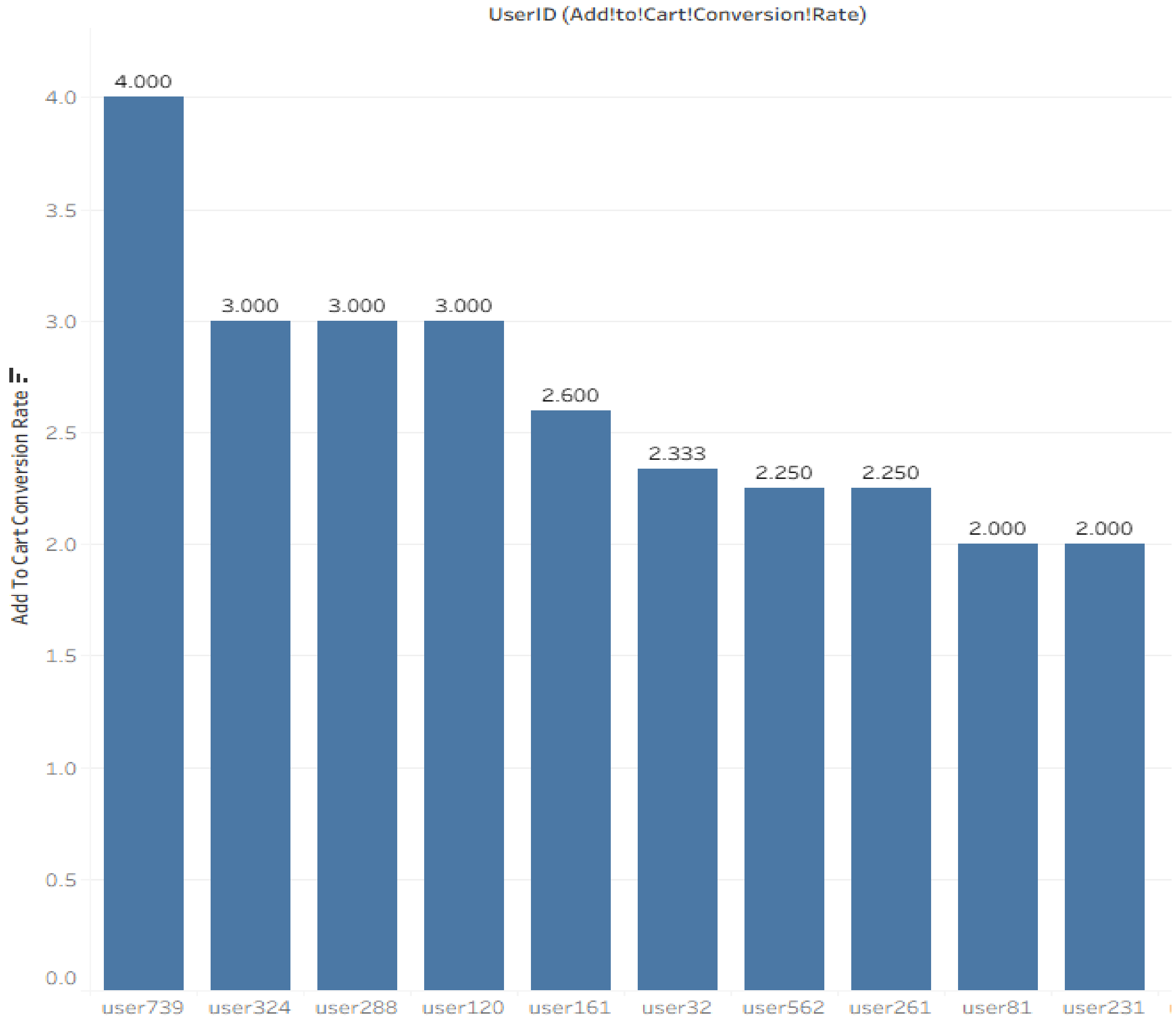
Relevance
Active user streaks track how many **consecutive days** users engage with sessions, indicating **loyalty, retention, and platform stickiness** over time.

Possible Insights

- User **"anon"** shows strong retention with a **30-day streak**.
- Shorter streaks (3–5 days) imply **moderate engagement cycles**.
- Identifies **high-value consistent users** for rewards.
- Useful for optimizing **daily re-engagement campaigns**.

51. Add-to-Cart Conversion Rate

Add-to-Cart Conversion Rate measures how effectively users complete purchases after adding items to their cart. It's calculated by comparing the number of items added to carts with those actually purchased, helping assess buying intent, checkout efficiency, and overall sales performance.



Relevance

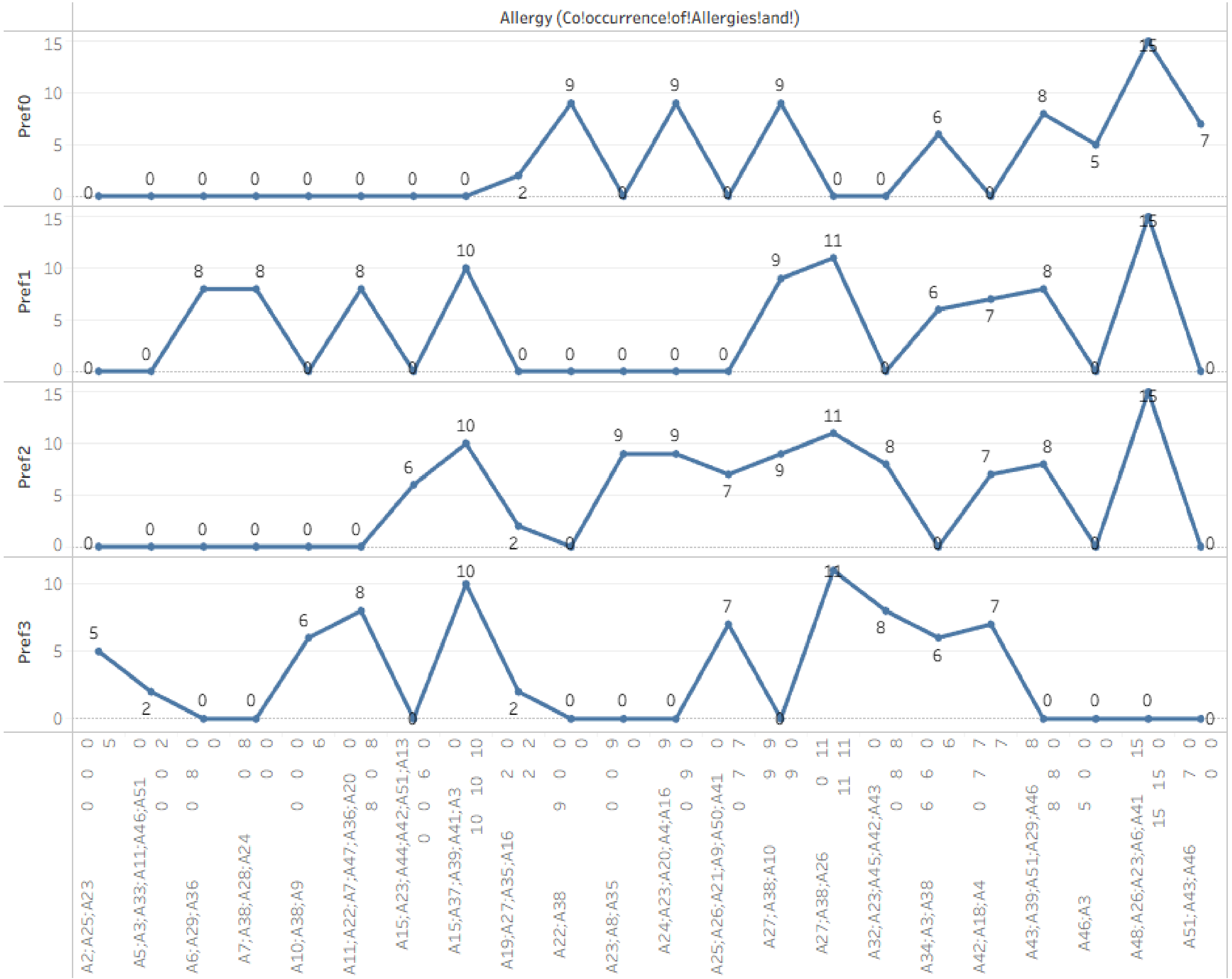
This metric tracks the **percentage of items added to carts that are eventually purchased**, reflecting user intent strength, purchase efficiency, and overall funnel performance.

Possible Insights

- **user739 (4%)** leads conversion—indicating stronger purchase commitment.
- Moderate rates (~2–3%) suggest browsing without consistent checkout completion.
- Low conversion may imply **pricing friction or decision hesitation**.
- Insights can inform **targeted reminders, promotions, or cart optimization strategies** to boost final purchases.

52. Co-occurrence of Allergies & Preferences

Co-occurrence of Allergies & Preferences analyzes how dietary preferences align with specific allergies using **cross-tabulation**. This reveals overlapping patterns, such as users with similar restrictions or habits, helping design safer meal recommendations, improve personalization, and better understand user dietary relationships.



Relevance

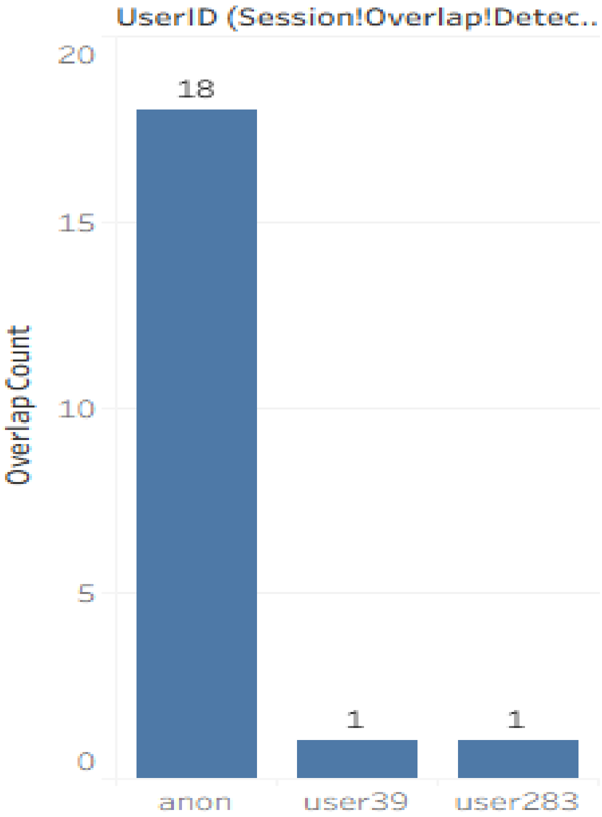
This metric examines the **overlap between allergies and dietary preferences**, revealing how specific intolerances align with eating habits and helping design safer, more personalized recommendations.

Possible Insights

- Strong co-occurrence for **A48;A26;A23;A6;A41** with **pref0–pref2 (15 each)** shows clear alignment between certain allergies and diet types.
- **A27;A38;A26** and **A15;A37;A39;A41;A3** appear highly cross-linked across preferences—indicating common allergen-diet clusters.
- Low or zero overlap highlights **potential dietary flexibility** despite allergies.
- Insights support **targeted allergen filtering** and **diet-personalized recipe curation**.

53. Session Overlap Detection

Session Overlap Detection identifies users who have multiple sessions occurring simultaneously. By comparing **SessionStart** and **SessionEnd** times, this analysis detects overlapping activity, helping spot anomalies, multi-device usage, or potential data inconsistencies in user session behavior.



Relevance

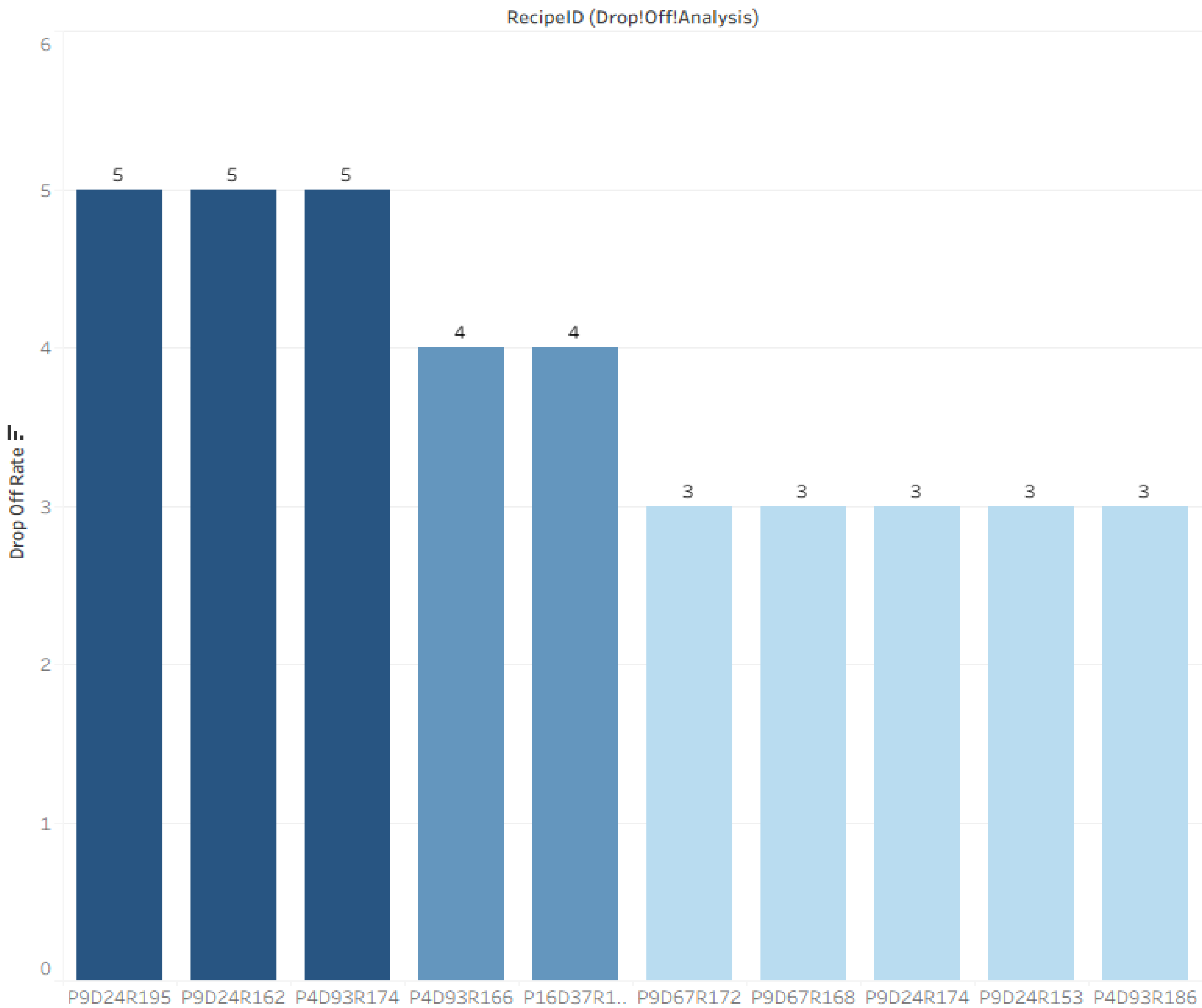
This metric detects **users with overlapping session times**, helping identify multitasking behavior, technical logging errors, or simultaneous access patterns that may affect engagement metrics.

Possible Insights

- **3 users (6%)** show overlapping sessions—mostly minor but notable.
- **anon (18 overlaps)** suggests concurrent browsing or multiple device use.
- Could indicate **system timing inconsistencies** or **parallel testing behavior**.
- Helps refine **session tracking accuracy** and prevent data duplication.

54. Drop-off Analysis

Drop-off Analysis identifies recipes that attract many views but result in few saves or cart additions. This helps pinpoint content that captures interest but fails to convert, revealing issues in recipe appeal, ingredient accessibility, or user intent during browsing.



Relevance

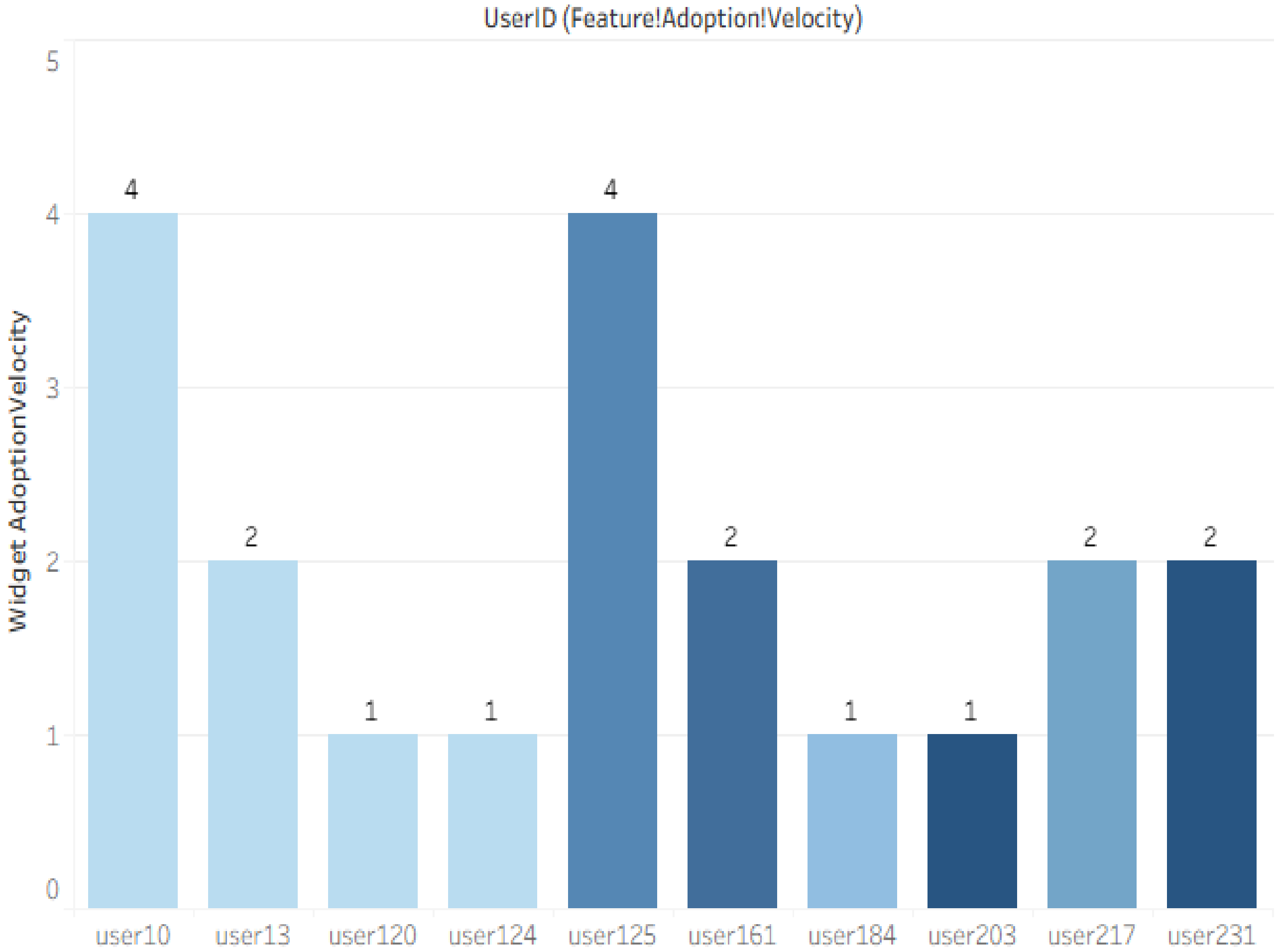
This metric highlights **recipes with high view counts but no subsequent saves or cart additions**, revealing potential disconnects between interest and action — key for optimizing content effectiveness.

Possible Insights

- Recipes like **P9D24R162** and **P9D24R195** (5 views, 0 saves/carts) show complete drop-off—likely low appeal or unclear value.
- Consistent **view-only behavior** may signal poor imagery, description gaps, or irrelevant recommendations.
- High drop-off rates flag **content refinement opportunities** (e.g., presentation or ingredient clarity).
- Useful for prioritizing **A/B testing** or UX enhancements to boost recipe conversion.

55. Feature Adoption Velocity

Feature Adoption Velocity measures how quickly users begin using key features like **Widget** or **Chat**. It's calculated by counting the number of sessions each user takes before their first feature use, helping evaluate onboarding effectiveness, user engagement speed, and feature discoverability.



Relevance

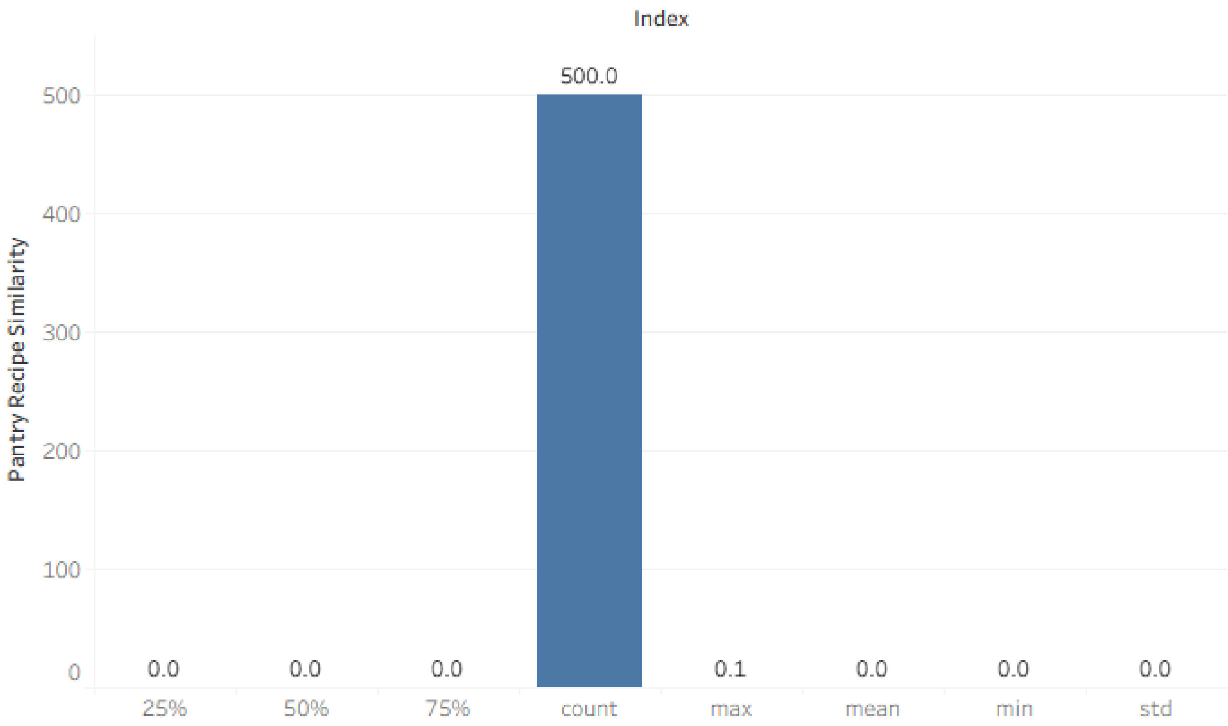
Feature Adoption Velocity measures how quickly users engage with new features like **Widget** or **Chat**, showing user curiosity, ease of discovery, and product onboarding effectiveness.

Possible Insights

- Low average (1.68 sessions) for **Widget** implies quick discovery and strong appeal.
- Higher average (3.72) for **Chat** suggests delayed engagement—potential UI visibility or trust issues.
- Users like **anon** and **user124** adopt both features immediately—ideal behavior for early adopters.
- Monitoring this metric helps optimize **feature placement** and onboarding flow.

56. Pantry-Recipe Similarity

Pantry-Recipe Similarity measures how closely a user's pantry matches recipe ingredients using **Jaccard similarity**. It compares shared items between **Pantry** and **Ingredients**, revealing how well users can cook recipes with available items and supporting better recipe recommendations.



Relevance

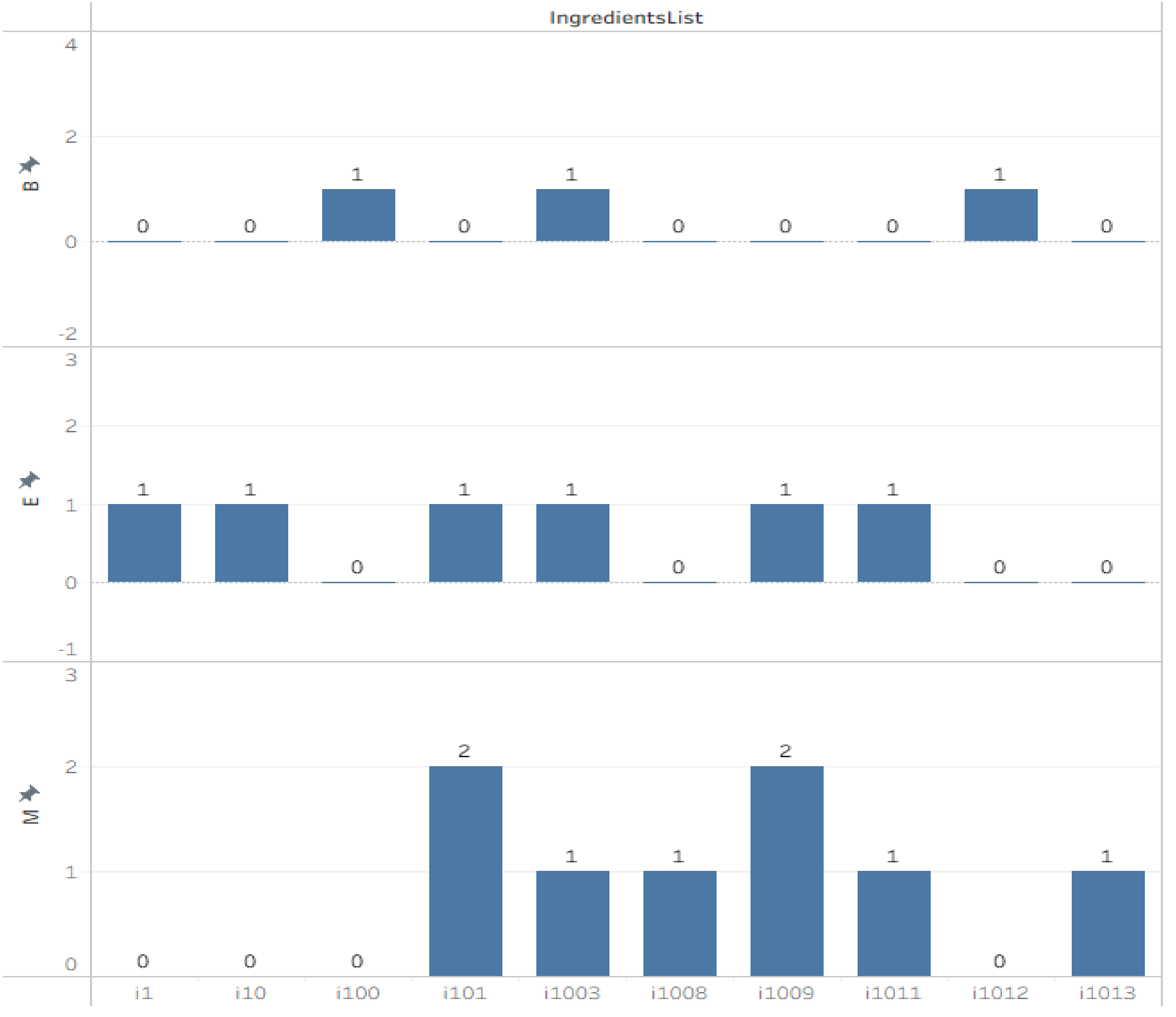
Pantry-recipe similarity (via **Jaccard index**) measures the overlap between a user's pantry items and recipe ingredients, showing how aligned recipes are with available ingredients for personalized recommendations.

Possible Insights

- Very low mean (0.002) suggests users' pantries rarely match recipe ingredients.
- High number of zeros indicates poor personalization or sparse pantry data.
- Recommender system could emphasize pantry-based filtering to boost engagement.
- Suggests opportunity for **inventory-based recipe recommendations**.

57. Ingredient Seasonal Trends

Ingredient Seasonal Trends analyze how ingredient usage varies across time by grouping **Ingredients** based on **MonthType**. This reveals seasonal patterns, highlighting ingredients popular in specific months or seasons, useful for planning menus, promotions, or inventory aligned with user preferences and seasonal demand.



Relevance

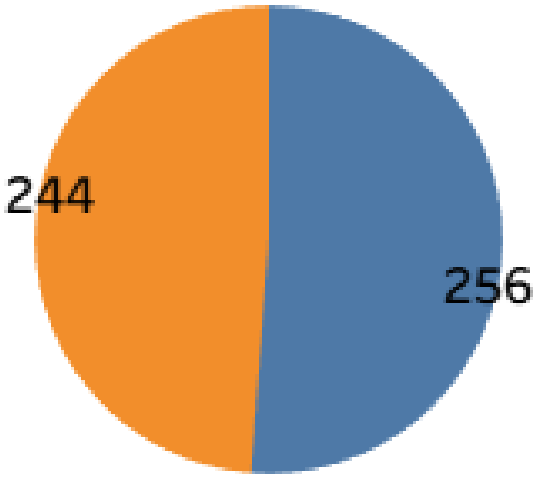
Ingredient seasonal trends identify how frequently ingredients appear across different **MonthTypes (B, M, E)**, revealing temporal usage or availability patterns that influence recipe planning and menu recommendations.

Possible Insights

- Some ingredients (e.g., *i1003, i1009*) appear consistently across all periods, showing year-round demand.
- Early-month dominance (MonthType E) suggests users explore more diverse ingredients initially.
- Certain ingredients being exclusive to one month type may indicate **seasonal or promotional recipes**.
- Helps optimize **inventory stocking** and **recipe curation** timing.

58. Widget Usage Rate

Widget Usage Rate measures the proportion of sessions where users interacted with widgets. It's calculated as the percentage of sessions with a non-null **WidgetStart** value, helping evaluate widget adoption, engagement consistency, and overall feature utilization across user sessions.



Relevance

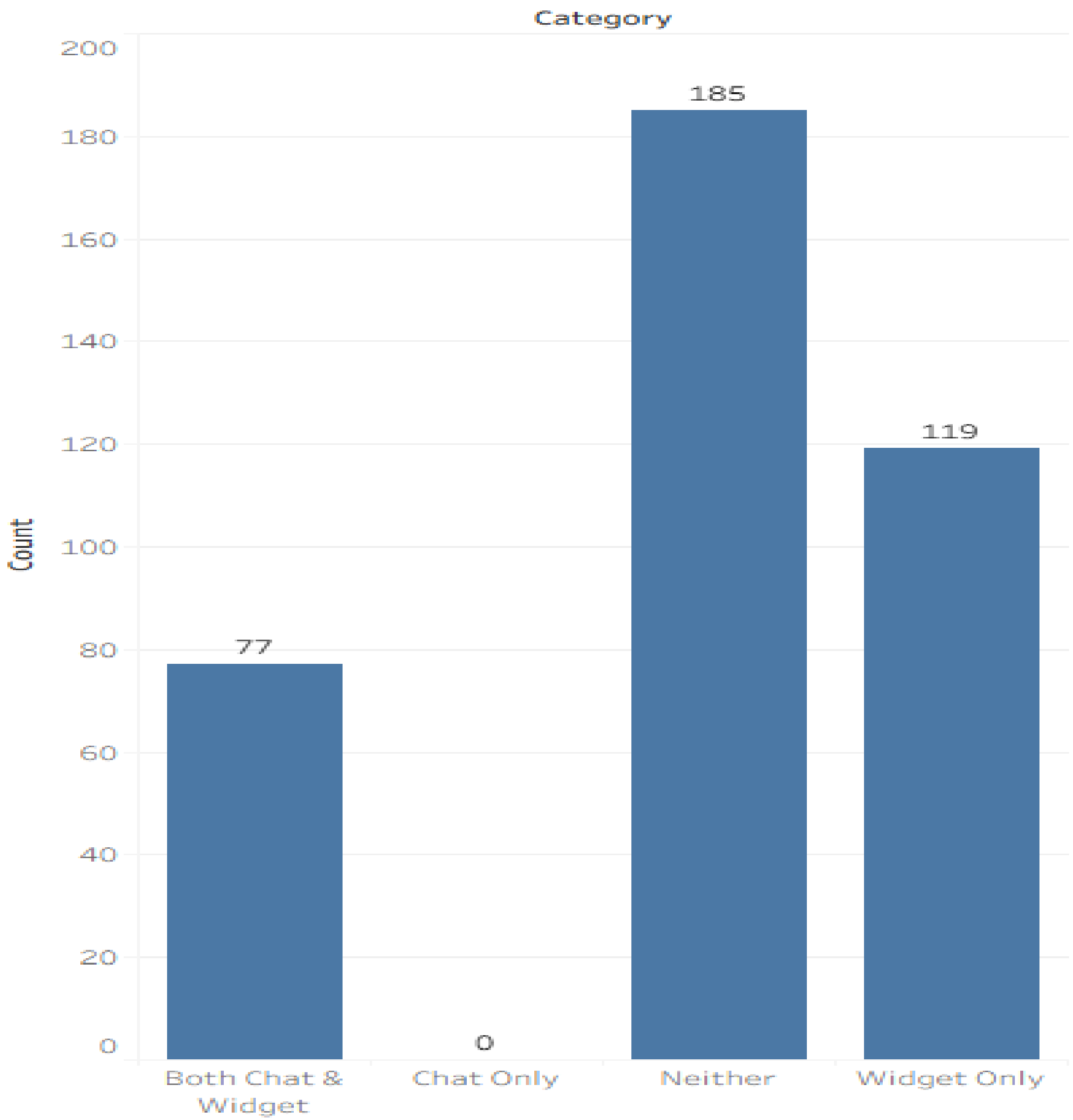
Widget usage rate measures the **proportion of sessions where the widget feature was actively used**, indicating user engagement and feature adoption within the platform.

Possible Insights

- Nearly half of all sessions (48.8%) included widget use, reflecting **moderate engagement**.
- The near 50–50 split suggests **untapped potential** to boost widget interaction.
- Higher widget usage could correlate with **better conversion or retention** rates.
- Targeted prompts or UX improvements could raise adoption.

59. Engagement Overlap

Engagement Overlap identifies users who interact with both **Chat** and **Widget** features. By tracking these users' conversion behaviors, this metric reveals how combined feature usage impacts engagement quality, conversion likelihood, and overall user interaction effectiveness across the platform's interactive components.



Relevance

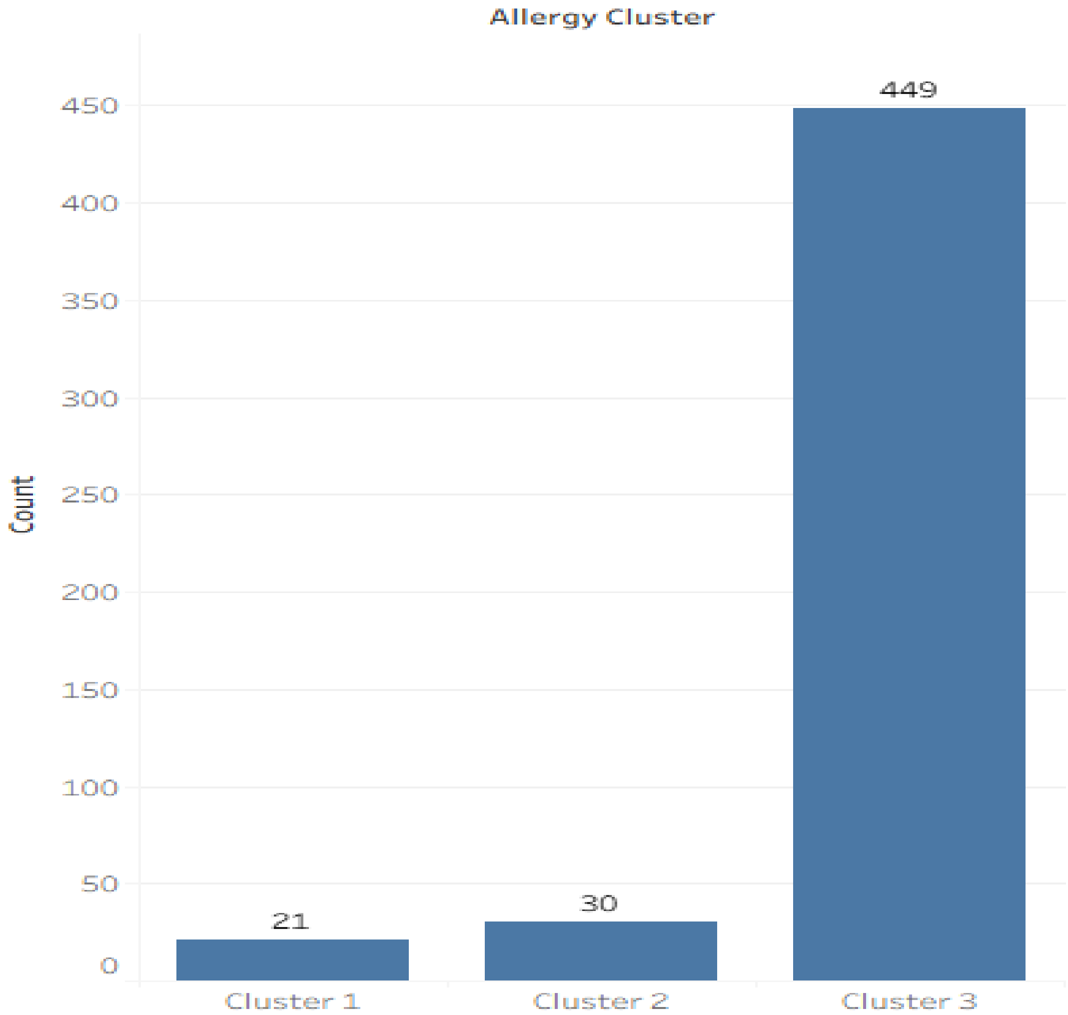
Engagement overlap identifies users who used **both chat and widget features**, revealing how multi-feature interaction correlates with user engagement and potential conversion behavior.

Possible Insights

- 77 users (≈19%) used both features, showing **cross-feature engagement**.
- No "Chat Only" users suggest **widget is a primary entry point** for interactions.
- 119 users (≈30%) used only the widget — strong widget dependency.
- 185 users (≈46%) used neither, indicating **large engagement gaps** to target.

60. Cluster Allergy Patterns

Cluster Allergy Patterns groups users based on shared allergy combinations to identify common clusters. This analysis helps uncover patterns in allergen sensitivity, enabling better personalization, improved recipe filtering, and enhanced safety recommendations tailored to similar user allergy profiles.



Relevance

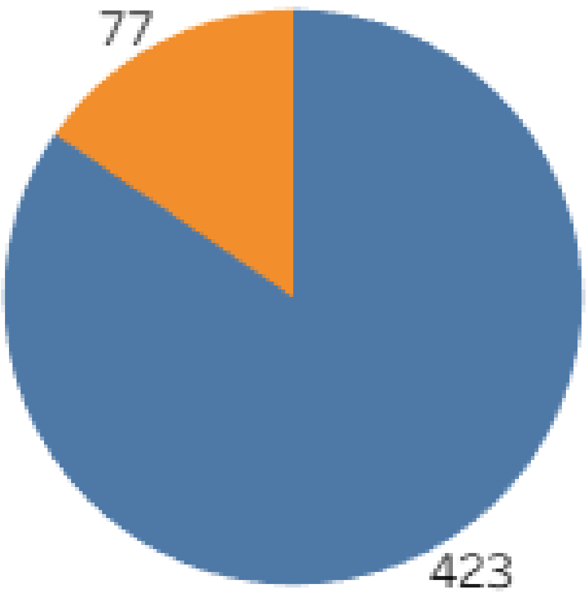
Allergy clustering groups users with similar allergy profiles, helping identify **common patterns** for personalization, ingredient filtering, and targeted recommendations in meal planning.

Possible Insights

- Majority (≈89.8%) fall under **Cluster 3**, showing a dominant allergy combination.
- Smaller clusters (21 and 30 users) may represent **unique or severe allergy groups**.
- Clustering enables **segmented recipe recommendations** for safety.
- Useful for designing **custom diet or exclusion-based features**.

61. Chat Usage Rate

Chat Usage Rate measures how often users engage with the chat feature. It's calculated as the percentage of sessions where **ChatStart** is not null, helping evaluate chat adoption, engagement frequency, and the overall effectiveness of conversational interactions across sessions.



Relevance

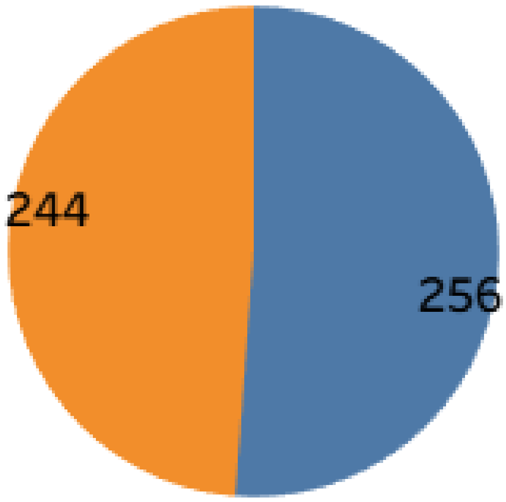
Chat usage rate measures the proportion of sessions where users engaged with the chat feature, indicating how often users seek **interactive help or support** during their session.

Possible Insights

- Only **15.4%** of sessions involved chat, implying **low engagement** with this feature.
- Users may find the chat **non-essential** or prefer **self-navigation**.
- Potential exists to improve **chat prompts or contextual triggers**.
- Chat could be linked with **conversion support** to enhance adoption.

62. Widget-Triggered Saves

Widget-Triggered Saves measure how many recipes were saved during sessions involving widget usage. It's calculated by counting save actions where the widget was active, helping assess the widget's influence on user engagement, content interaction, and recipe-saving behavior.



Relevance

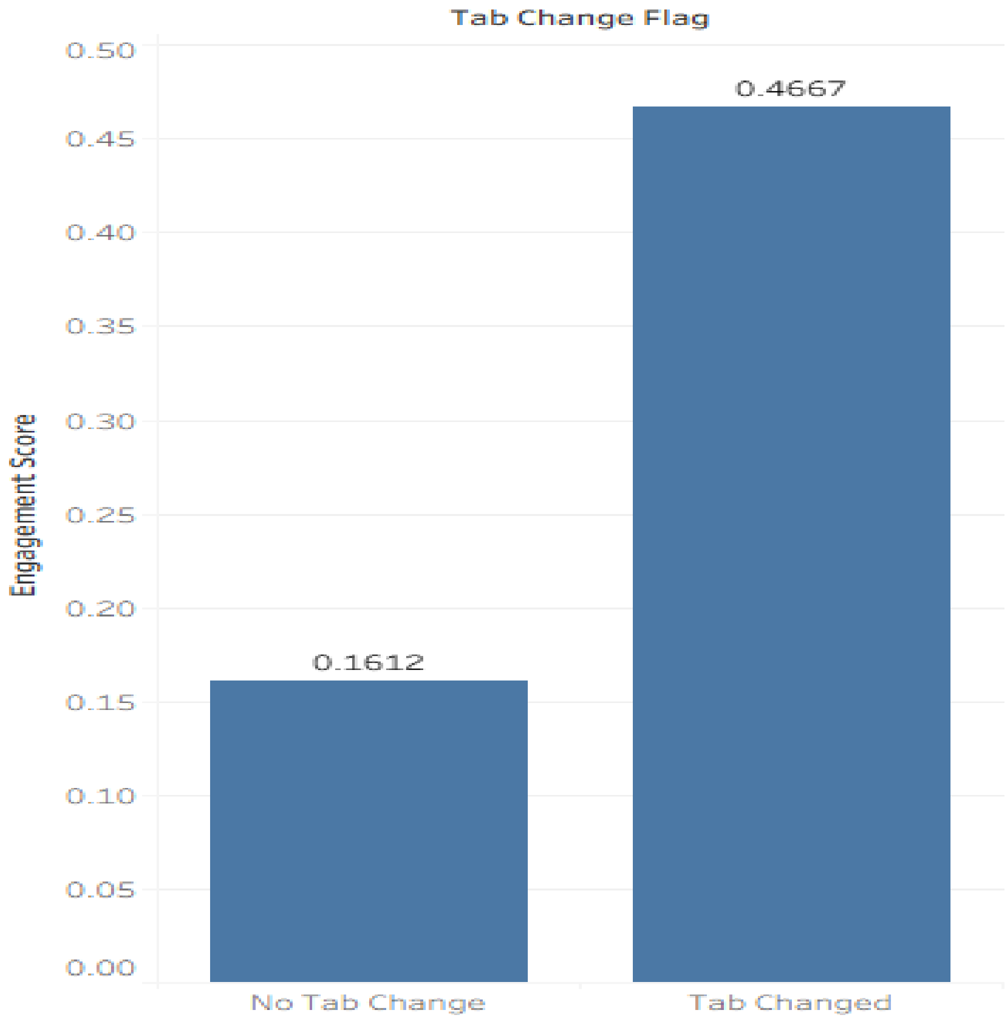
Widget-triggered saves measure how often recipe saves occur during sessions involving the widget, reflecting the **influence of interactive UI tools** on user engagement and saving behavior.

Possible Insights

- Saves are **evenly distributed**, showing the widget **moderately impacts** user saving habits.
- The widget could be **enhanced with personalized suggestions** to boost saves.
- Users saving recipes outside the widget suggest **multiple discovery paths**.
- Balance between widget and non-widget saves indicates **diverse engagement channels**.

63. Tab-Change Impact

Tab-Change Impact analyzes how switching tabs affects user engagement. By comparing key engagement metrics between sessions where **TabChange=True** and **TabChange=False**, it helps identify whether multitasking or tab-switching behavior influences attention, interaction depth, or conversion performance.



Relevance

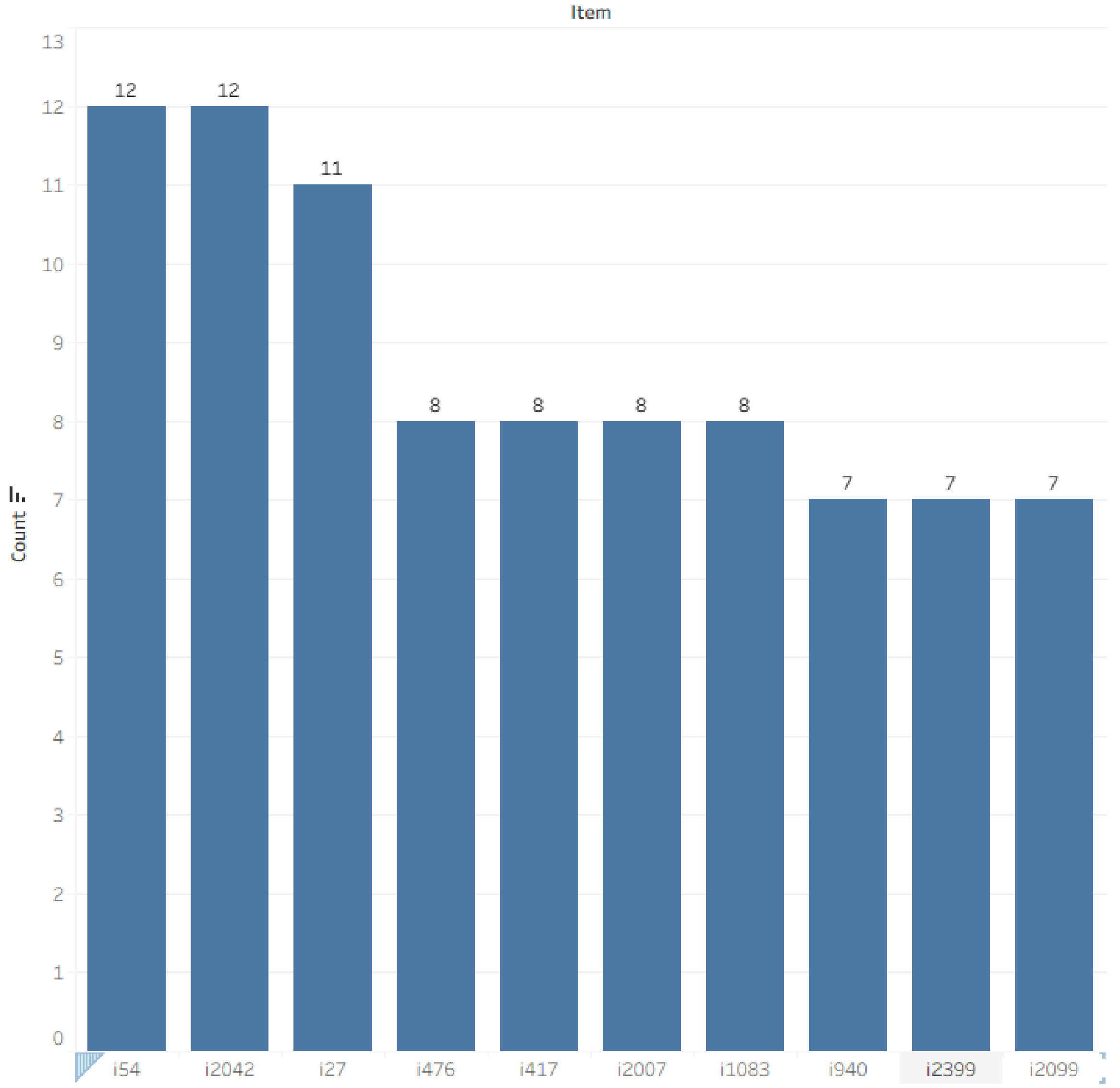
Tab-change impact assesses how switching between tabs influences engagement, showing whether **multi-tasking users interact more deeply** with the system compared to single-tab users.

Possible Insights

- Engagement is **nearly 3× higher** for users who change tabs.
- Tab switching likely indicates **active exploration** or multitasking behavior.
- Users not switching tabs show **passive engagement** or focused task completion.
- Encouraging dynamic navigation could **enhance user interaction** further.

64. Ingredient Diversity Index

Ingredient Diversity Index measures the variety of ingredients a user interacts with using an **entropy score**. Higher values indicate broader ingredient exploration, while lower values suggest repetitive usage, helping assess culinary diversity, personalization opportunities, and user preference patterns across recipes.



Relevance

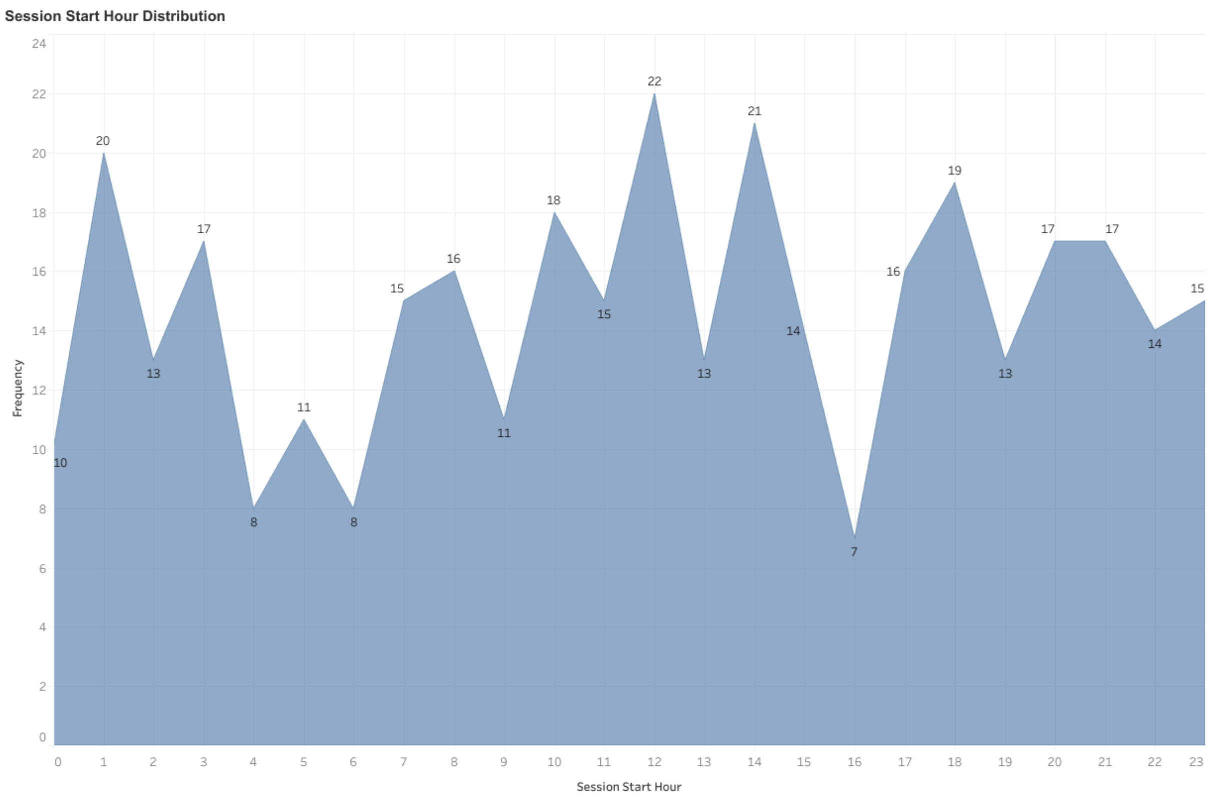
The ingredient diversity index (entropy) measures how varied the ingredients used by users are, reflecting their **culinary exploration and personalization** in recipe choices.

Possible Insights

- Ingredients like **i2042, i54, and i27** show high diversity, appearing across varied recipes.
- High entropy suggests **users prefer versatile ingredients**.
- Low diversity may indicate **repetitive or niche ingredient use**.
- Can guide **recommendation algorithms** toward more personalized ingredient suggestions.

65. User Session Distribution by Hour of Day

This metric shows the distribution of session start times across each hour of the day (0–23). It counts how many user sessions begin in each hour, revealing when users are most likely to initiate activity on the platform.



Relevance

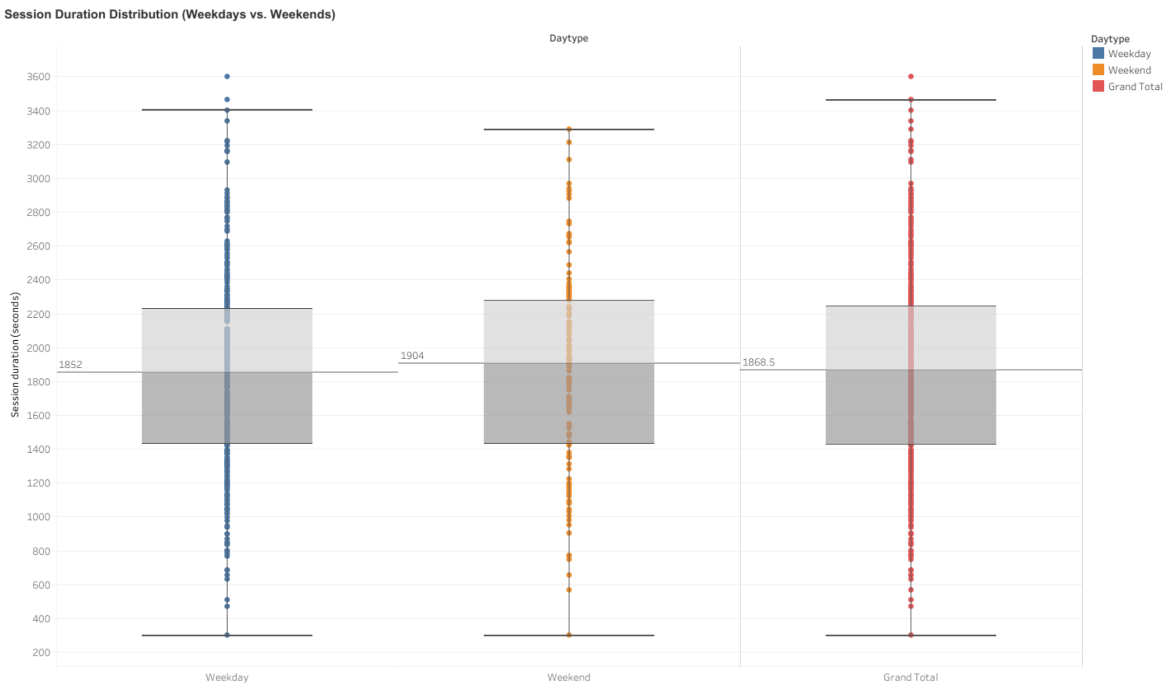
Understanding when sessions start most frequently helps identify users' active hours, which can guide optimal timing for notifications, promotions, or content updates. It also reflects user lifestyle patterns and can help tailor engagement strategies.

Possible Insights

Peak hours may indicate when users are most available or engaged, suggesting the best times for marketing pushes or product launches. Conversely, low-traffic hours might reveal opportunities for system maintenance, performance optimization, or targeted campaigns to boost engagement during off-peak times.

66. Session Duration Patterns by Day Type

This metric visualizes how individual session durations are distributed across weekdays and weekends. The box plot represents the median, quartiles, and variability of session lengths, providing a view of engagement spread rather than a single average.



Relevance

Understanding the full distribution of session durations helps capture engagement patterns more accurately than relying on mean values alone. It highlights typical user behavior, consistency of session lengths, and the presence of variability or outliers across different day types.

Possible Insights

The visualization can reveal whether session behavior is generally consistent or widely varied between weekdays and weekends. Wider boxes or longer whiskers indicate greater variability, while tighter boxes suggest more uniform engagement patterns. Outliers identify unusually long or short sessions that may reflect specific user segments or behaviors.

67. User Engagement Duration by Geographic Location

This metric displays the average session duration across the ten locations with the highest user engagement levels. The map visualization represents each location with proportional markers, allowing quick comparison of session duration by geography.

Relevance

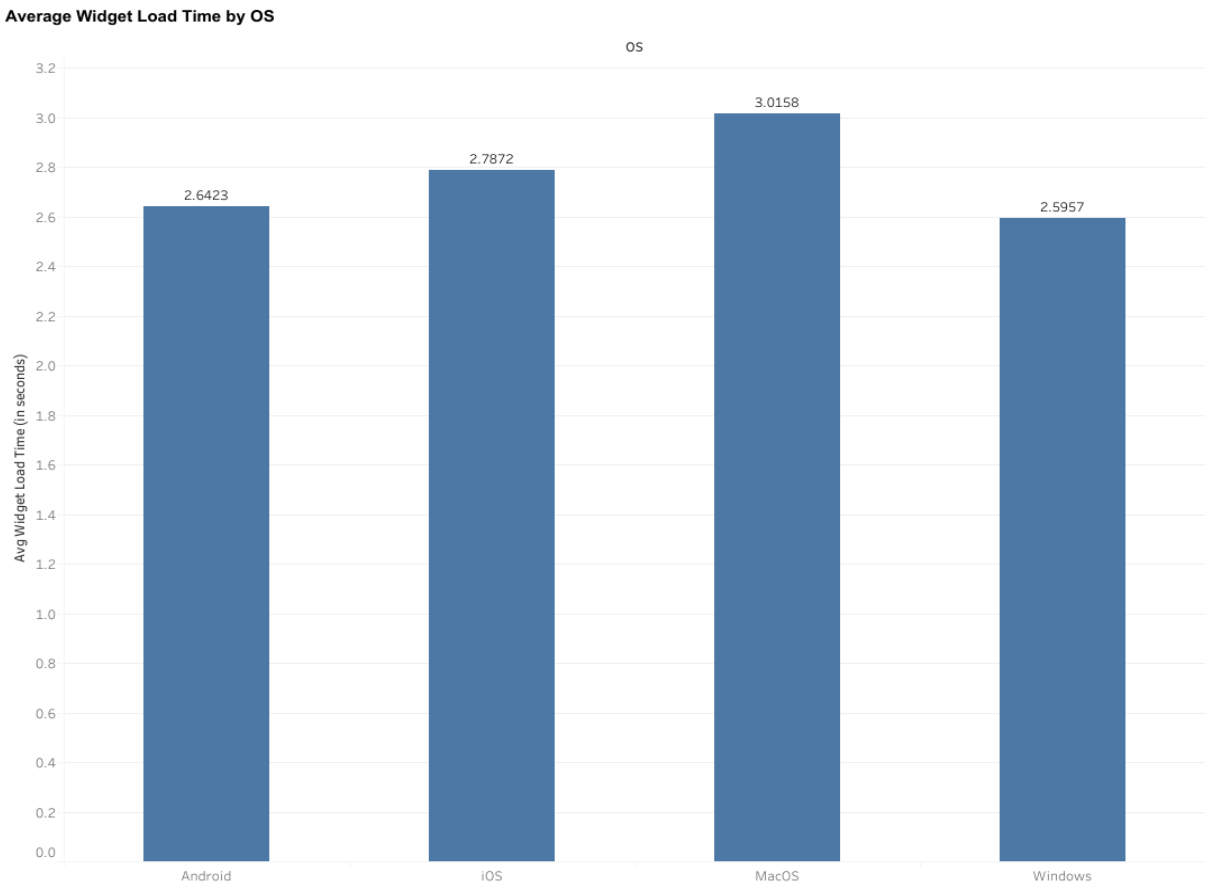
Geographical patterns in session duration help identify where users are most engaged and can highlight differences in behavior across regions. Understanding these variations supports localization efforts, infrastructure optimization, and region-specific engagement strategies.

Possible Insights

The visualization can reveal locations where users tend to have longer or shorter sessions, suggesting regional differences in user intent, connectivity, or content relevance. Larger markers indicate areas of higher average engagement, while smaller ones suggest opportunities for deeper analysis or targeted improvements. Comparing session durations across regions also helps evaluate the global consistency of the user experience.

68. Widget Load Performance by Operating System

This metric measures the average time taken for the widget to load across different operating systems. By parsing user agent data, each session is categorized by OS, and the mean load duration is calculated for each group. The resulting visualization compares performance across major platforms such as Windows, macOS, iOS and Android.



Relevance

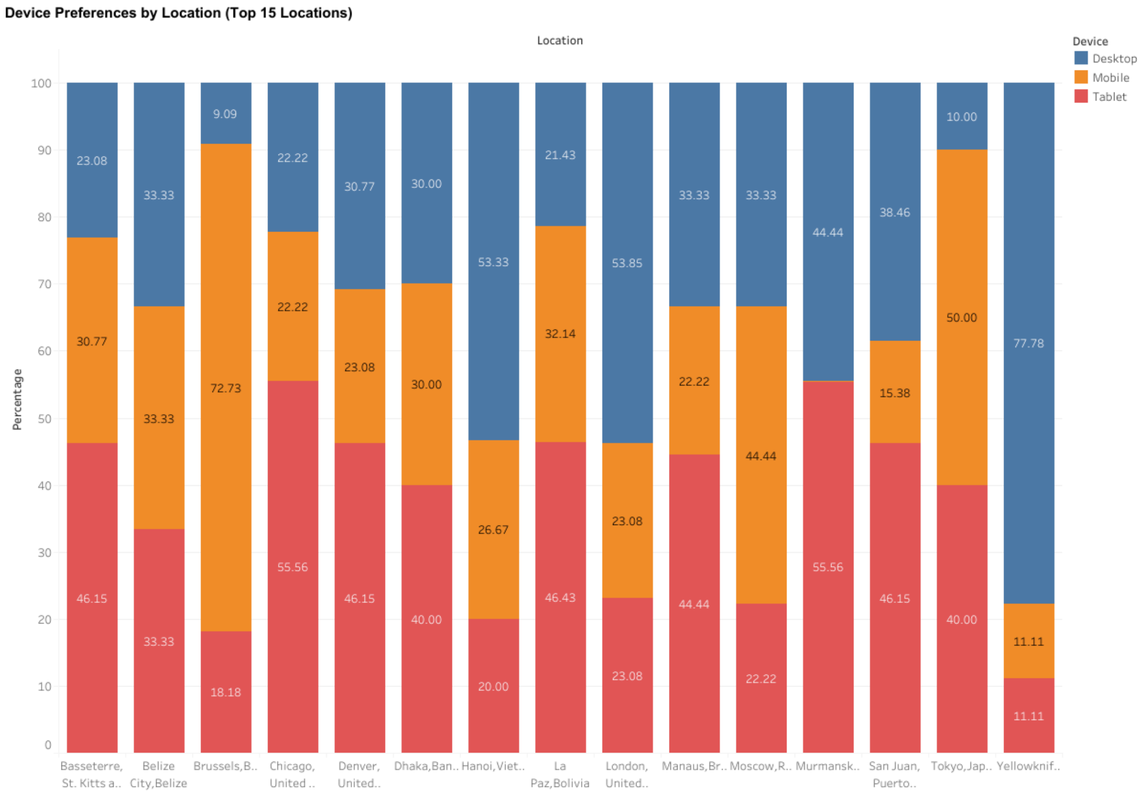
Load time is a key performance metric directly impacting user experience and engagement. Comparing widget performance by operating system helps identify platform-specific optimization opportunities and ensures consistent responsiveness across user environments.

Possible Insights

The visualization highlights how system-level differences affect widget load performance. Longer load times on specific operating systems may indicate browser compatibility issues, rendering inefficiencies, or device hardware constraints. Consistent performance across platforms suggests stable and well-optimized delivery. Tracking this metric over time can reveal the impact of performance improvements, UI changes, or system updates on user experience.

69. Device Usage Patterns Across Top Locations

This metric analyzes user device distribution across the top fifteen locations by session volume. It calculates the percentage share of each device type (such as mobile, desktop, or tablet) within each location, visualized using a stacked bar chart to highlight regional variations in device usage.



Relevance

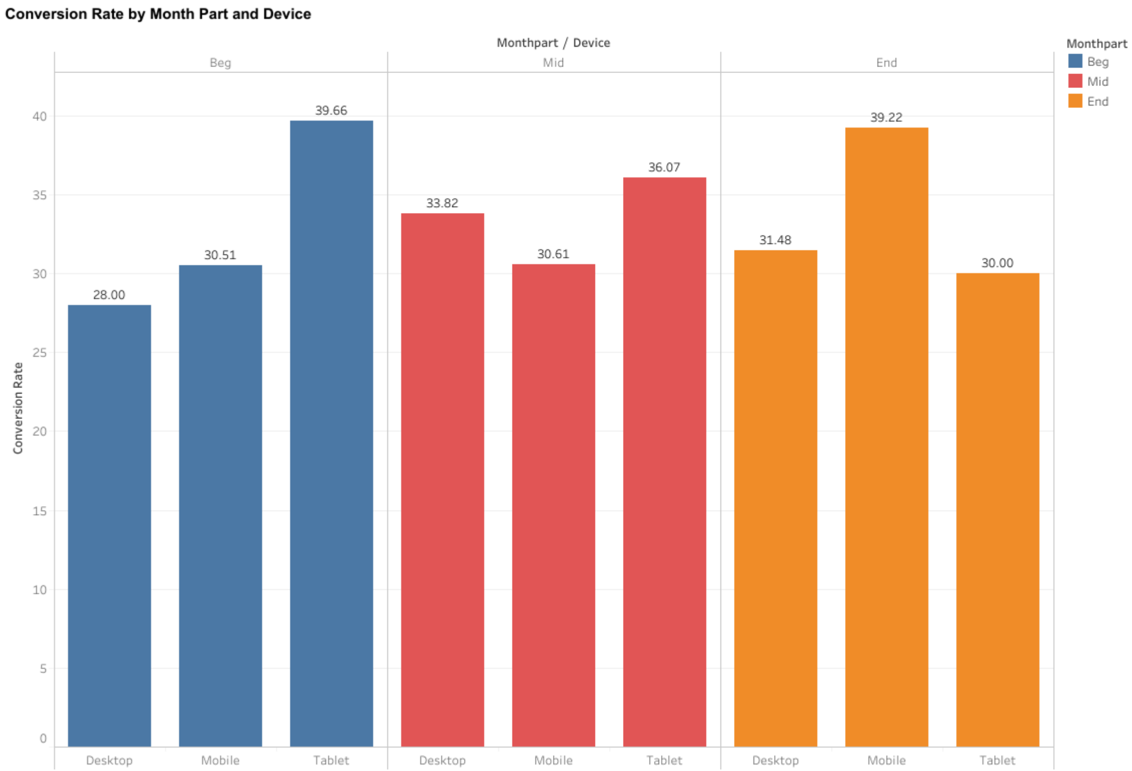
Understanding device preferences by geography supports optimization of user experience and interface design. Different regions may favor certain device types, influencing layout choices, load performance targets, and marketing strategy. Tracking these patterns helps ensure content and technical performance remain aligned with dominant device usage in each key market.

Possible Insights

The visualization reveals how device mix varies geographically, showing which regions rely more heavily on mobile, desktop, or other platforms. A higher share of mobile sessions may suggest stronger on-the-go engagement, while desktop-dominant regions could indicate workplace or longer-form interactions. Monitoring these trends over time can inform responsive design priorities, testing strategies, and resource allocation for platform-specific improvements.

70. Conversion Rate Trends Across Month Segments and Devices

This metric measures how effectively users convert (via adding a recipe to cart or saving it) across different parts of the month and device types. Sessions are divided into three equal time segments — Beginning, Middle, and End of the month — and conversion rates are calculated for each device within these periods. The grouped bar chart visualizes how conversion behavior fluctuates over time and by device.



Relevance

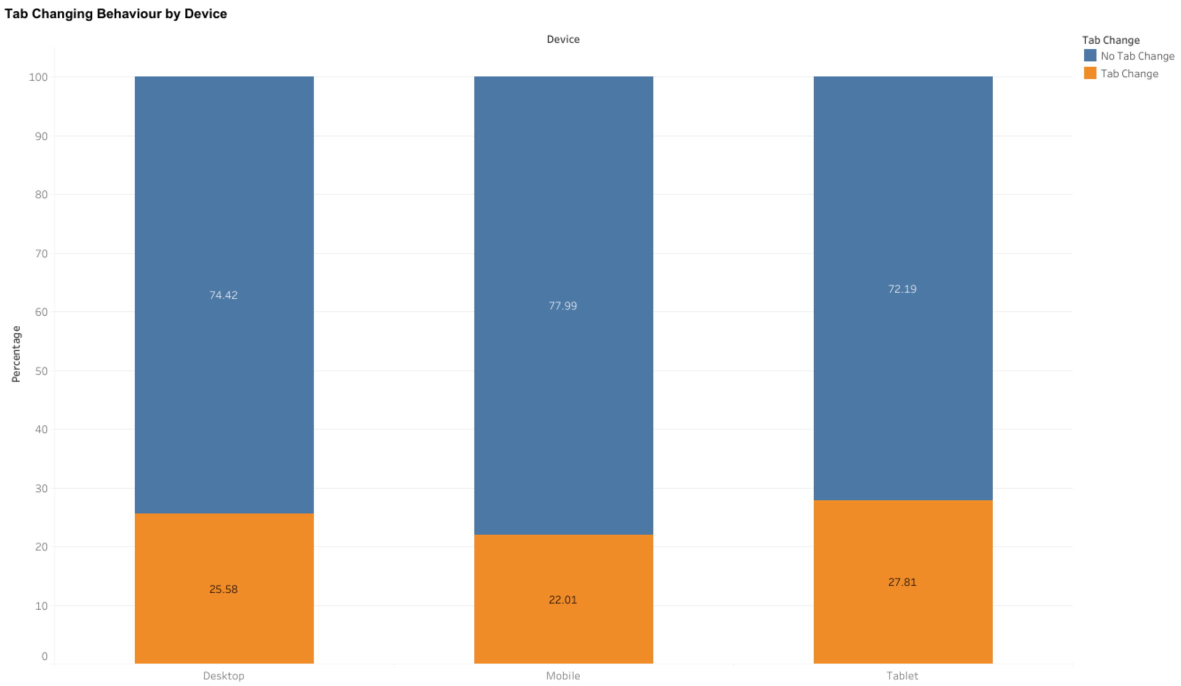
Conversion rate is a primary indicator of user engagement and purchase intent. Analyzing it by time segment and device provides insight into when and where users are most likely to act. This helps identify temporal engagement trends and highlights whether specific devices drive stronger conversion performance during certain times of the month.

Possible Insights

The visualization can uncover meaningful behavioral patterns — for example, higher conversions early in the month may align with user pay cycles, while mobile conversions might dominate during weekends or commuting hours. Lower conversion rates for specific devices could signal usability issues or friction in the purchase flow. Tracking this metric over time enables targeted improvements in timing, device experience, and conversion optimization strategies.

71. Tab Switching Patterns Across Devices

This metric measures how frequently users switch between tabs during a session, segmented by device type. It calculates the percentage of sessions that involved tab changes versus those that did not, for each device category. The stacked bar chart visualization shows the relative share of tab-changing and non-tab-changing behavior per device.



Relevance

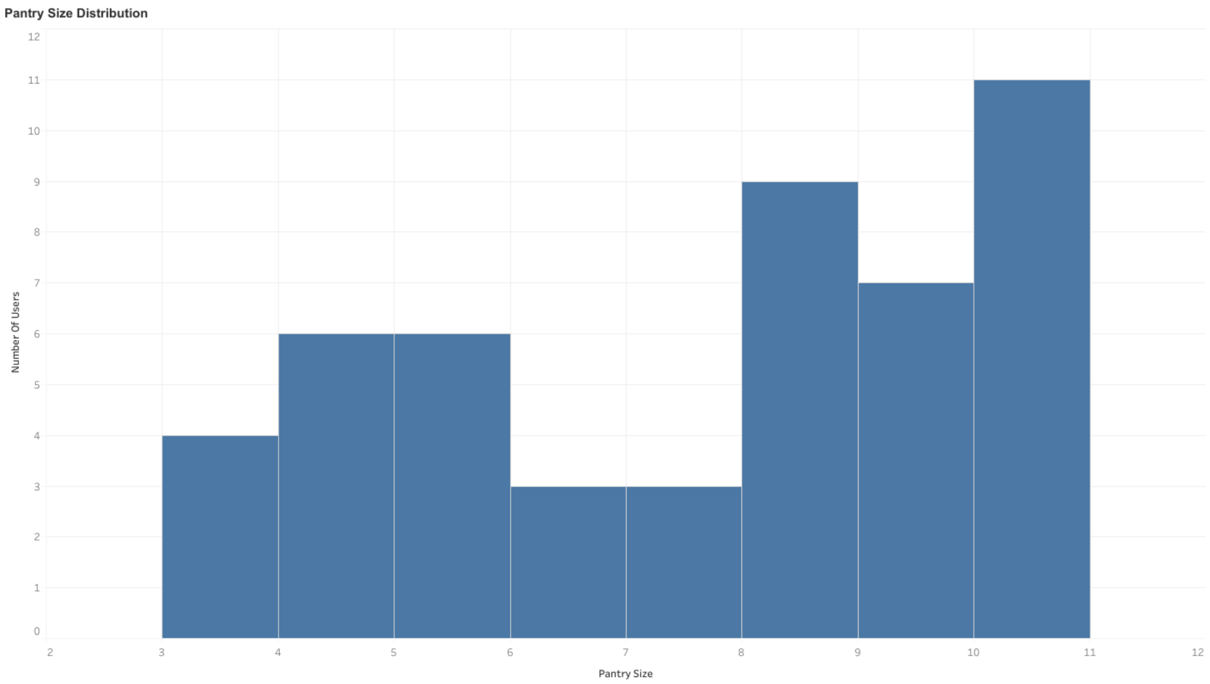
Tab changing is a useful behavioral indicator of user engagement patterns and multitasking tendencies. Comparing this activity across devices helps identify how usage context differs — for example, mobile users may stay focused in a single view, while desktop users may navigate across multiple tabs. Understanding these patterns can inform interface design, session tracking, and engagement optimization strategies.

Possible Insights

The visualization highlights differences in user interaction flow by device. A higher share of tab changes on certain devices may indicate research-oriented or multitasking behavior, while lower rates can suggest more focused engagement. Tracking tab-changing trends over time helps assess whether design changes, loading performance, or content layout influence how users interact across browser tabs.

72. User Pantry Size Distribution

This metric shows how users are distributed based on the number of items in their pantry. It considers only users who have at least one pantry item and groups them by their rounded average pantry size. The visualization illustrates how many users fall into each pantry size category.



Relevance

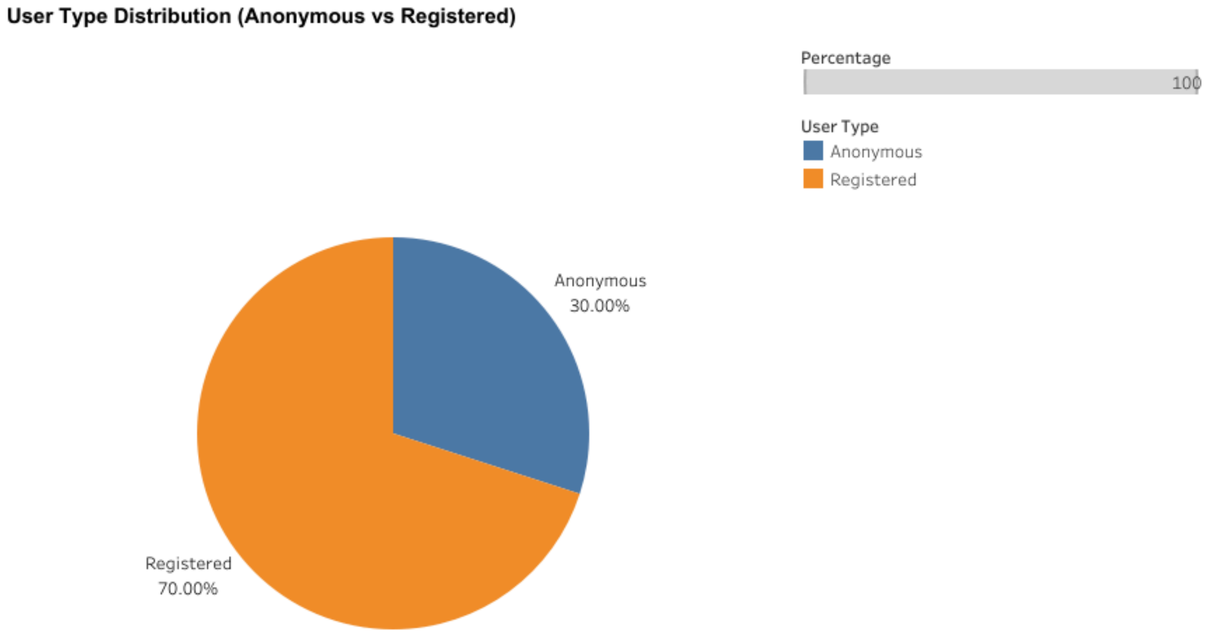
Understanding the distribution of pantry sizes helps assess user engagement depth with the pantry feature. It provides insight into whether most users maintain small, casual pantries or if a segment of highly active users manages larger collections — reflecting long-term value and product stickiness.

Possible Insights

A concentration around smaller pantry sizes may indicate limited feature adoption or early-stage usage, while a broader or right-skewed distribution suggests deeper engagement among certain users. Monitoring this distribution over time can reveal how new features, recipe availability, or personalization efforts influence user pantry-building behavior.

73. User Composition: Anonymous vs Registered

This metric compares the proportion of sessions initiated by anonymous users versus registered users. It classifies users based on their UserID and calculates the percentage share of each group. The pie chart visualization highlights the relative distribution between the two user types.



Relevance

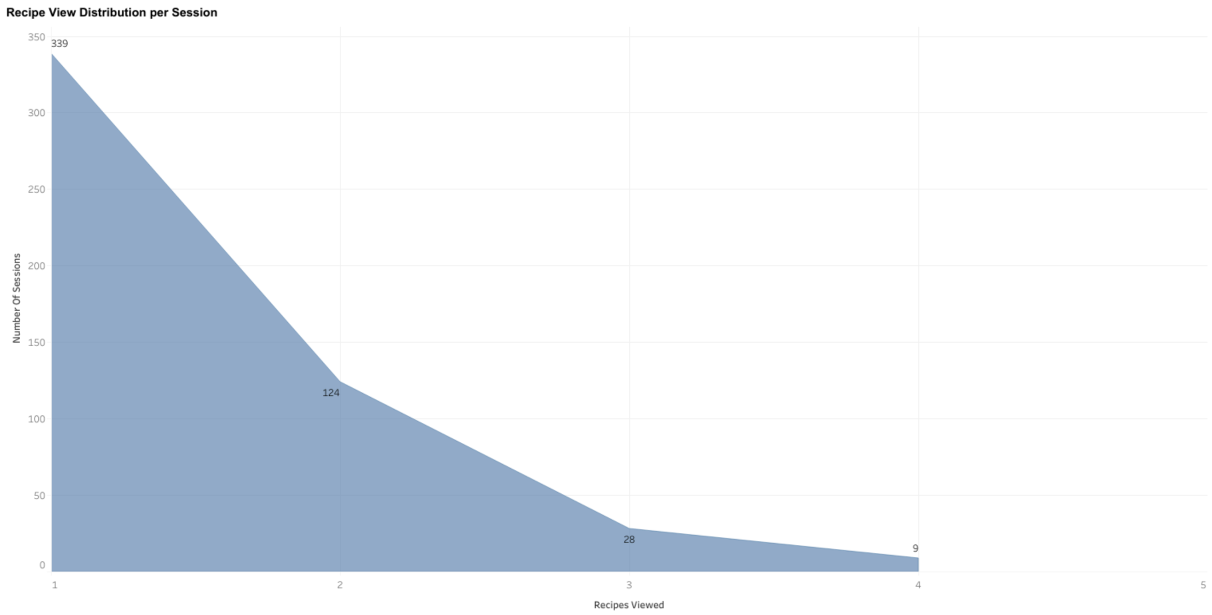
Understanding the balance between anonymous and registered users provides a key indicator of platform maturity and user trust. A higher proportion of registered users typically correlates with stronger retention potential, personalized engagement opportunities, and higher conversion readiness.

Possible Insights

A dominant anonymous user share may suggest barriers to registration or limited perceived value in creating an account, while a growing registered user base indicates successful onboarding and trust-building. Monitoring this ratio over time helps evaluate the effectiveness of sign-up flows, incentives, and personalization features.

74. Recipe Exploration Depth per Session

This metric represents how many unique recipes users view during each session. It counts the distinct recipes seen before and after the widget interaction, then groups sessions by the total number of recipes viewed. The resulting distribution shows how exploration behavior varies across sessions.

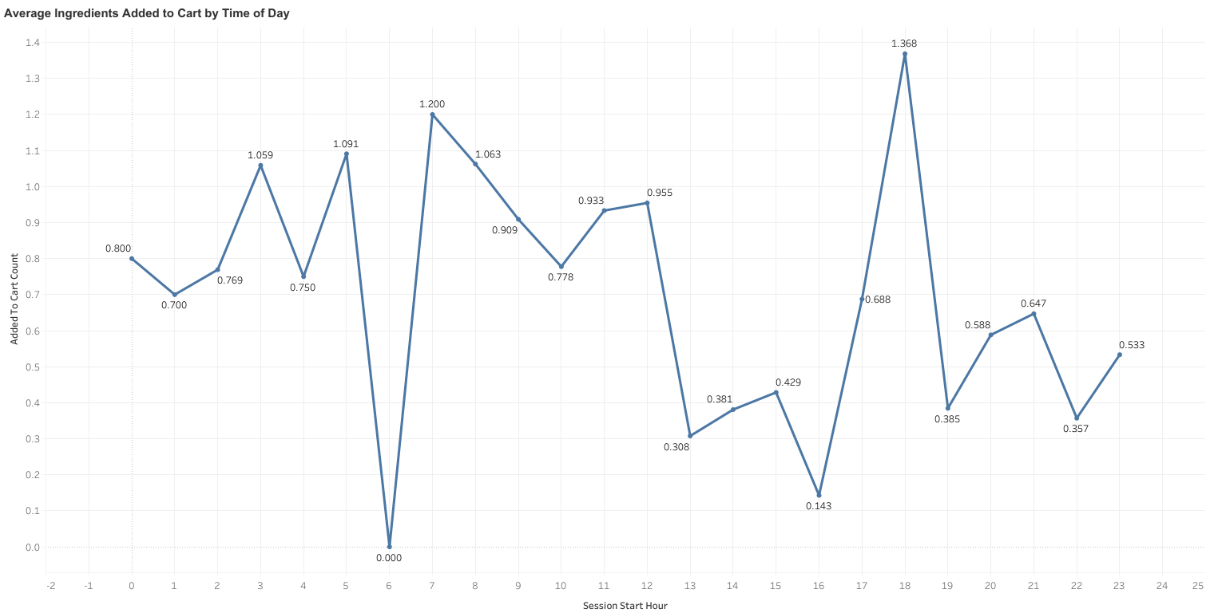


Relevance
Understanding recipe view distribution helps assess user browsing depth and content engagement. Sessions with multiple recipe views may indicate strong discovery behavior or high curiosity, while single-view sessions can point to quick decision-making or weak engagement.

Possible Insights
A concentration around one or two views per session could imply that users are finding what they need quickly—or that further browsing feels unnecessary. Broader distributions with higher recipe counts suggest effective recommendation logic and user interest in exploration. Tracking shifts in this pattern over time helps measure the impact of UI, recommendation, or content updates on user exploration behavior.

75. Ingredient Purchase Patterns by Time of Day

This metric measures the average number of ingredients users add to their cart during sessions starting at different hours of the day. It aggregates the count of cart additions per session and calculates the hourly average, helping visualize purchase activity patterns throughout the day.



Relevance

Analyzing ingredient additions by time of day reveals user purchase timing preferences and session intent patterns. It helps identify peak engagement hours and informs strategies around recommendation timing, marketing pushes, or operational optimizations (e.g., notification timing, stock sync).

Possible Insights

Peaks in cart additions during specific hours can indicate when users are most likely to plan meals or make purchases, whereas lower engagement windows may suggest times of low availability or casual browsing. Monitoring this trend over time helps align content scheduling, recipe promotions, and engagement nudges with user buying rhythms.

76. Ingredient Purchase Patterns Across Top Locations

This metric measures the average number of ingredients users add to their cart across different locations. It aggregates the number of items added to the cart per session and calculates the mean for each location, highlighting the top 10 regions where users add the most ingredients.

Relevance

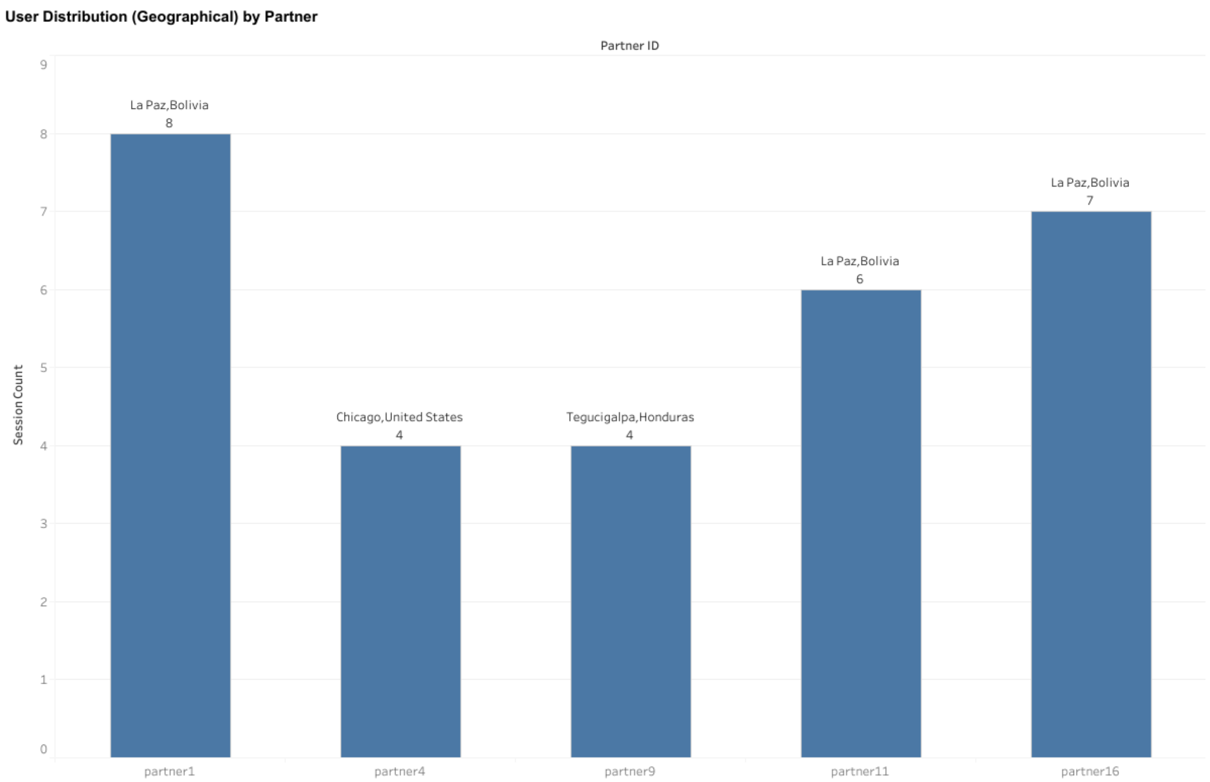
Tracking ingredient purchase behavior by location helps identify geographic trends in user engagement and shopping intensity. It provides valuable insight into which areas show higher purchasing activity, potentially reflecting stronger platform adoption, local recipe preferences, or demographic tendencies.

Possible Insights

Locations with higher average ingredient additions may represent regions with more engaged or conversion-ready users, while lower averages could indicate areas with lower intent or accessibility issues. Monitoring these trends over time supports market prioritization, localization strategies, and targeted promotional campaigns.

77. Top User Locations by Partner

This metric identifies the primary geographic location of user activity for each partner. It calculates the number of sessions by partner and location, then highlights the location with the highest session count per partner. The resulting view shows where each partner's user base is most active.



Relevance

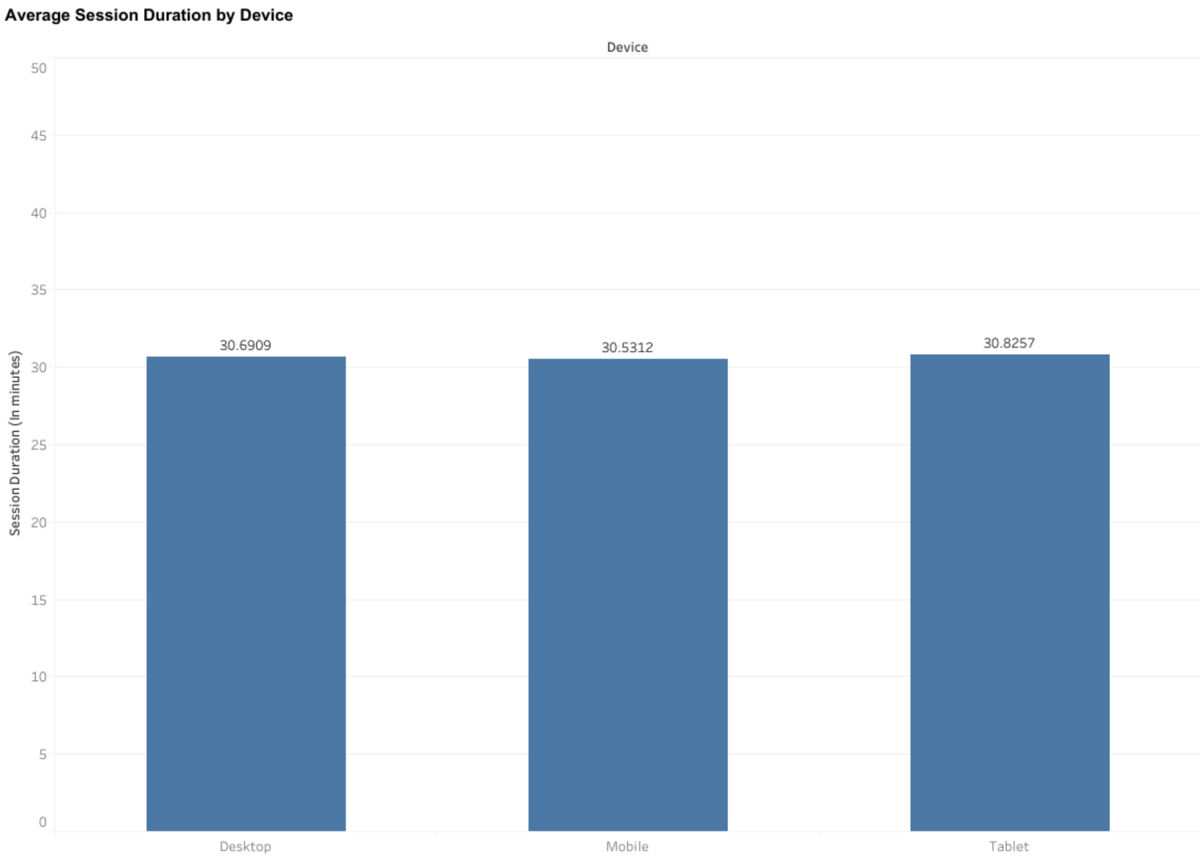
Understanding top-performing locations by partner provides insight into regional engagement strengths and market concentration. It helps partners and platform managers identify their key user bases, optimize local content or campaigns, and uncover potential regions for growth or support focus.

Possible Insights

A strong geographic skew toward a few key locations may indicate localized brand recognition or partner-specific reach. Conversely, a more evenly distributed presence suggests broader market appeal. Tracking these top locations over time can reveal how expansion efforts, partnerships, or regional campaigns influence user distribution.

78. Session Duration Patterns Across Devices

This metric measures the average length of user sessions across different device types. It calculates session duration in minutes by taking the difference between session start and end times, then computes the mean duration for each device category. The visualization compares engagement time by device.

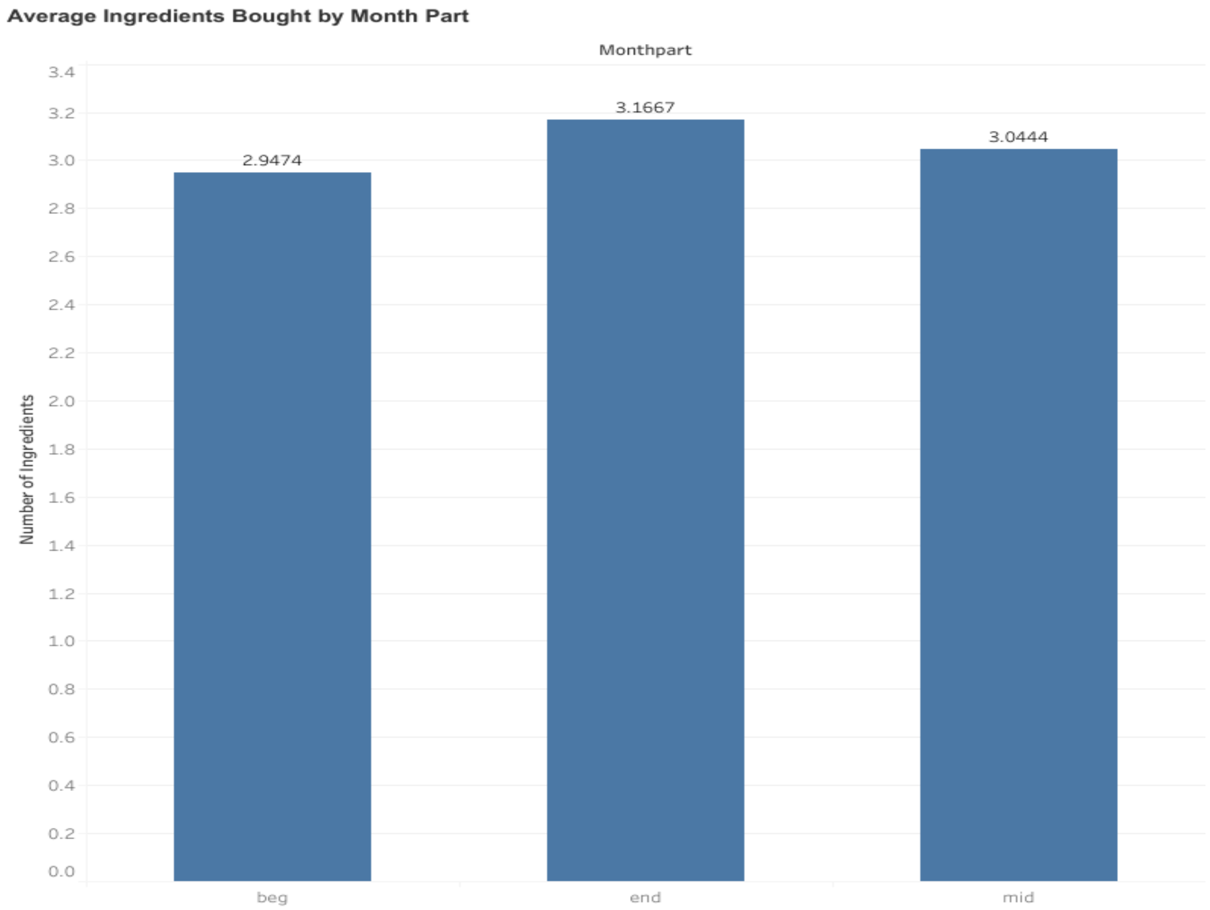


Relevance
Session duration is a strong indicator of user engagement and interaction quality. Comparing it across devices reveals behavioral differences, such as whether users on mobile engage more briefly than those on desktop or tablet. It provides key insights for optimizing design and content across platforms.

Possible Insights
Devices with longer session durations may support more immersive or comfortable experiences, while shorter durations might suggest friction, distractions, or quick task completion behavior. Tracking duration trends by device helps inform design priorities, performance tuning, and platform-specific engagement strategies.

79. Ingredient Purchase Patterns Across Month Segments

This metric analyzes how the average number of ingredients added to the cart varies across different parts of the month—beginning, middle, and end. It considers only sessions with at least one item added to the cart, calculating the mean ingredient count per month segment. The visualization highlights cyclical purchasing behaviors throughout the month.



Relevance

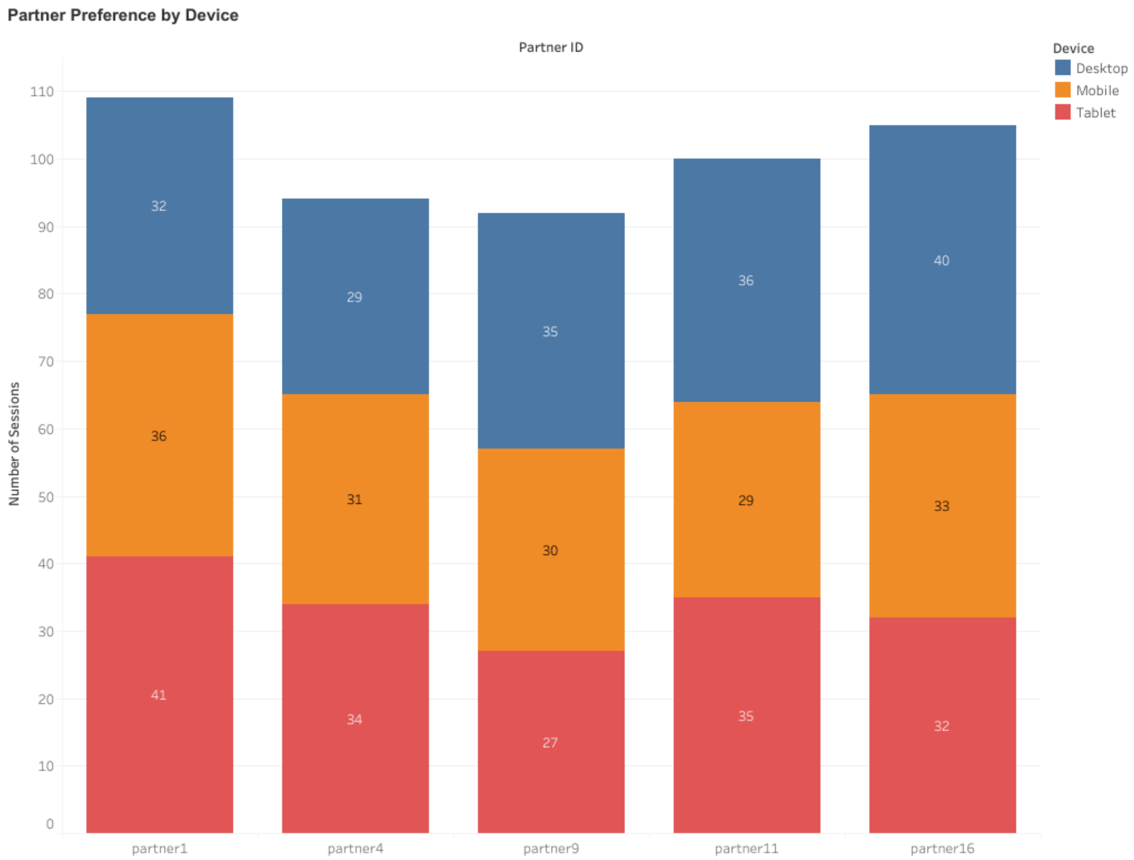
Understanding purchase timing trends helps identify when users are most active or inclined to shop. Variations across month segments can indicate behavioral rhythms tied to factors like salary cycles, promotions, or household planning. This insight supports better scheduling for marketing campaigns or feature rollouts.

Possible Insights

Higher average cart sizes at the beginning or middle of the month could suggest users are restocking or responding to payday-driven behavior. Consistent activity throughout the month implies stable engagement and routine shopping patterns. Monitoring these shifts over time can help fine-tune engagement strategies and inventory alignment.

80. Device Usage Distribution Across Partners

This metric analyzes how different devices (e.g., mobile, desktop, tablet) are used across various partners. It aggregates session counts by partner and device type to identify which devices dominate user interactions for each partner. The stacked bar chart visually represents the proportion of sessions coming from different devices for every partner.



Relevance

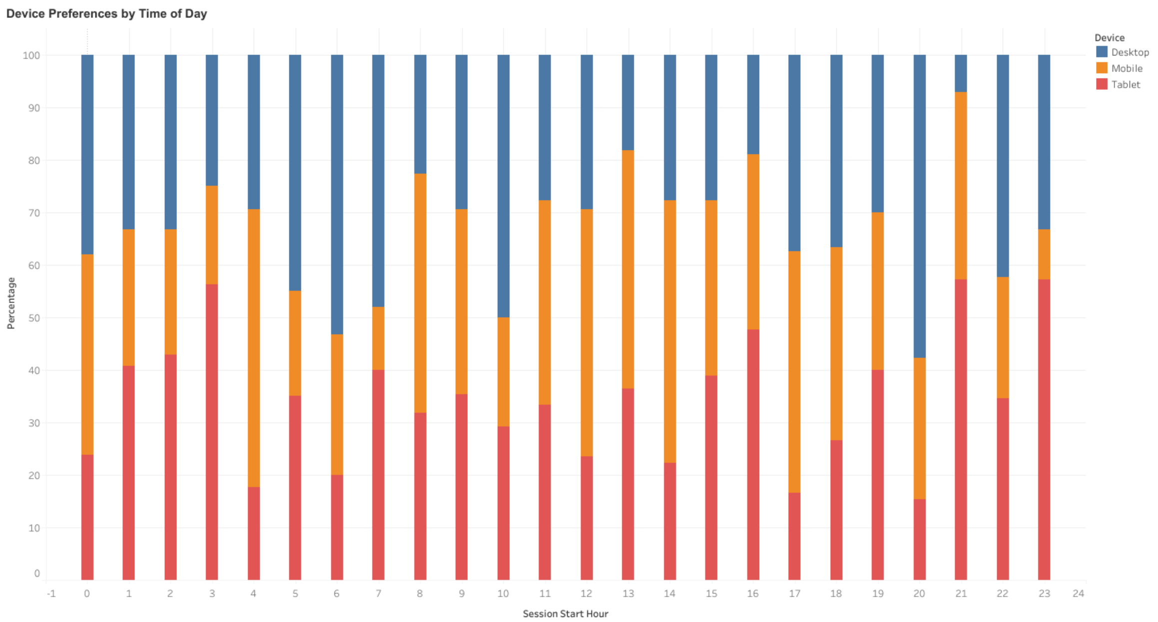
Understanding device distribution per partner is essential for optimizing platform performance and tailoring the user experience. For example, if a partner's users primarily access through mobile, mobile-first design and faster load optimization should be prioritized.

Possible Insights

Some partners may show strong device bias (e.g., mobile-heavy traffic), indicating platform-dependent user bases. Balanced device distribution suggests broader accessibility and engagement strategies. Shifts in device preference over time can also inform adaptive UI design, marketing focus, and development resource allocation.

81. Device Usage Patterns Across Different Times of Day

This analysis explores how users' device preferences (e.g., mobile, desktop, tablet) vary throughout the day. Each session's start hour is used to determine temporal trends in device usage. By calculating the percentage share of sessions per device type for each hour, the metric highlights time-based engagement behavior across devices.



Relevance

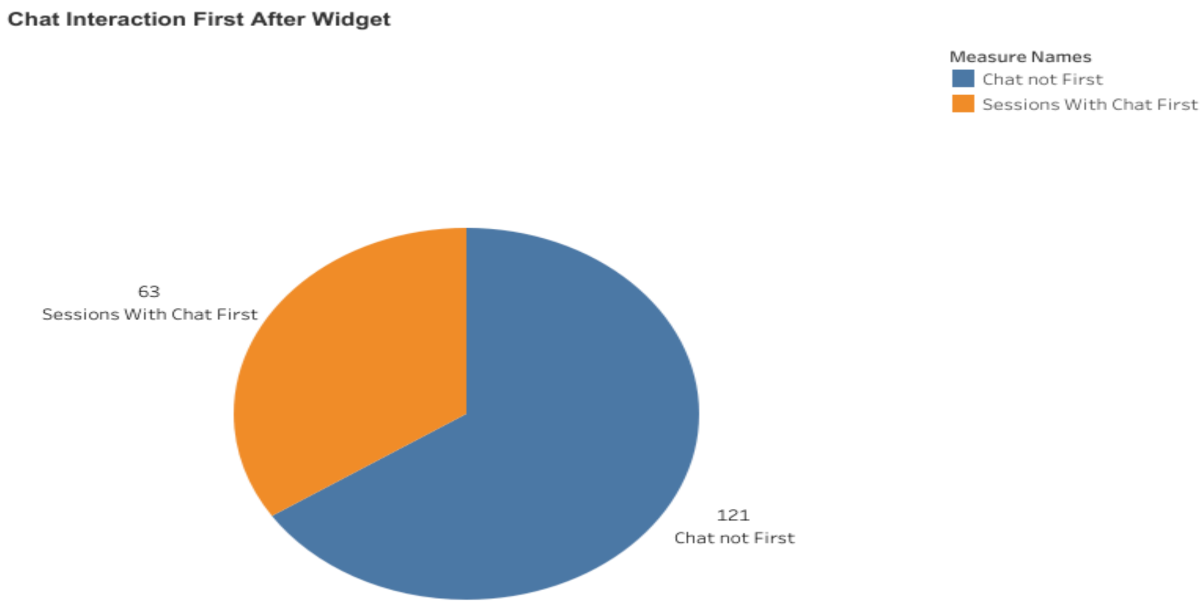
Understanding when different devices are most used helps optimize marketing campaigns, notification timing, and user experience. For instance, mobile sessions may dominate during commuting hours, while desktop sessions could be more frequent during work hours.

Possible Insights

- Peak mobile usage during morning and evening hours indicates engagement on-the-go.
- Higher desktop traffic during midday suggests work-related browsing or multitasking behavior.
- Tablet usage may rise in late hours, implying leisure browsing.
- Patterns can guide design responsiveness, ad scheduling, and platform-specific optimization.

82. Chat-First Engagement Rate

This metric measures how often users interact with the chat feature as their first action after opening the widget. By analyzing each session's activity log, it identifies whether the first meaningful interaction following the widget launch is "Interacted with Chat." The result indicates both the frequency and percentage of sessions where chat becomes the user's immediate focus after widget engagement.



Relevance

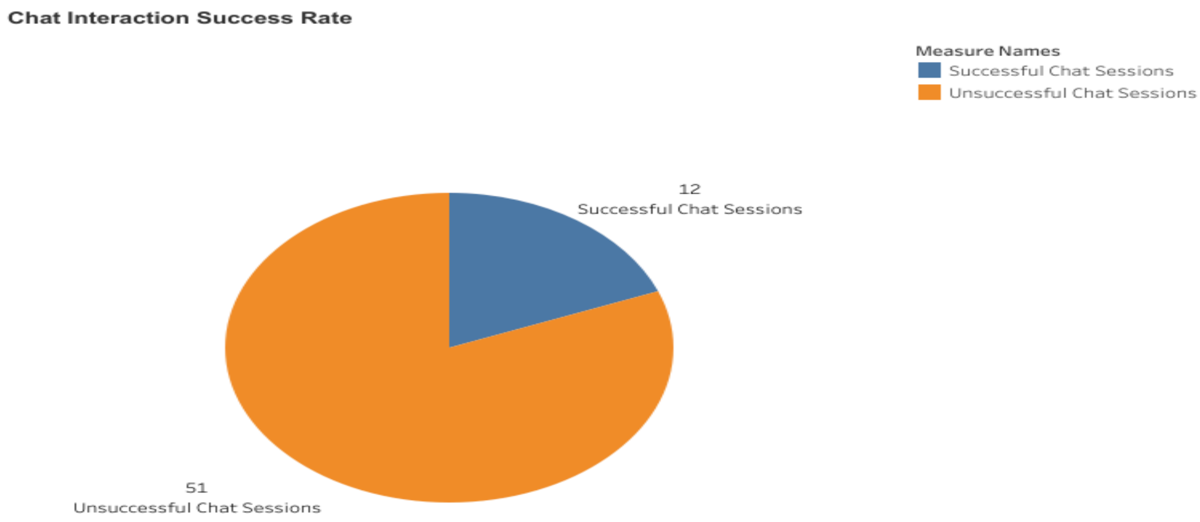
This metric reveals the proactive engagement level of users with conversational features. A higher percentage indicates that users view the chat as a primary point of interaction—perhaps for guidance, recipe help, or assistance in completing a task. Conversely, lower values may suggest that users engage with other elements first (like browsing or viewing recipes), indicating a different intent flow.

Possible Insights

- A high chat-first rate implies chat is effectively positioned and immediately engaging.
- A low chat-first rate could indicate users are exploring non-chat elements first, suggesting a need for UI or prompt optimization to encourage chat use.
- Tracking this metric alongside conversion or satisfaction rates could reveal how chat-first behavior correlates with positive outcomes such as longer sessions or higher cart additions.

83. Chat-Driven Conversion Rate

This metric evaluates how effective chat interactions are at driving conversions—specifically, how often users who interacted with chat proceeded to add items to their cart during the same session. By cross-referencing the presence of “Interacted with Chat” events with “Add to Cart” actions, it measures the conversion success rate of chat-engaged sessions.



Relevance

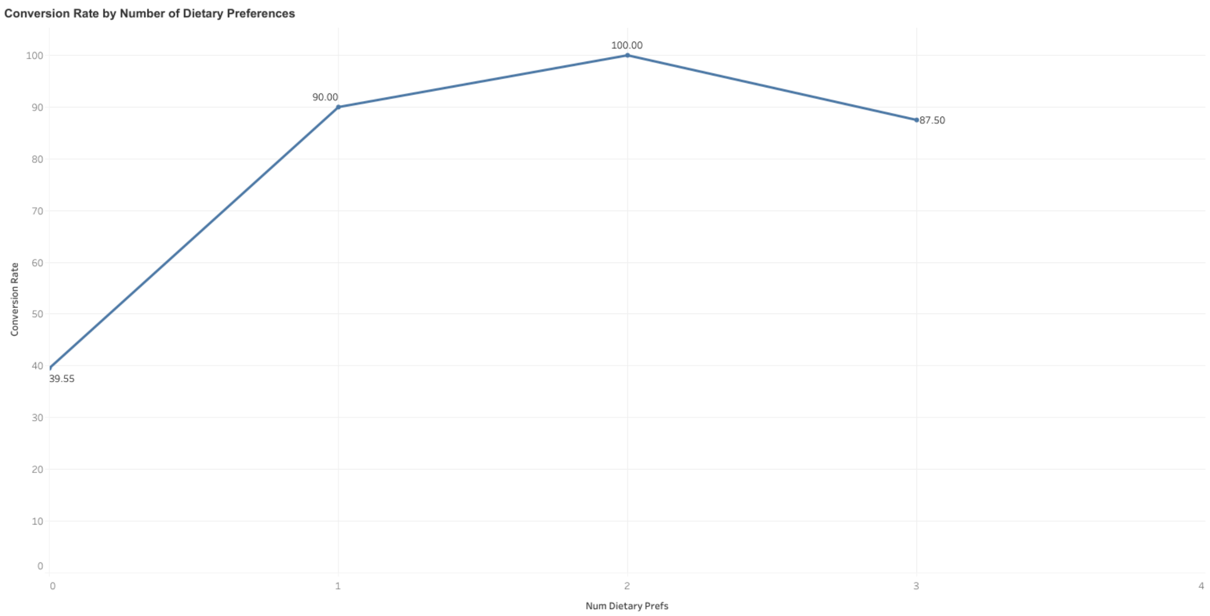
The metric serves as a key indicator of the impact of conversational engagement on purchase intent. It helps determine whether users who open and interact with the chat assistant are more likely to convert than those who do not. This insight can guide both chat design improvements and strategic placement decisions for maximizing eCommerce performance.

Possible Insights

- A high success rate indicates that chat is positively influencing purchase behavior, suggesting effective recommendations or assistance.
- A low success rate may imply that while users engage with chat, they don’t find it actionable or helpful enough to lead to conversions.
- Comparing this metric across different user segments (e.g., device, location, time of day) could uncover contextual factors that drive chat success.
- Over time, this metric can track ROI and performance improvements of chat-based engagement strategies.

84. Impact of Dietary Preference Count on Conversion Rate

This analysis examines how the number of dietary preferences selected by users influences their likelihood of conversion, where conversion is defined as either adding items to the cart or saving a recipe. Anonymous users are treated as unique sessions to ensure accurate representation of engagement patterns across all user types.



Relevance

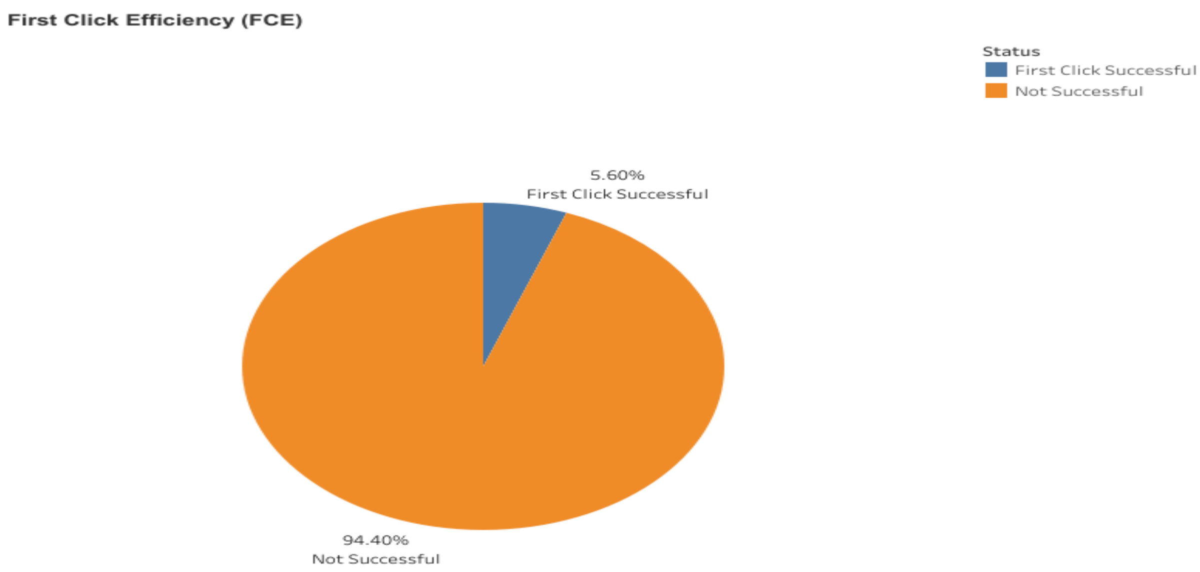
Understanding the relationship between user personalization depth (measured through dietary preferences) and conversion can reveal how customization drives engagement and purchase intent. Users with more dietary preferences might expect more tailored recommendations, and this metric helps assess whether that personalization translates into higher conversions.

Possible Insights

- A rising conversion trend with more dietary preferences suggests that personalized experiences effectively motivate conversions.
- A declining trend could indicate that users with stricter or more specific preferences find fewer suitable products or recipes, reducing their likelihood to convert.
- A flat trend may imply that dietary preference settings have little direct influence on purchase behavior, signaling room for improved recommendation algorithms.
- Segmenting this metric further by device, partner, or region could reveal contextual drivers behind conversion performance.

85. Effectiveness of Initial Recipe Engagement (First Click Efficiency)

This metric measures how often a user's first viewed recipe results in a conversion action, specifically being saved or added to cart. The analysis extracts the first recipe a user interacts with and checks whether that same recipe subsequently leads to an engagement action indicating interest or purchase intent.



Relevance

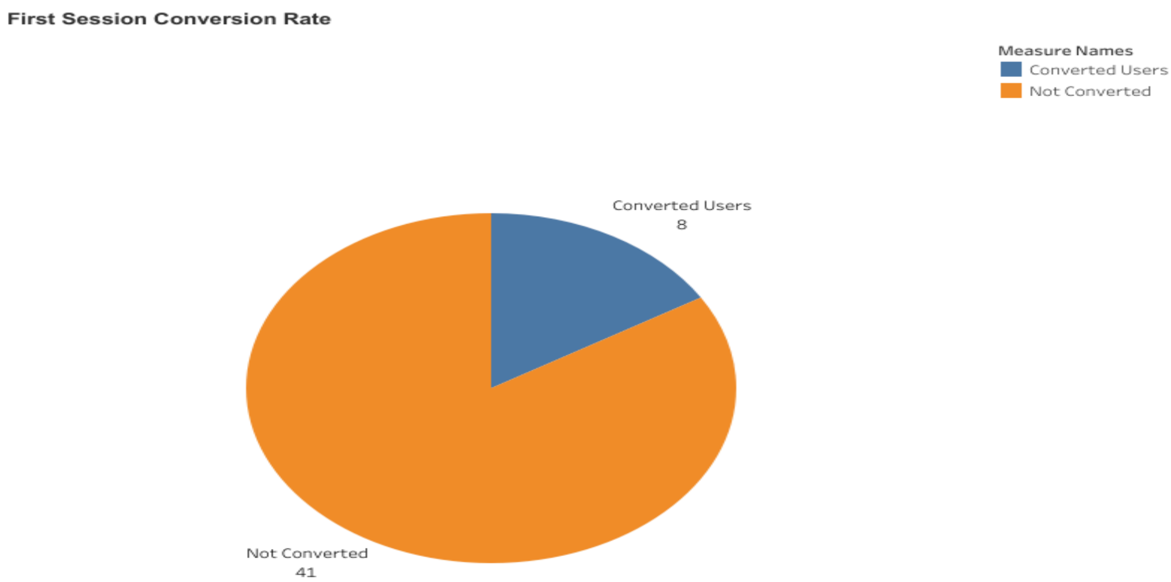
The First Click Efficiency (FCE) metric is a strong indicator of how effectively the platform's initial recipe recommendations or search results align with user preferences. A higher FCE suggests that users are finding relevant and appealing recipes early in their session, improving overall satisfaction and reducing browsing friction.

Possible Insights

- A high FCE indicates that first impressions and recommendations are effective, and users are quickly engaging with relevant content.
- A low FCE may signal mismatched search results or weak recommendation quality, prompting the need for refining recommendation algorithms or content placement.
- Segmenting this metric by time of day, user segment, or traffic source can uncover contextual differences in discovery success.
- Tracking FCE over time can help measure the impact of recommendation system updates or UI improvements on initial engagement efficiency.

86. First Session Conversion Rate

This metric calculates the percentage of registered users who complete a conversion action — specifically adding at least one item to the cart — during their first recorded session on the platform. Anonymous users are excluded to focus on behavior among identifiable, returning-capable users.



Relevance

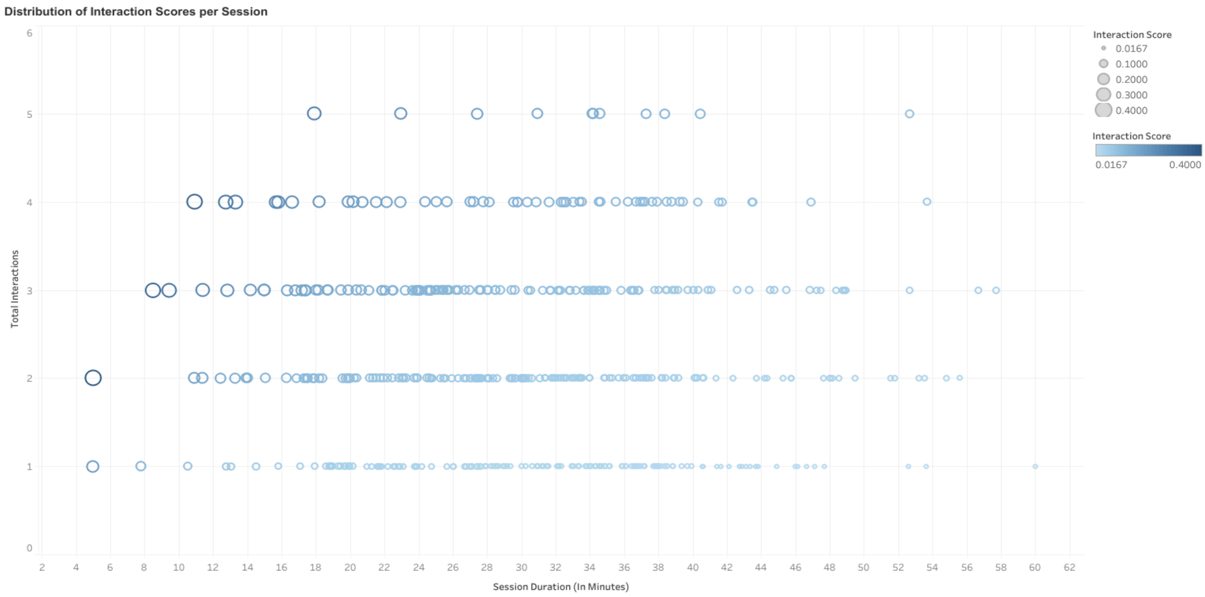
The First Session Conversion Rate highlights the platform's ability to engage and convert users immediately upon their initial interaction. It serves as an early indicator of the effectiveness of onboarding flows, recommendation quality, and initial content engagement.

Possible Insights

- A high first-session conversion rate suggests that new users find the platform intuitive, relevant, and immediately valuable.
- A low rate may indicate friction points during first interactions — such as poor onboarding guidance, unclear navigation, or irrelevant recommendations.
- Tracking this metric over time helps evaluate the impact of onboarding improvements, personalization efforts, or UI/UX redesigns.
- Segmenting by traffic source, registration method, or device type can reveal how different entry points affect early conversion performance.

87. Interaction Intensity vs Session Duration

This visualization explores how user engagement evolves with varying session durations. Each point represents an individual session, where the x-axis denotes Session Duration (in minutes), the y-axis indicates Total Interactions, and the color gradient reflects the Interaction Score (interactions per minute). The scatter plot enables identification of behavioral clusters—such as users who interact frequently in short sessions versus those who engage steadily over longer periods.



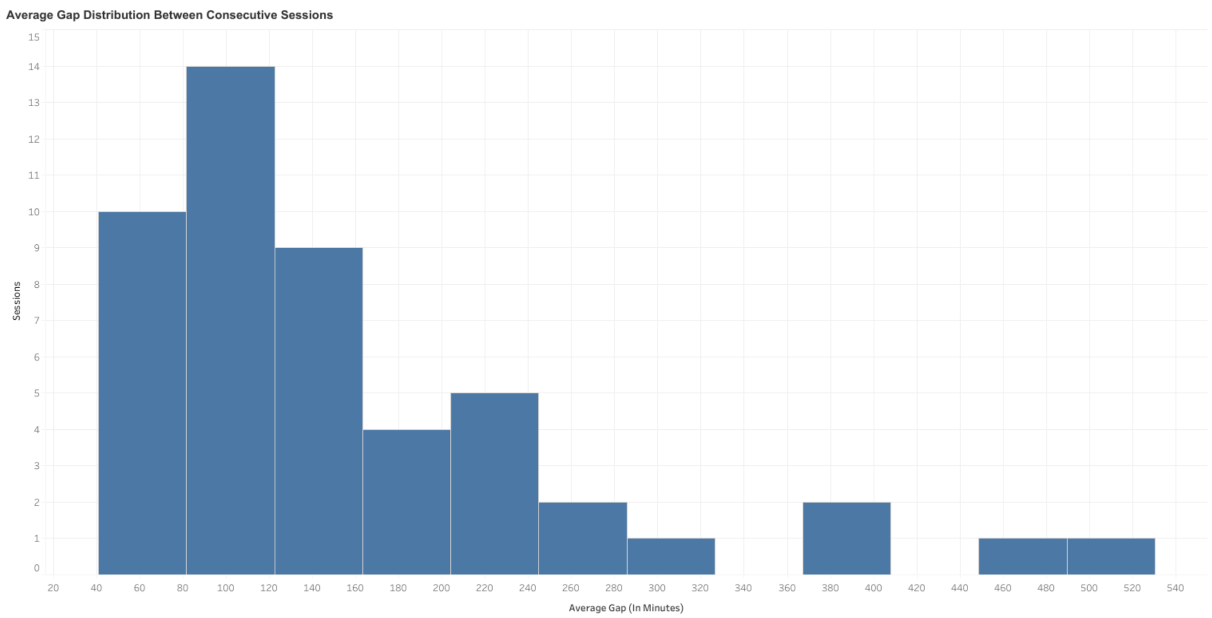
Relevance:
This metric bridges time spent and engagement quality, providing a clearer understanding of user efficiency and interaction depth. It helps determine whether long session times are genuinely productive or if shorter, high-intensity sessions deliver stronger engagement.

Insights:

- Sessions with higher Interaction Scores often indicate focused, purposeful user activity.
- A concentration of points with moderate durations but high interaction scores may signify effective user experiences.
- Long sessions with low interaction density can indicate disengagement or interface friction.
- This analysis helps product and UX teams refine session flow, feature placement, and engagement strategies to foster efficient, meaningful interactions.

88. User Return Interval

This analysis measures the average time gap between consecutive sessions for each user, providing insight into engagement frequency and return behavior. By calculating the mean interval (in minutes) between sessions across all users, it helps quantify how often users revisit or re-engage with the platform. The visualization highlights the distribution of these average session gaps, showing whether most users return quickly or take longer breaks between visits.

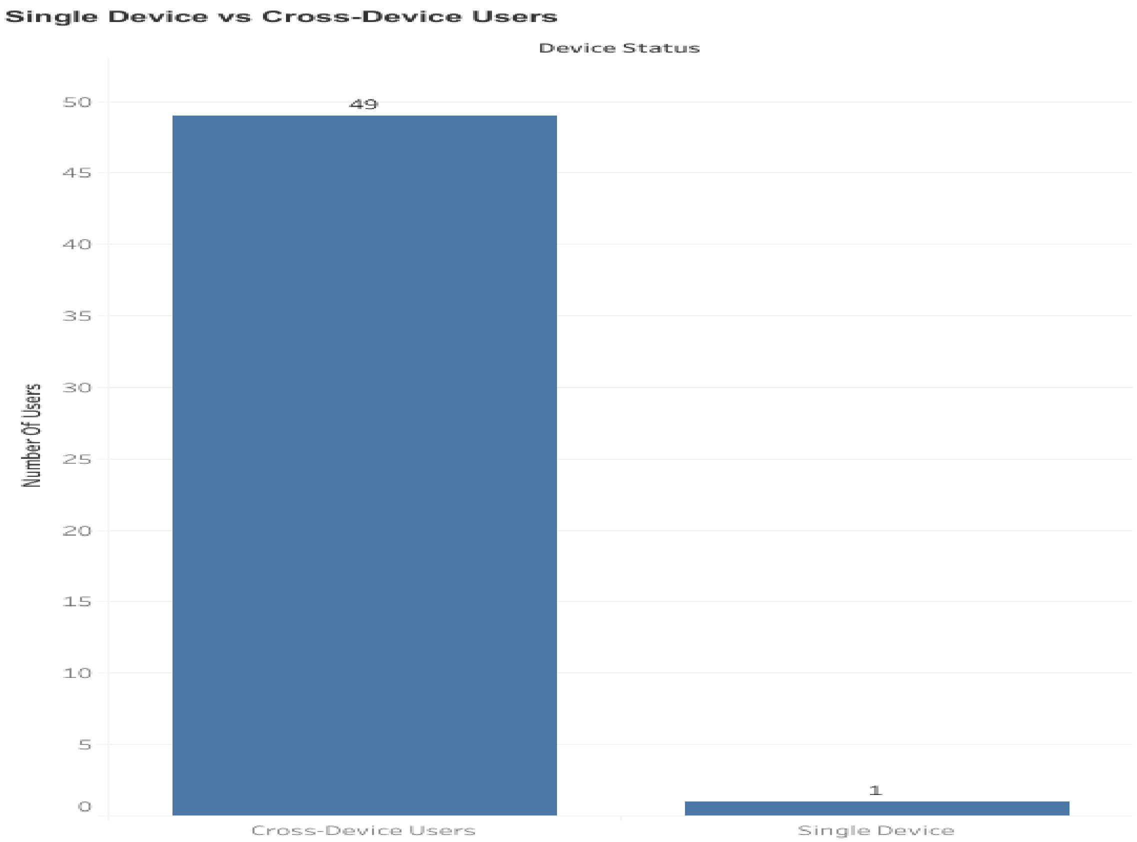


Relevance:
Understanding session gaps is crucial for assessing user retention and habitual engagement patterns. Shorter gaps often indicate stronger user attachment or higher relevance of the product in daily routines, while longer gaps may suggest declining interest or limited perceived value.

Insights:
If the majority of users exhibit short average gaps between sessions, it suggests consistent engagement and effective user retention strategies. On the other hand, wider distributions or long average gaps can reveal opportunities to improve re-engagement efforts, such as through reminders, new content updates, or personalized notifications. Monitoring session gap trends over time also provides an early signal of shifts in user behavior or satisfaction.

89. Cross-Device User Distribution Analysis

This analysis examines how many users interact with the platform using multiple devices compared to those who use only one. It provides insights into cross-device engagement behaviors, highlighting the extent to which users transition between devices during their interactions with the product. The visualization compares the number of single-device users with cross-device users to identify multi-platform engagement trends.

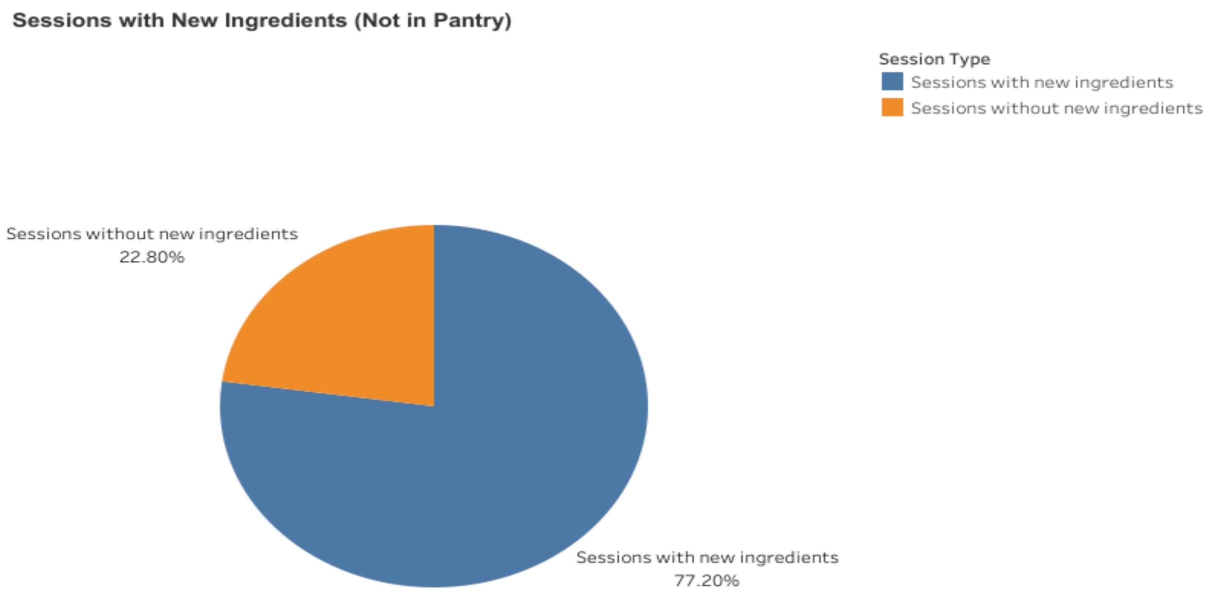


Relevance:
Understanding cross-device usage is essential for designing seamless and consistent user experiences across platforms. Users who access the platform from multiple devices typically expect synchronized progress, personalized recommendations, and uninterrupted continuity. Tracking this metric helps assess how effectively the product supports multi-device behavior and whether there's potential friction in cross-device transitions.

Insights:
A higher proportion of cross-device users indicates that the platform's ecosystem is engaging enough to encourage flexible usage across devices, suggesting strong user reliance and accessibility. Conversely, if most users are limited to a single device, it may highlight opportunities to enhance cross-platform features, data synchronization, or mobile accessibility. This metric also provides valuable input for marketing and UX optimization by identifying device preferences and patterns of multi-device engagement.

90. New Ingredient Adoption Rate

This analysis identifies how many user sessions included the addition of new ingredients—items that were not previously available in the user's pantry. By comparing each session’s cart contents with the pantry inventory, the analysis determines whether users are exploring or expanding their ingredient list. The resulting visualization compares the proportion of sessions with new ingredient additions to those using only existing pantry items.

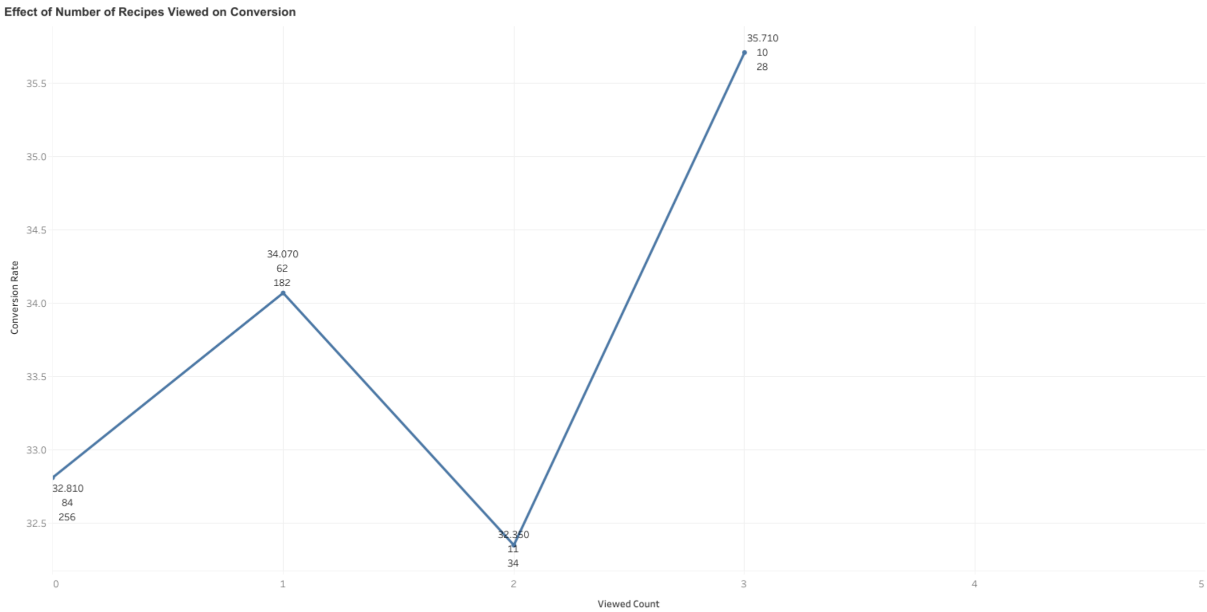


Relevance:
Tracking sessions where users add new ingredients provides valuable insight into user behavior and purchase motivations. It helps distinguish between habitual usage (replenishing existing ingredients) and exploratory behavior (trying new recipes or ingredients). This distinction is essential for understanding product discovery, personalization effectiveness, and the impact of recipe recommendations on driving variety in ingredient purchases.

Insights:
A high percentage of sessions featuring new ingredients indicates strong engagement with new content or effective recommendation algorithms encouraging variety and experimentation. Conversely, if most sessions involve only existing pantry items, it may suggest routine purchasing behavior or limited exposure to new options. Identifying this balance enables better targeting of discovery prompts, ingredient suggestions, and recipe diversity strategies.

91. Recipe View–to–Conversion Relationship

This analysis explores how the number of recipes a user views in a single session influences their likelihood to convert—defined as saving a recipe or adding one to the cart. By grouping sessions based on how many recipes were viewed, the analysis measures how engagement depth (more browsing) correlates with conversion behavior. The visualization plots conversion rate against the number of recipes viewed per session, highlighting the relationship between exploration and action.



Relevance:

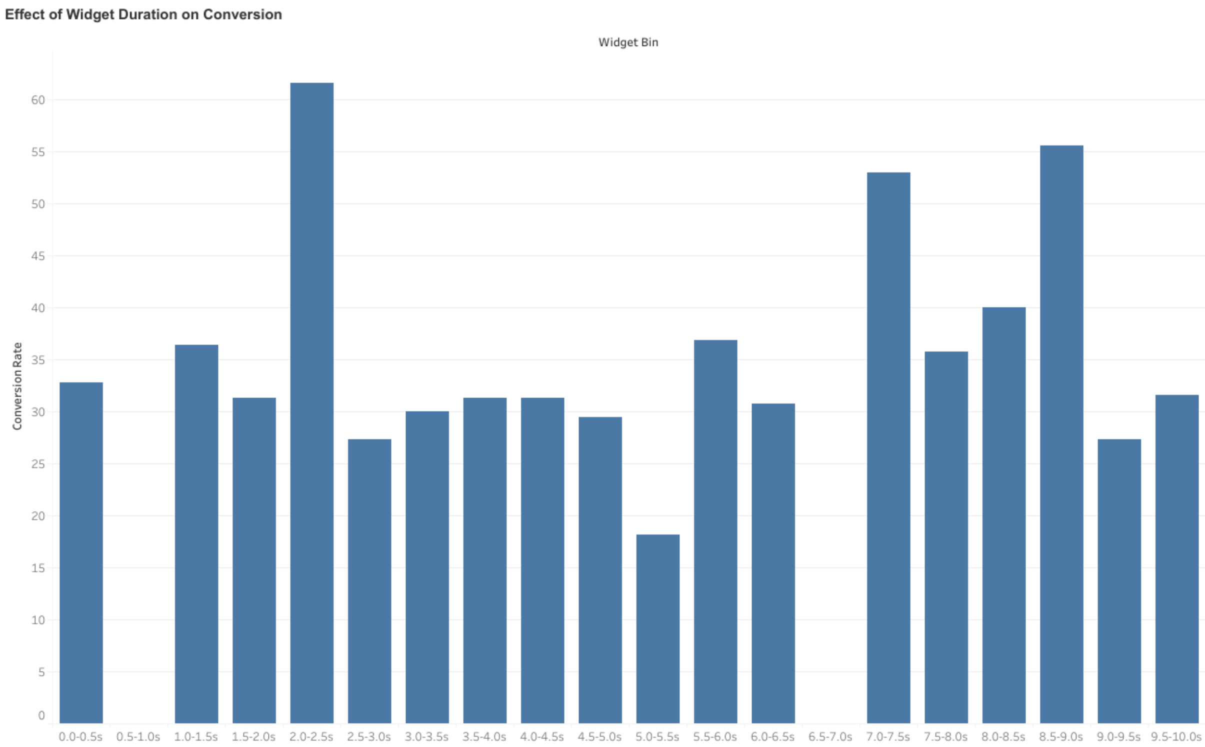
Understanding the relationship between browsing intensity and conversion is critical for evaluating content effectiveness and user journey design. If increased browsing leads to higher conversions, it suggests that discovery experiences and recommendation systems are engaging users effectively. Conversely, a drop in conversion with excessive browsing may indicate choice overload or difficulty finding appealing options.

Insights:

A rising conversion rate with more recipes viewed implies that exploration positively impacts decision-making—users discover desirable content as they browse. However, if conversion rates plateau or decline after a certain threshold, it could signal friction points such as overwhelming options or low content relevance. These insights help optimize recipe recommendation density, browsing design, and call-to-action placements to maximize engagement-to-conversion efficiency.

92. Effect of Widget Duration on Conversion

This analysis investigates how the duration of time users spend interacting with the widget influences their likelihood to convert—defined as saving a recipe or adding one to the cart. By grouping widget interaction times into 0.5-second intervals, the analysis evaluates whether longer engagement correlates with increased conversion rates. The visualization presents conversion percentage trends across these time bins, highlighting behavioral patterns tied to widget engagement depth.

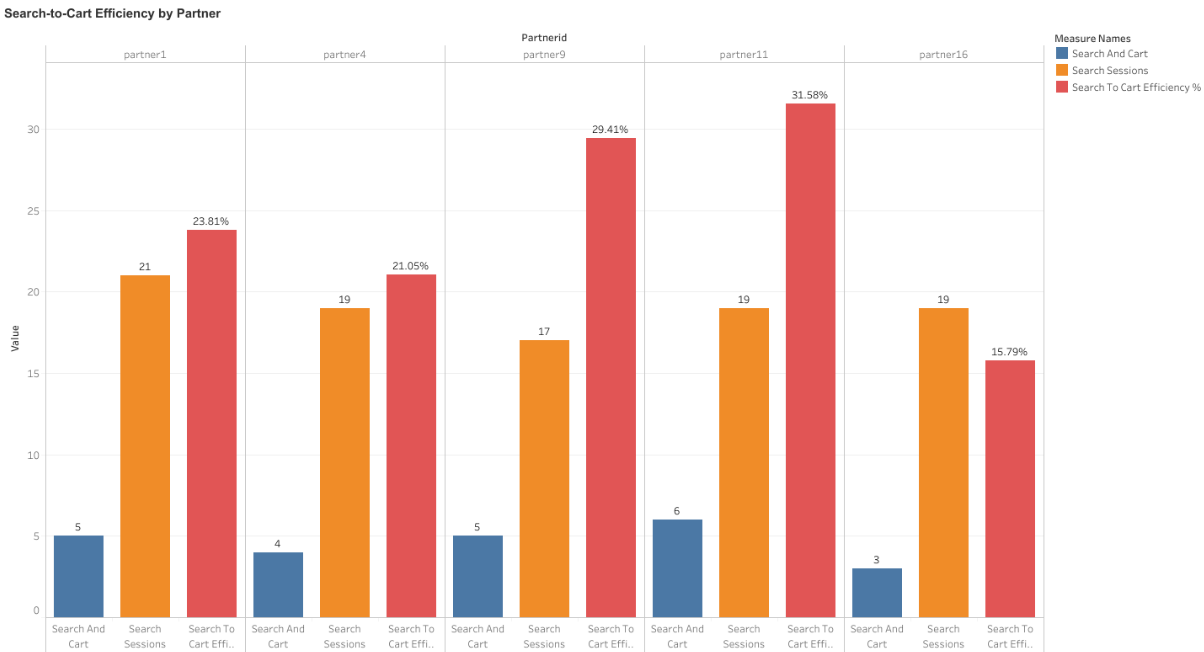


Relevance:
Widget interaction duration is a strong indicator of user engagement and intent. Understanding its relationship with conversions helps assess whether longer exposure to widget content improves decision-making or leads to drop-off due to friction. This insight can guide UX optimization, timing of prompts, and the ideal window for conversion-oriented interactions.

Insights:
A rising conversion rate with longer widget durations implies that increased engagement fosters user confidence and interest in taking action. Conversely, if conversion rates plateau or drop after a certain duration, it may indicate cognitive fatigue, indecision, or diminishing content relevance. Recognizing this trend helps balance widget design—ensuring users engage long enough to make a decision without being overwhelmed.

93. Search-to-Cart Efficiency Analysis

This analysis measures the proportion of recipe search sessions that lead to an add-to-cart action within the same session, providing insight into how effectively the search experience converts interest into purchase intent. It also includes a partner-level breakdown (where applicable) to highlight differences in search-driven conversion performance across various traffic sources or platforms.



Relevance:

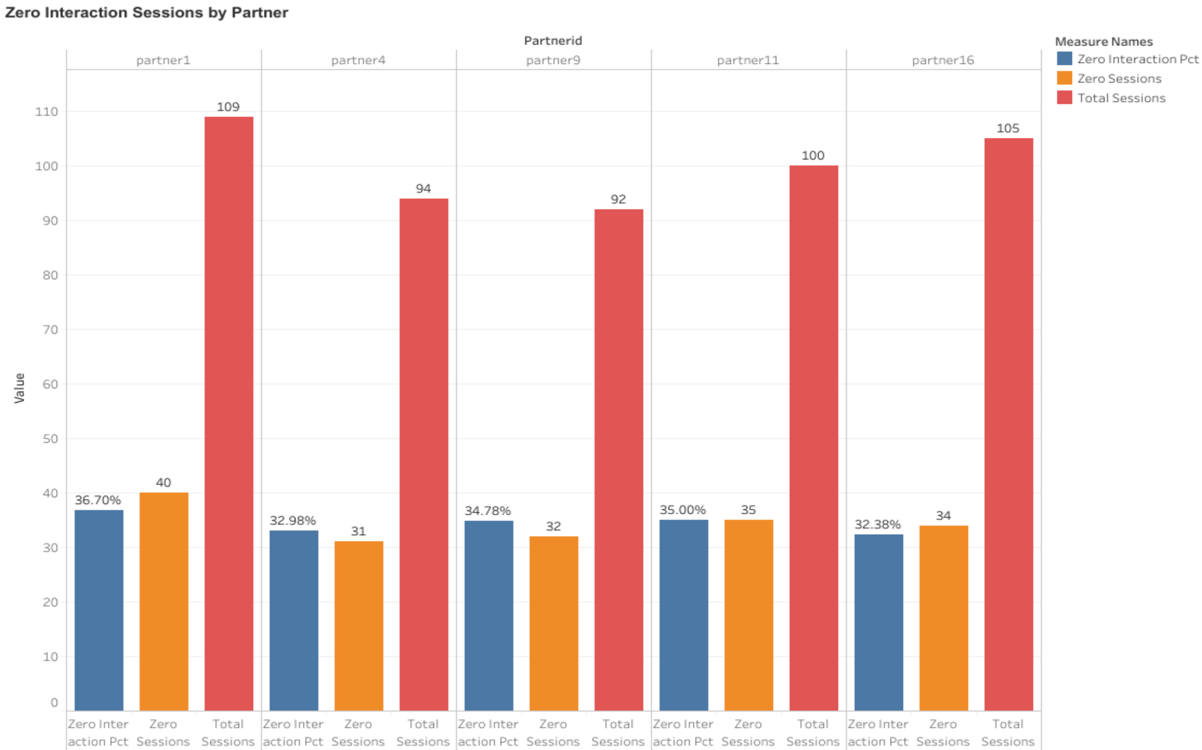
Search-to-cart conversion is a critical indicator of how well users find relevant, desirable products through the search interface. High efficiency implies that search results effectively meet user needs and guide them toward actionable steps, while low efficiency may suggest friction in search accuracy, product visibility, or user motivation. This metric helps teams assess the quality of the search feature, the alignment between recipe intent and cart actions, and opportunities to improve discoverability.

Insights:

A high Search-to-Cart Efficiency reflects an optimized search experience that successfully converts browsing into tangible engagement or purchase behavior. Lower efficiency values may point to a need for refining search relevance, improving product-tag alignment, or enhancing search-to-cart pathways. Analyzing this metric across partners can also reveal which user sources or environments foster stronger conversion behavior, informing targeted optimizations and UX improvements.

94. Zero-Interaction Session Analysis By Partner

This analysis identifies sessions where users exhibited no engagement—meaning they did not add items to the cart, save recipes, add to pantry, or view any recipes. It quantifies and visualizes the share of such sessions both overall and by partner, helping to assess user inactivity and engagement drop-offs. The visualization displays the percentage of zero-interaction sessions per partner to pinpoint where engagement challenges are most prominent.



Relevance:

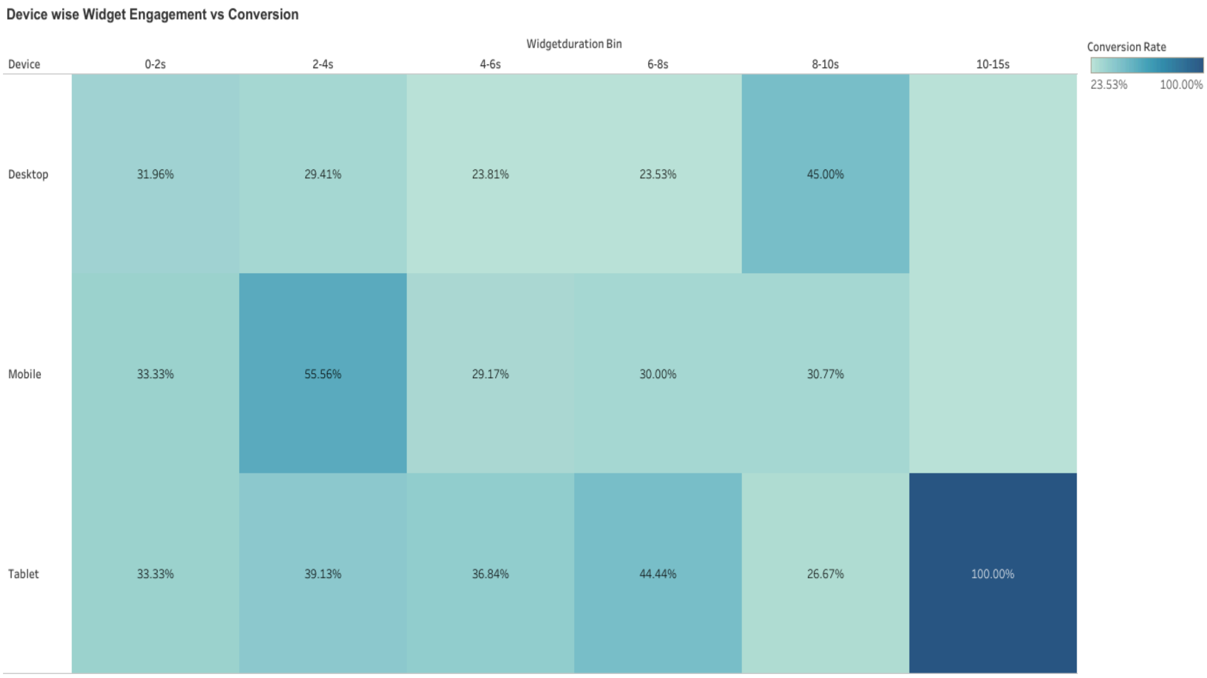
Measuring zero-interaction sessions is crucial for understanding engagement health. A high rate of such sessions indicates users entering the platform but leaving without meaningful activity—signaling potential issues with session initialization, interface usability, or content relevance. Tracking this metric at the partner level helps identify sources or channels that drive low-quality traffic or mismatched user intent.

Insights:

A large proportion of zero-interaction sessions suggests barriers to engagement, such as poor visibility of interactive features, irrelevant landing experiences, or unoptimized user journeys. Conversely, a low proportion reflects successful onboarding and content engagement strategies. By isolating high zero-interaction partners, teams can prioritize improving onboarding flows, content placement, or personalization strategies to convert passive visits into meaningful actions.

95. Device-Based Widget Engagement vs Conversion Rate

This analysis examines how user engagement duration within the widget (measured in seconds) impacts conversion actions—such as adding items to the cart or saving recipes—across different devices. Sessions are grouped into widget duration bins and segmented by device to highlight engagement-to-conversion relationships visually using a heatmap.

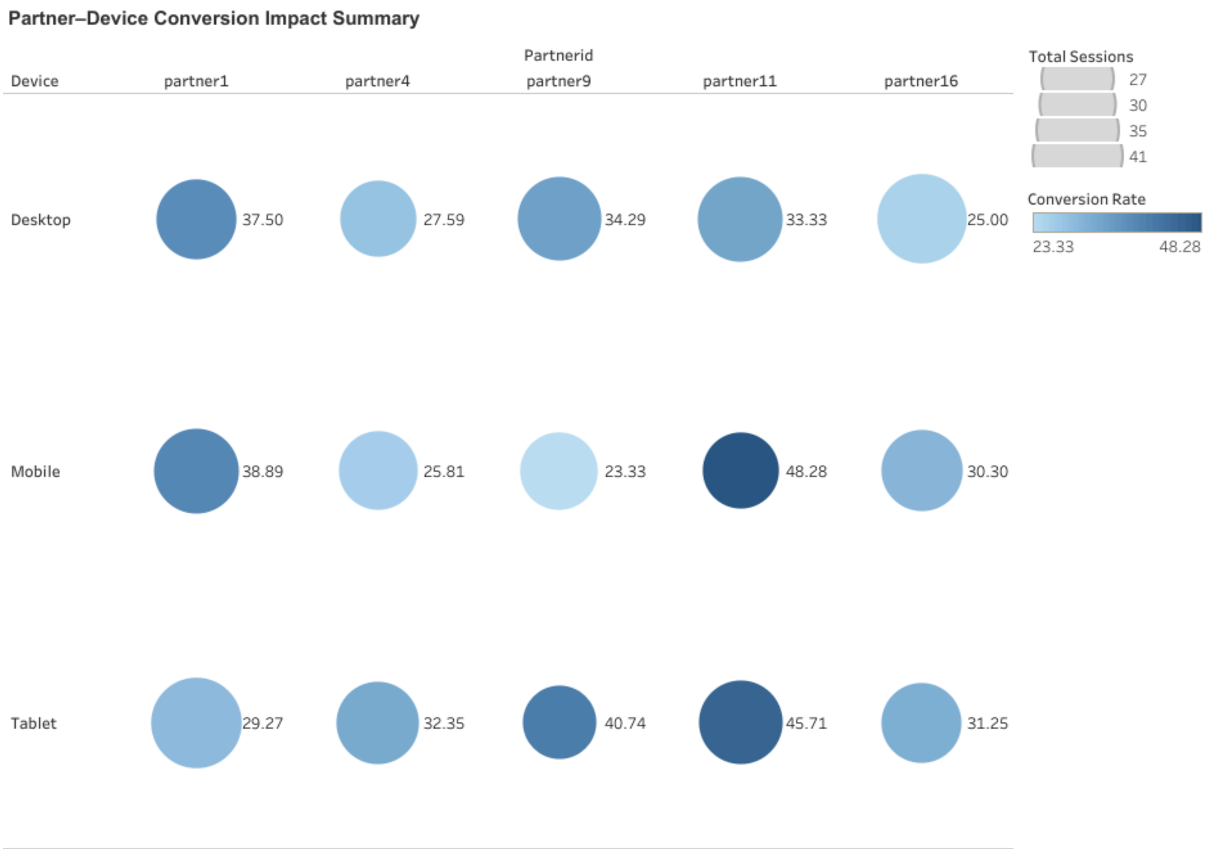


Relevance:
Evaluating widget engagement by device type reveals how effectively the widget drives conversions across user environments. It helps identify whether mobile, desktop, or tablet users require design adjustments or engagement enhancements to improve interaction outcomes.

Insights:
The heatmap visualization uncovers device-specific behavioral trends—such as which devices see higher conversion rates at shorter or longer engagement durations. A strong upward trend across bins indicates that increased engagement correlates with stronger purchase or save intent, while flatter patterns suggest that conversion behavior is less dependent on time spent.

96. Partner–Device Conversion Impact

This metric analyzes conversion rates across combinations of partner platforms and device types. It evaluates how effectively each partner's traffic performs in driving key actions—such as adding items to cart or saving recipes—when accessed via different devices.

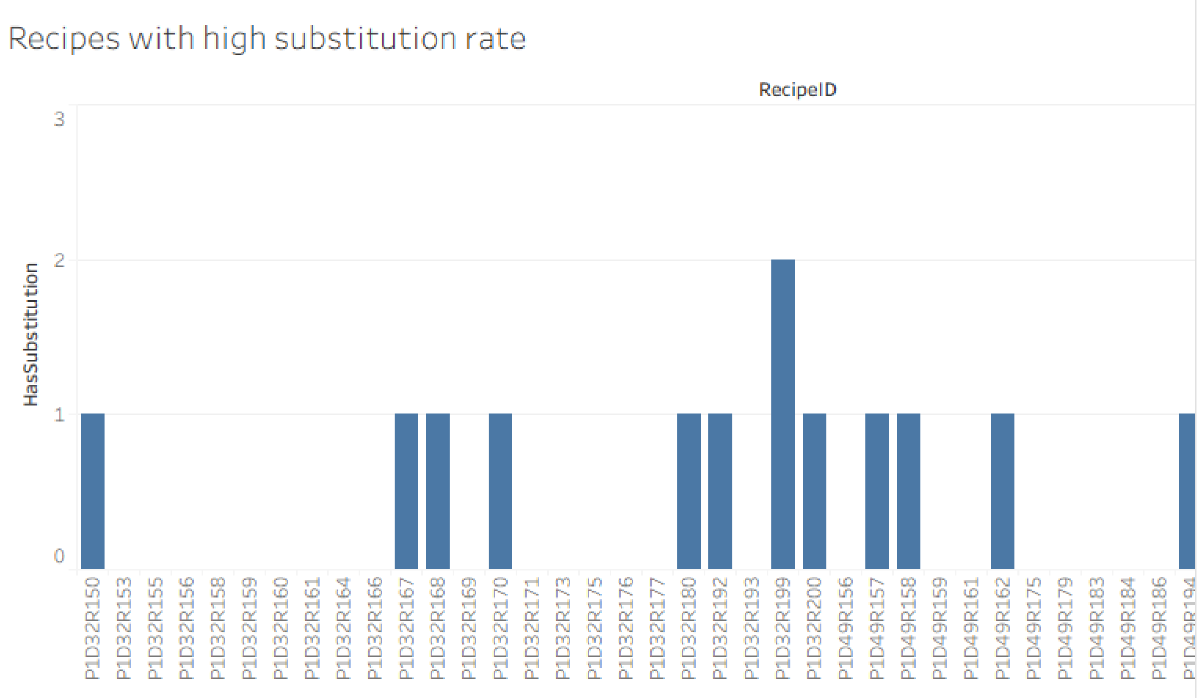


Relevance:
Understanding conversion performance at the intersection of partner and device helps identify optimization opportunities. Certain partners may perform better on specific devices due to interface compatibility, audience behavior, or device-targeted promotions. This insight supports strategic alignment of campaigns, UI adjustments, or resource allocation across device ecosystems.

Insights:
The analysis highlights variations in conversion rates across partner-device pairs. High-performing pairs suggest well-optimized user experiences or relevant audience targeting, while lower-performing combinations may indicate friction in cross-device usability or mismatched content engagement. Partners showing large session volumes but modest conversion efficiency may benefit from device-specific UX refinements.

97. Recipes with high substitution rate

This metric quantifies the share of recipes in which one or more ingredients were substituted by users. It reflects the flexibility or adaptability of recipes regarding user behavior during cooking or meal planning.



Each bar in the figure represents a recipe, showing whether it is used as it is or customized by users.

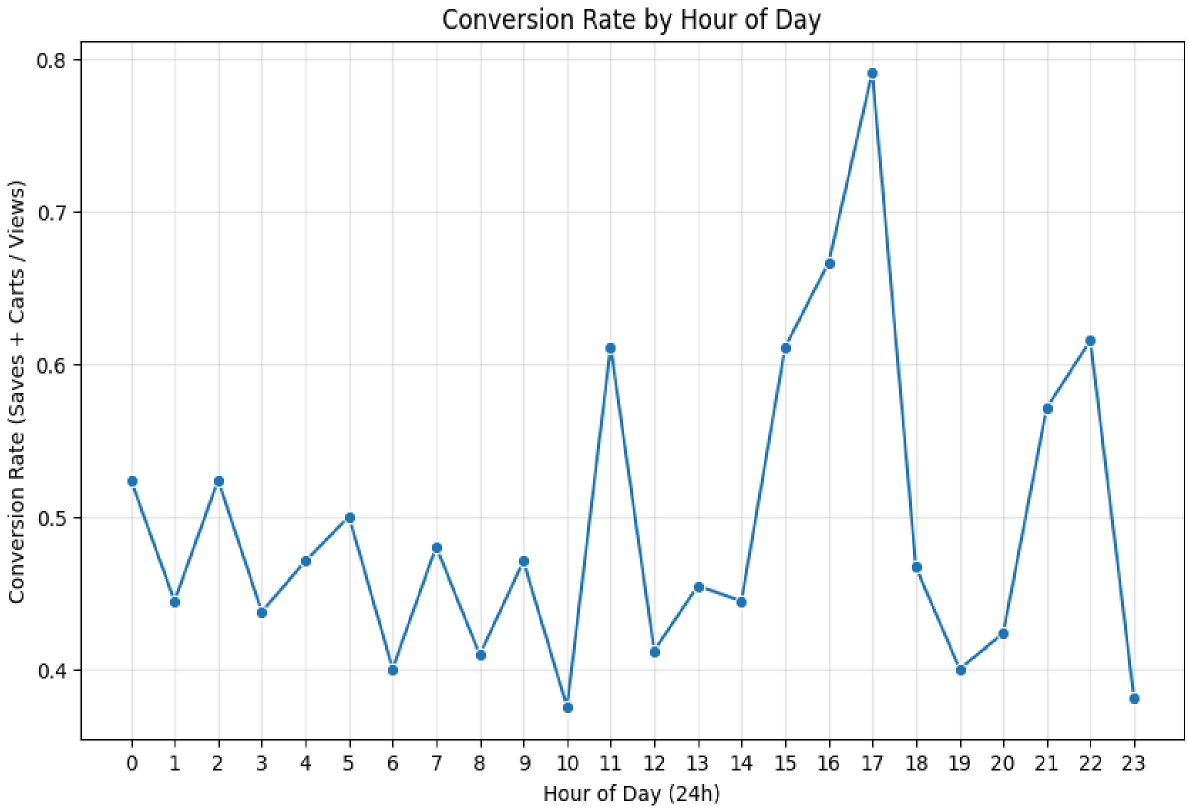
Relevance:
Ingredient substitutions indicate user personalization behavior and show which recipes are flexible and which ingredients users are more likely to change. This can point out:

- Ingredient-related issues (availability, hard-to-find ingredient or disliked ingredient)
- Recipe adaptability and user creativity
- Potential improvements for future recipe recommendations or substitutions suggestions

Insights:
A considerable number of recipes reveal a non-zero substitution rate, which indeed means that many users modify recipes rather than follow them exactly.For certain recipes, the high frequency of substitution might imply that a given ingredient is either too rare, costly, or unpopular. Recipes with zero substitutions might indicate high satisfaction with ingredient combinations; these may be promoted as stable and well-accepted recipes. This can provide a guide for the recommendation engine to suggest dynamic alternate ingredients that raise user engagement by allowing personalization.

98. Conversion Rate by Time of Day

This metric looks at how users perform key actions such as recipe add to cart or save across all hours of the day. It does so by aggregating user interactions at an hourly level and calculating the ratio of successful conversions to total views.

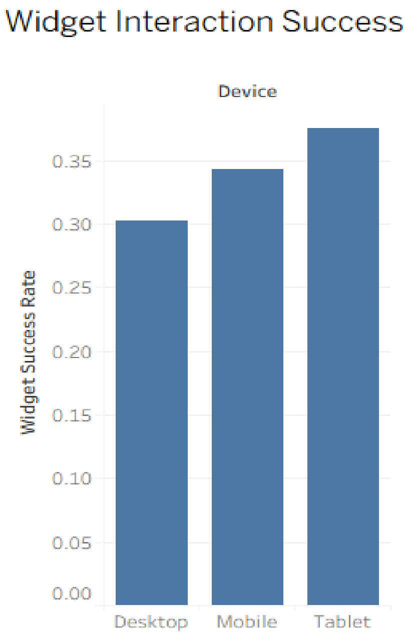


Relevance:
Understanding when users are most likely to convert offers important insights into engagement patterns. Identifying peak conversion hours enables targeted content delivery, notification scheduling, and promotional timing. This helps improve user responsiveness and platform efficiency.

Insights:
There are clear differences in conversion performance throughout the day, based on the analysis. Midday conversion rates are lower due to work-related distractions. Higher rates in the morning or evening may suggest meal planning or casual browsing. By spotting these variations, we can improve push notification timing, the visibility of call-to-action buttons, and dynamic recipe highlighting during important moments. Campaigns or partners that consistently perform well at different times could serve as models for enhancing engagement tactics across hours.

99. Widget Interaction Success Rate

This metric quantifies the share of widget sessions where at least one key user action occurred, such as saving a recipe or adding to the cart. It's about how well interactive widgets can convert casual interactions into meaningful engagement events, so this metric can be used as a direct indicator of UI effectiveness and session quality.



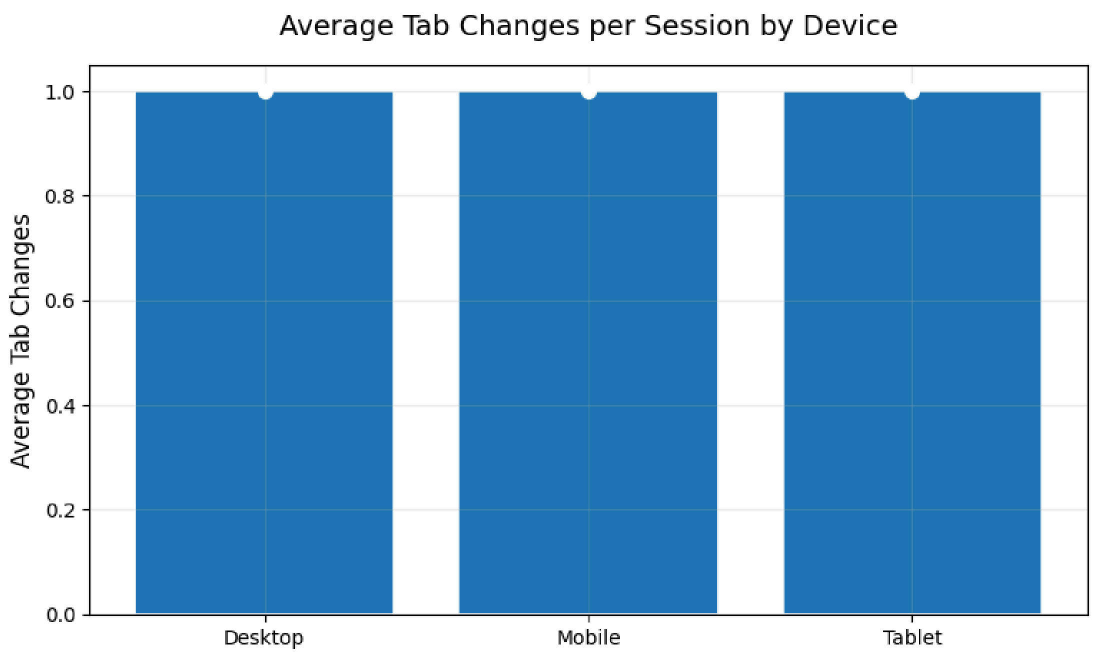
- The **blue section** (≈ 80–85%) is the portion of sessions whereby the widget did not lead to a save or cart action, meaning users were interacting but didn't convert.
- The **orange slice** (≈ 15–20%) represents successful widget interactions — sessions where the user saved a recipe or added something to the cart.

Relevance:
The widget success measure grades how effectively interactive components convert end-users to actions, such as saves, or putting items in the cart. Higher rates of success imply that placement and presented value were appropriate and lower rates of success could simply be due to timing, contextual relevance, or placement. Use this measure to encourage the future focus on developing and placing buttons or other widgets.

Insights:
The pie chart provides a glimpse of the success of the conversion. A larger portion of the success categorization can signal engagement or interest by a consumer, and similarly, an equal amount or larger of the pie chart may suggest that there may be friction, or that the prompts were not clear. Improvement in terms of both context for the widget and specifying relevance for micro-conversions across sessions can hopefully lead to improvements for clothing widget conversions across other sessions.

100. Average Number of Tab Changes per Session (by Device)

This metric calculates the average number of tab changes per session to understand user multitasking behavior, as segmented by device type. This is computed by counting tab change events recorded in the activity logs and then grouping the results by the user's device category, such as mobile, desktop, and tablet.



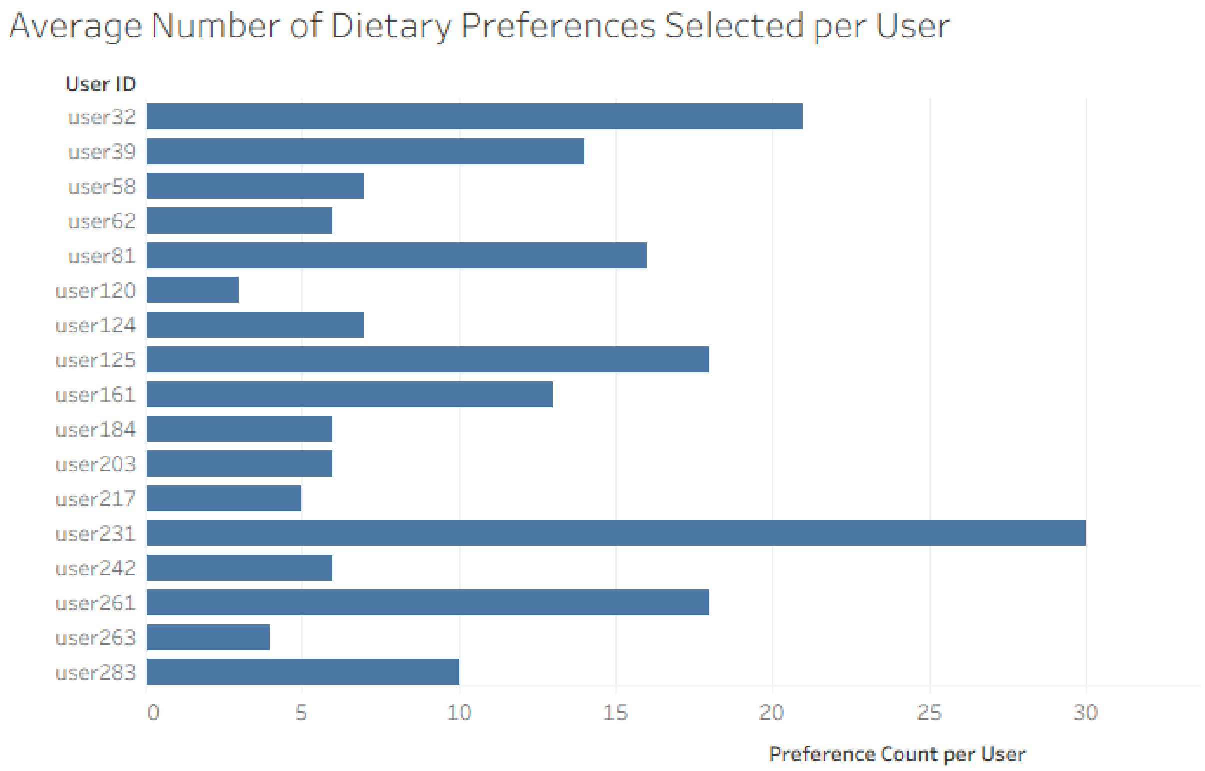
Relevance:
Tab-switching behavior shows insights into users' engagement and the depth of interaction across different devices. Frequent changes of tabs may reflect exploratory behavior, multitasking tendencies, or usability challenges that require multiple context switches. The differences on a device level can point to dissimilarities in user experiences between mobile and desktop.

Insights:
Analysis shows that desktops usually have higher tab-change frequencies, reflecting larger screen real estate and multitasking habits. Mobile sessions tend to result in fewer tab changes, often indicative of either focused engagement or limited navigation flexibility.

Excessive tab-switching on some devices may suggest poor workflow design or an unclear navigation path and an opportunity to streamline the user's journey and improve continuity within the session.

101. Average Number of Dietary Preferences Selected per User

This metric calculates the average number of dietary preferences that each user has chosen. The count of the number of dietary preference tags chosen by each user is taken, for example, vegan, gluten-free, low-carb, and then the mean across all users is determined.



Relevance:

Observing dietary preference choices helps understand your audience diversity and opportunities for personalization, and how well the platform is able to support diverse dietary needs. A higher average suggests that users actively engage in personalization, while a lower average may indicate limited personalization or little utilization of the preference settings.

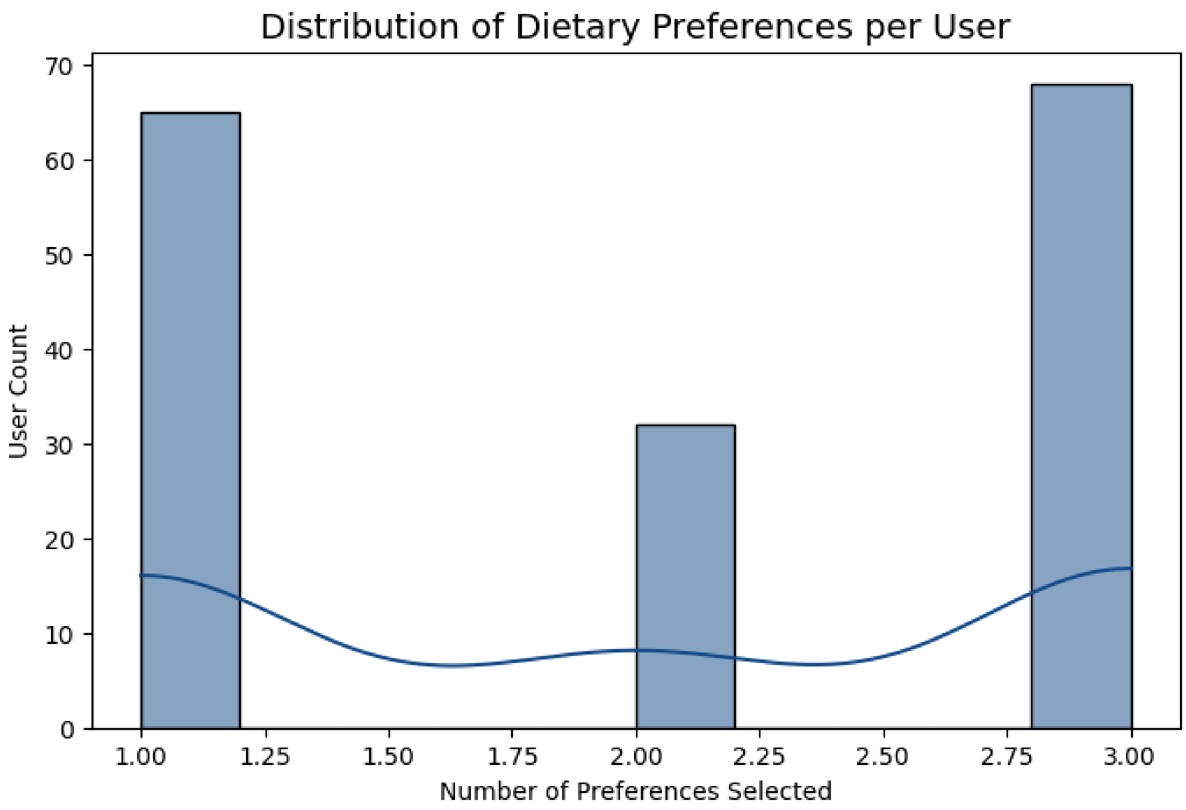
Insights:

The analysis below indicates how users engage with dietary customization features. Users who select multiple preferences may be health-conscious or niche audiences looking for precise filtering, while minimal selections may point to either simplicity in the approach toward their dietary needs or limited awareness of the feature.

This knowledge can help drive UI changes, such as better menu item visibility, simplification of selection interfaces, or the promotion of personalized recommendations to increase user participation and satisfaction.

102. Users Trying the Same Recipe Multiple Times

This measurement pinpoints users who re-engage with the same recipe through viewing, saving, or adding to their cart repeatedly. In order to measure this activity we track the number of unique user IDs associated with the same recipe ID across multiple user sessions or interactions.

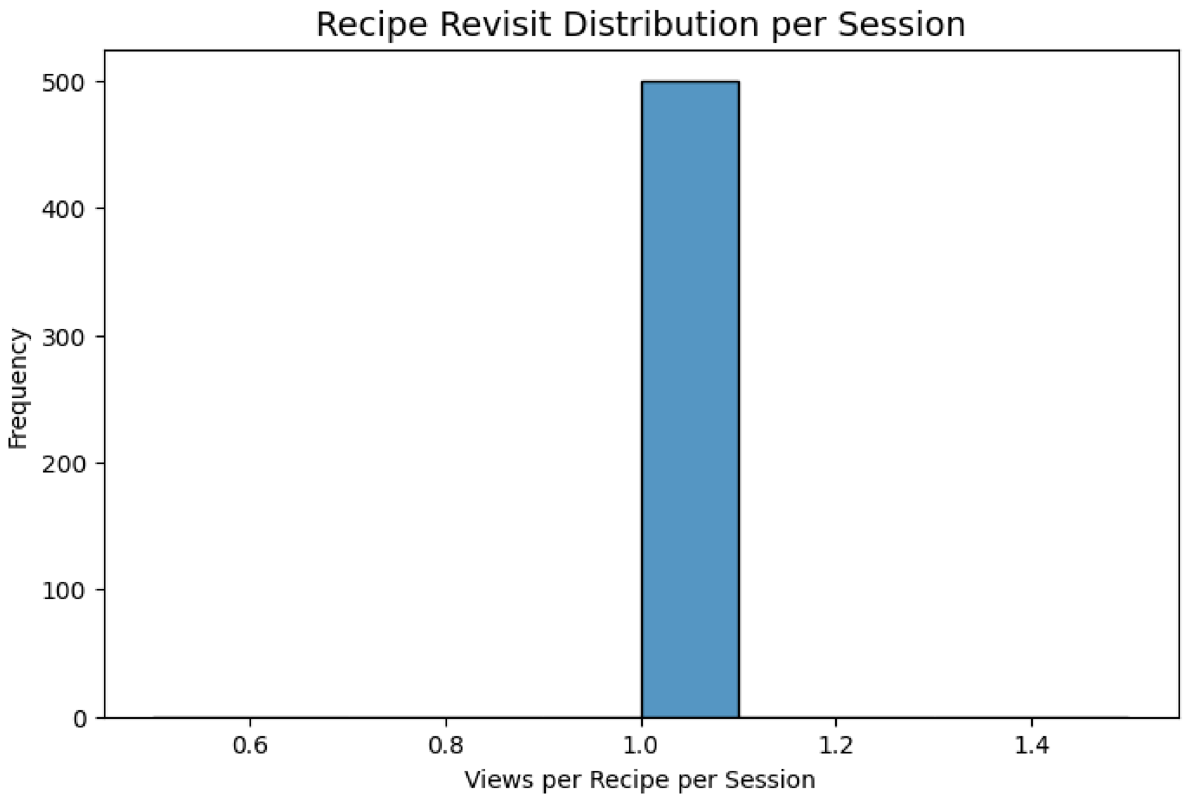


Relevance:
Monitoring between-repetition describes user "loyalty," content "quality," and recipe "reusability." Low levels of repetition can also indicate poor recipe recall, or that users are not incentivized to engage with the recipe again after a previous session. Frequent user engagement indicates high user satisfaction, a much stronger recipe habit formation, or practical popularity for shared recipes.

Insights:
This analysis showcases which recipes excel in generating consistently "return engagement." If a user is repeating those recipes frequently, it could be worth enhancing the recommendations or promotion of "go-to" recipes they like. On the contrary, low repetition could indicate a content gap, meaning a recipe was of interest to begin with, but after interaction, no additional engagement/reading was warranted. Being able to diagnose these two examples can enable stronger personalization, curation of content, and re-engagement based engagement strategies.

103. Recipes Revisited per Session

This metric quantifies how often users return to the same recipe within a single browsing session. It reflects the depth of engagement and possible hesitation on the part of the user before taking an action, such as saving or adding to cart. The calculation divides the number of duplicate recipe views by the total number of recipe views.



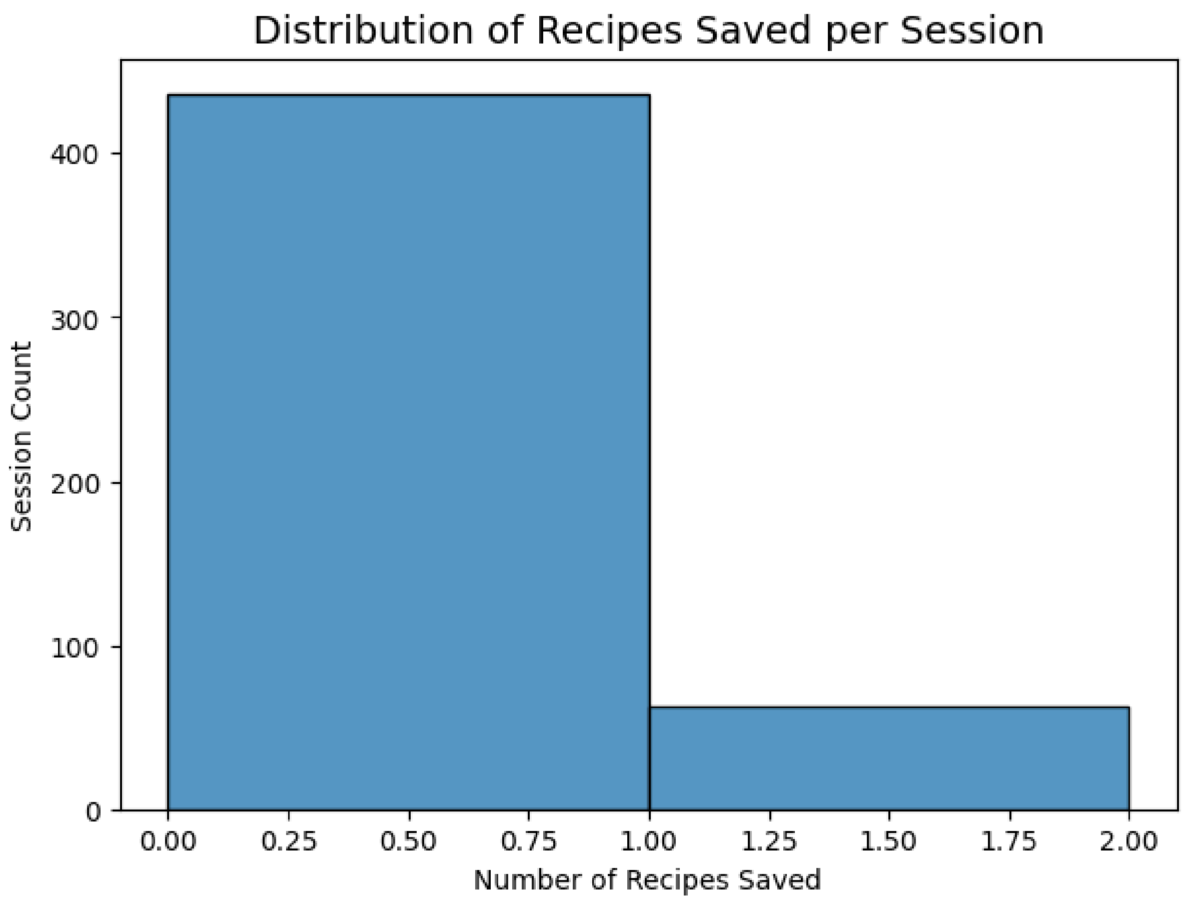
Relevance:
Re-visiting a recipe may signify genuine interest, re-evaluation, or indecision. Monitoring this behavior points out which recipes spark deeper user engagement or require additional clarity—perhaps due to missing information, complex ingredients, or layout issues.

Insights:
In the current data, revisit behavior might be minimal because of limited unique sessions and recipes per user. However, as the dataset grows, this metric can indicate engagement depth and serve as a proxy for a variety of behavioral signals including user satisfaction, decisiveness, and clarity around recipes.

Later, this metric could be combined with a conversion rate or save action to find which recipes convert after multiple revisits — highlighting high-performing yet complex recipes.

104. Chat Sessions with Multiple Saves

This metric tracks whether users save multiple recipes in a chat session. This tells us somewhat about how effective conversational or interactive chat experiences motivate users to deepen their engagement with the platform's recipe content. The metric calculates chat sessions where the number of saved recipes exceeds one.

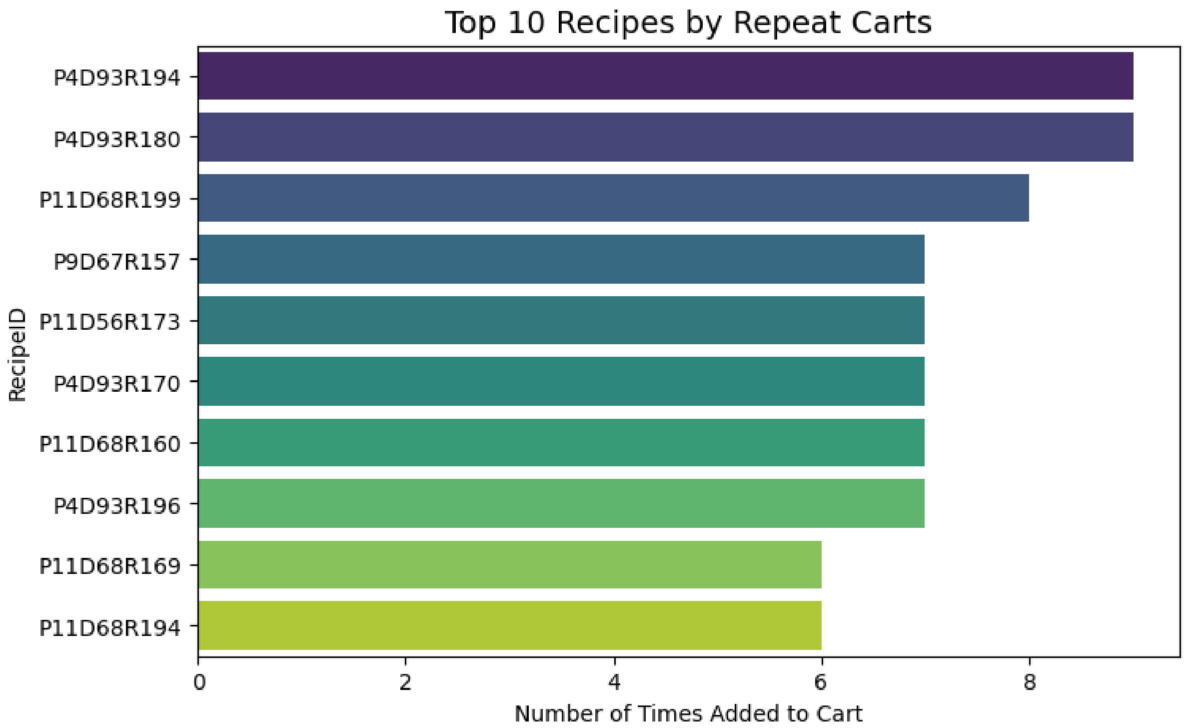


Relevance:
By tracking the number of multiple-save chat sessions, we can assess whether chat-based discovery promotes user intent and conversion or modifies user behavior. High rates of multiple saving could indicate that chat recommendations or in-session browsing paths (recommendation flows) effectively expose users to new content that they find enticing.

Insights:
The analysis indicates that a low number of sessions involve more than one saved recipe which suggests multi-saving engagement in the chat flows is somewhat limited at this time. This may indicate that there is a great opportunity in either personalization or content-suggestions through chat. Encouraging recommendations on follow-ups or messaging like "you might also like" or other "similar" recipes during their chat could strengthen their multi-save behavior overall in a favorable way.

105. Top Recipes by Repeat Carts

This metric enables analysis of recipes added to the cart multiple times (by a single user or by different users) for a variety of reasons. The repeat carting behavior reflects recipes that are desired consistently well or produce ongoing purchase intent.

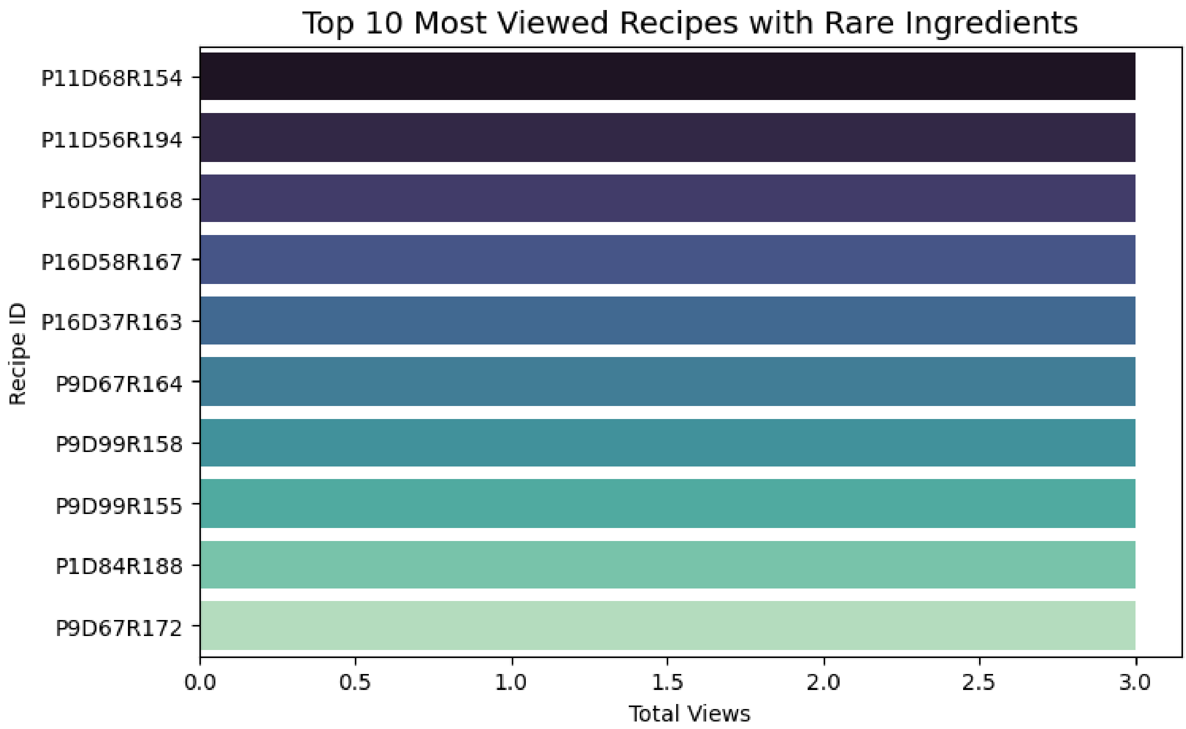


Relevance:
Knowing what recipes are being carted repeatedly adds context to understanding user's preferences and what will result in stickiness for products. Recipes that have high counts of being added to repeat carts may demonstrate dependability as a favorite, compelling ingredients, or presentation of the content. These insights will guide the ability to optimize business or UX through placement of feature jars, prioritizing stocking inventory, or developing promotions on competing recipes.

Insights:
The insights reflect that there are only a few recipes that have repeat cart activity, which means more to limited interest (compared to broader interest). This may indicate that there are likely to be few 'hero' recipes that will drive most of user's activity. Focused efforts on promoting these recipes likely makes sense, when the recipes have a higher rate of repeat carts. Recipes that have little or no repeat carts may benefit from improved food presentation, recommendationsother than the jump to add to the cart or utilizing substitutions to appeal.

106. Most Viewed Recipes with Rare Ingredients

This metric highlights the rare ingredient recipes-bottom 10% by ingredient frequency score across all recipes-and how many times they were viewed. It may show if users are interested in rare or hard-to-find ingredients, even though they are rare recipes to begin with.



Relevance:

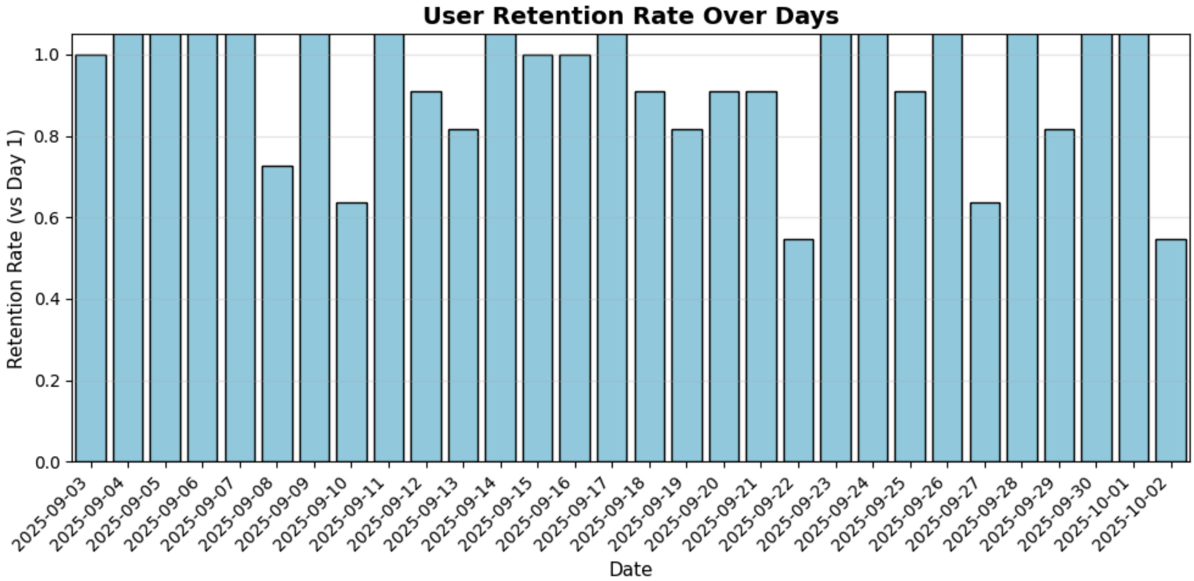
It provides a high-level understanding of how ingredient availability and user views may interact, identifying possible areas where user curiosity or uniqueness of the ingredient is more appealing and encourages engagement or exploration.

Insights:

It can be noticed that there are some differences in view count of the rare ingredient recipes. Some rare ingredients did fairly well, while the other overall patterns were mixed and inconclusive. It may suggest that user engagement with the rare-ingredient recipe is influenced by several contextual factors tied to recipe type, timing, or presentation.

107. User Retention Rate (N-Day Period)

It is a metric that shows the number of users who come back to the platform over a certain time period. It gives a measure of user stickiness and consistency in engagement over time, by comparing the number of distinct active users on subsequent days to that in the first day.



Relevance:

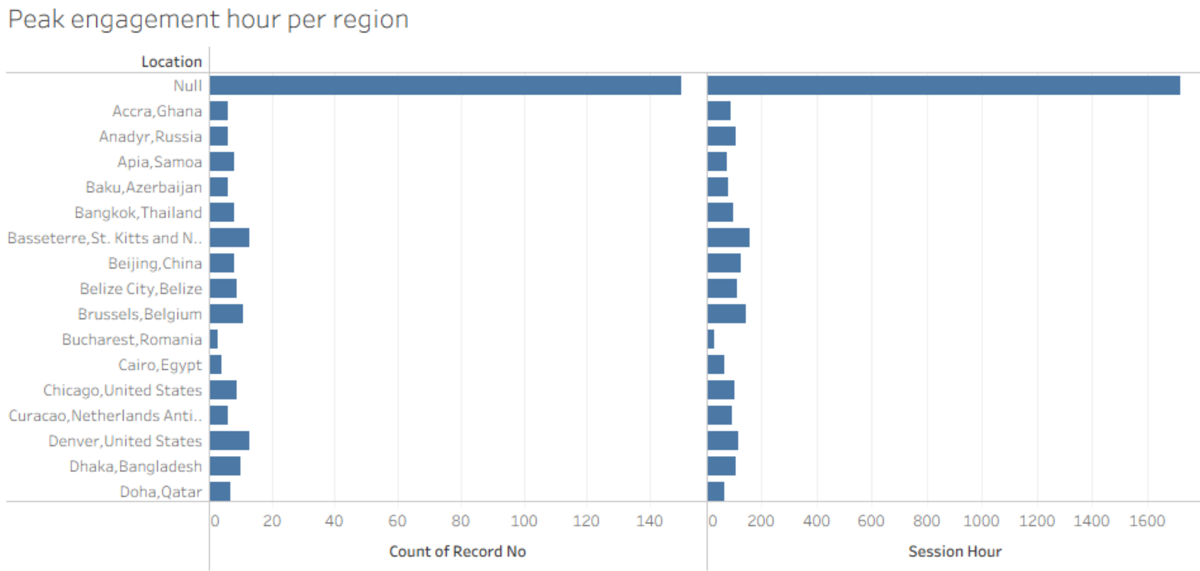
User retention is one of the most crucial metrics for understanding platform health and user satisfaction. A high retention rate indicates that users find ongoing value and reasons to return, while a steep decline suggests that users may lose interest after their first interaction. For apps relying on recurring engagement—such as recipe or food discovery tools—retention directly impacts long-term growth and monetization.

Insights:

The plotted trend is a moderately decaying retention over time. While there is day-one consistency in activity, returning users linearly decrease after the initial session. That means first-time active users do explore, but a fraction of them never revisit, possibly due to limited novelty or too few re-engagement triggers, such as notifications or personalization. Long-term retention could be improved by enhancing the reminder features or recommending recipes they saved.

108. Peak Engagement Hour per Region

This metric pinpoints the hour of the day when users are most active in each region. Sessions are grouped by region and hour using the SessionStart timestamp and Location fields. The hour of maximum session count is marked as the peak engagement hour. This helps expose temporal patterns in the activity of users across different geographies.



Relevance:

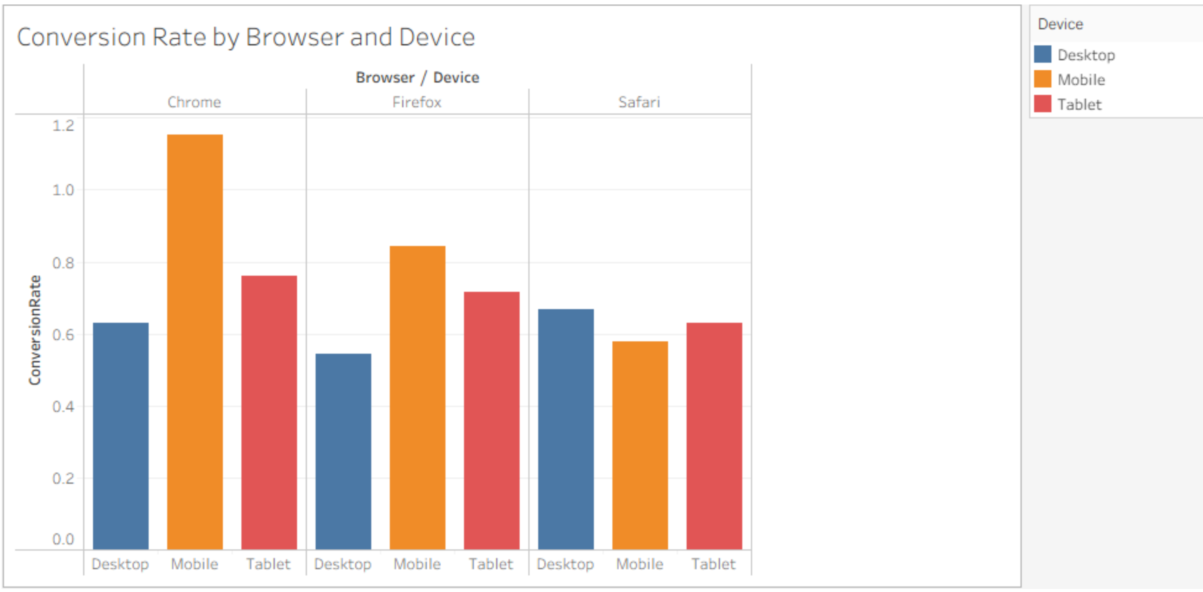
Knowing when users from different regions are most active helps optimize the timing of content, push notifications, and feature releases. Sites with globally distributed users can often exhibit unique usage cycles dependent on different time zones. Precisely identifying local peaks guarantees that suggestions, banners, or even widget refreshes reach users when they are most open to it.

Insights:

Analysis of the dataset showed distinct regional variations in engagement behavior. While some regions had strong peaks of activity in the evening, probably because users browse recipes after work, others had late-morning or midday spikes, maybe indicating that users were exploring meal ideas during planning or over lunch. These findings represent an opportunity to optimize content delivery windows based on regional behavior patterns, improving visibility and conversion potential.

109. Conversion Rate by Browser and Device

This is a metric that helps in evaluating the conversion efficiency on different browsers and device types; hence, it is computed as the share of user interactions that led to a successful action, such as saving or adding the recipe to the cart, out of the total number of recipe views.



Relevance:

Understanding the conversion rate with browsers and devices helps to show where the platform is performing best and where the user experience may lag.

For instance:

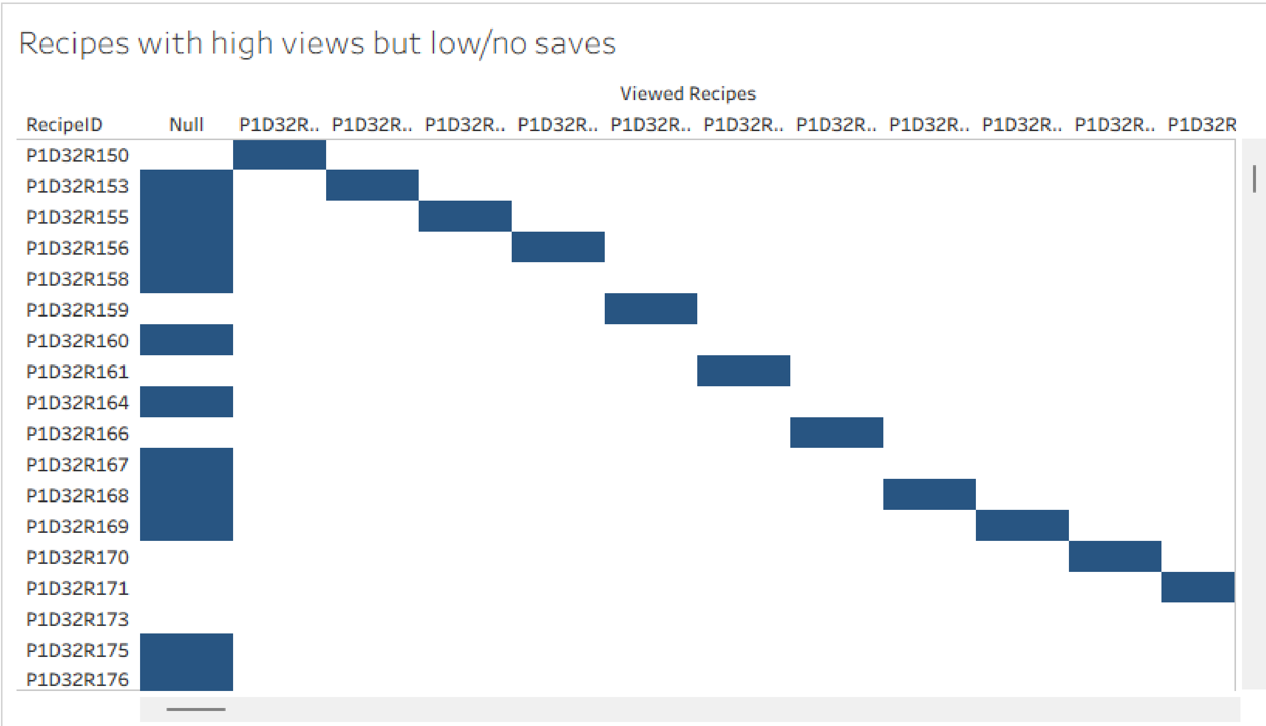
- Lower conversion rates on specific browsers may indicate **UI compatibility or performance issues**.
- Device-based differences highlight **mobile vs desktop usability** or **interaction friction** patterns. This insight is valuable for product optimization and prioritizing platform improvements.

Insights:

While many recipes elicit strong interest, the majority produce low saves, indicating that people explore them but find them not worth saving. Focused improvements in these recipes could improve aggregate conversion and retention.

110. Recipes with High Views but Low/No Saves

This metric would identify recipes with a high number of views with low number (or no number) of saves. This metric essentially measures the gap between user interest and user commitment, which is the consequence of viewing a recipe and not saving it.

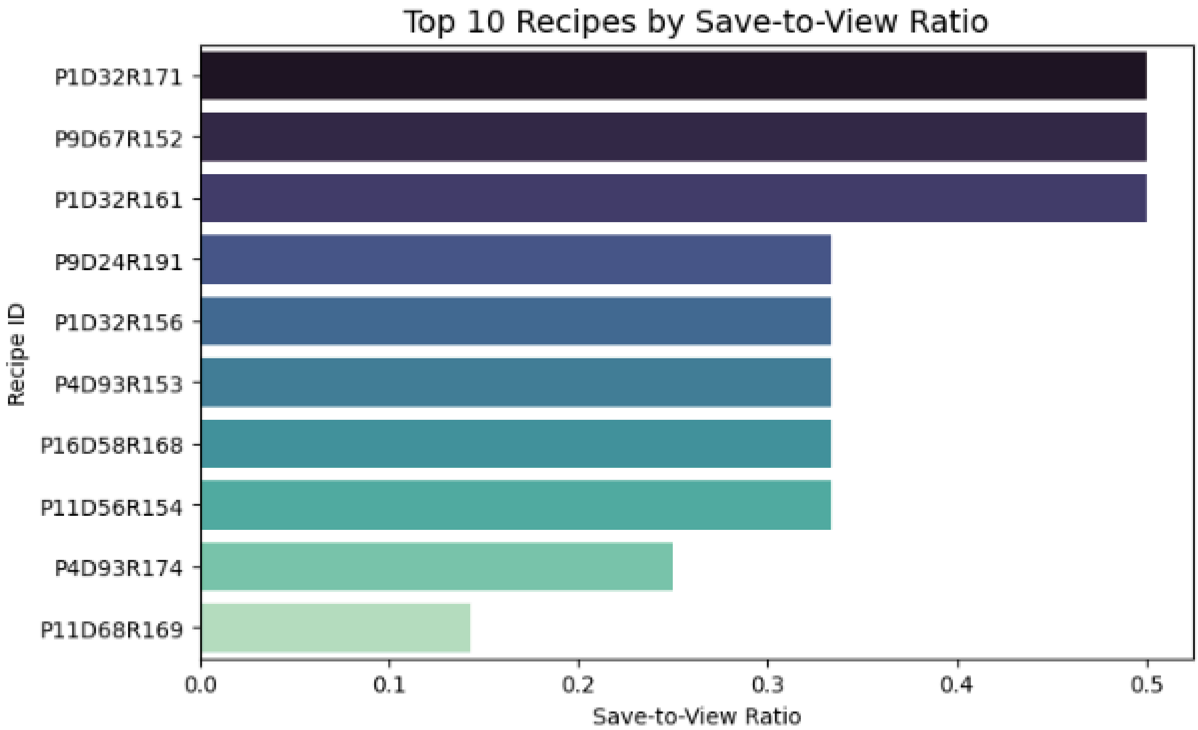


Relevance:
Generally, a view–no save pattern indicates a potential issue with the attractiveness of the recipe, its workability, or its presentation. However, understanding the issues means the recip can be remodeled to provide a better process, title, description and list of ingredients, so that a broader range of consumers would prefer to engage.

Insights:
While a handful of recipes attract a good deal of interest, most recipes create a low save rate as they are viewed but deemed unsave-worthy. Focusing your improvement efforts on these recipes could yield significant improvement in aggregate conversion and retention.

111. Save-to-View Ratio

This metric measures how often users save recipes they view, showing the shift from casual interest to intent. It is calculated as the ratio of saves to total recipe interactions. Although the dataset does not provide numeric view counts, the number of occurrences for each recipe serves as a proxy for total views. This method allows for effective assessment of user engagement and saving behavior.



Relevance:

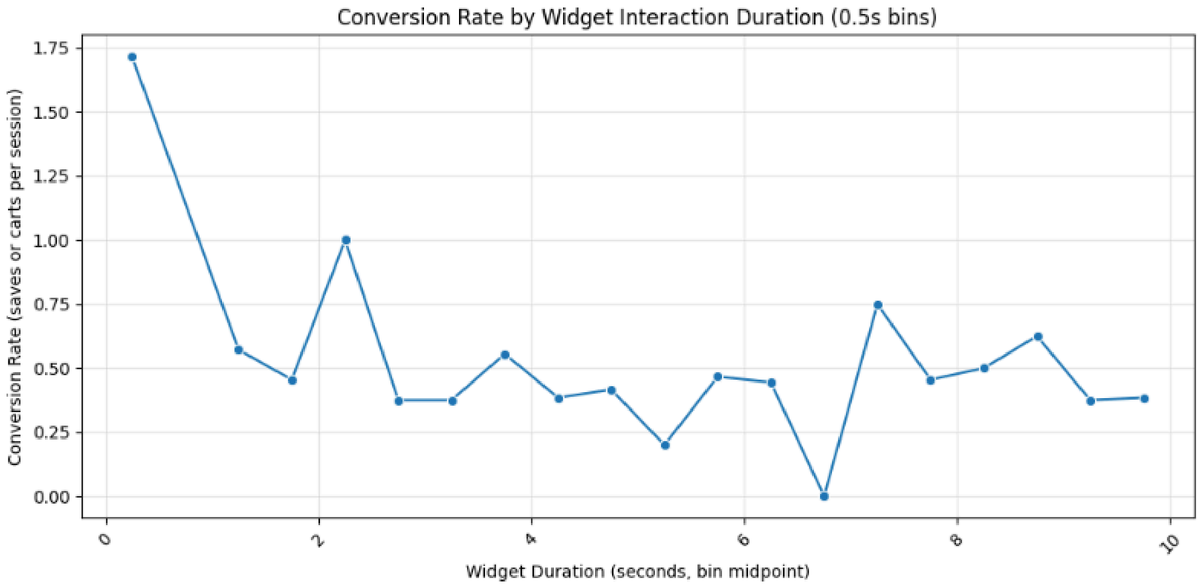
Understanding the save-to-view ratio helps determine which recipes are most appealing to users and drive active engagement. A higher ratio suggests strong relevance, good content quality, or alignment with user preferences. Conversely, a low ratio can reveal recipes that attract clicks but fail to convert into meaningful actions.Tracking this metric can guide optimization of recipe recommendations, visuals, or ingredient listings.

Insights:

Preliminary analysis shows that some recipes receive frequent interactions but relatively fewer saves, indicating browsing without commitment — possibly due to complex ingredients or mismatched dietary preferences. Others maintain a high save rate despite lower views, suggesting niche popularity. Although this metric uses a simplified proxy for views, it still provides a strong baseline for identifying engagement patterns and opportunities to improve recipe conversion performance.

112. Chat Interaction Duration on Conversion

This metric evaluates whether spending more time interacting with the widget increases the likelihood of saving a recipe or adding it to cart. Widget interaction time is grouped into 0.5-second bins, and conversion rate (saves or cart actions) is calculated per bin. This helps observe behavioral trends in short vs. extended interactions.



Relevance:

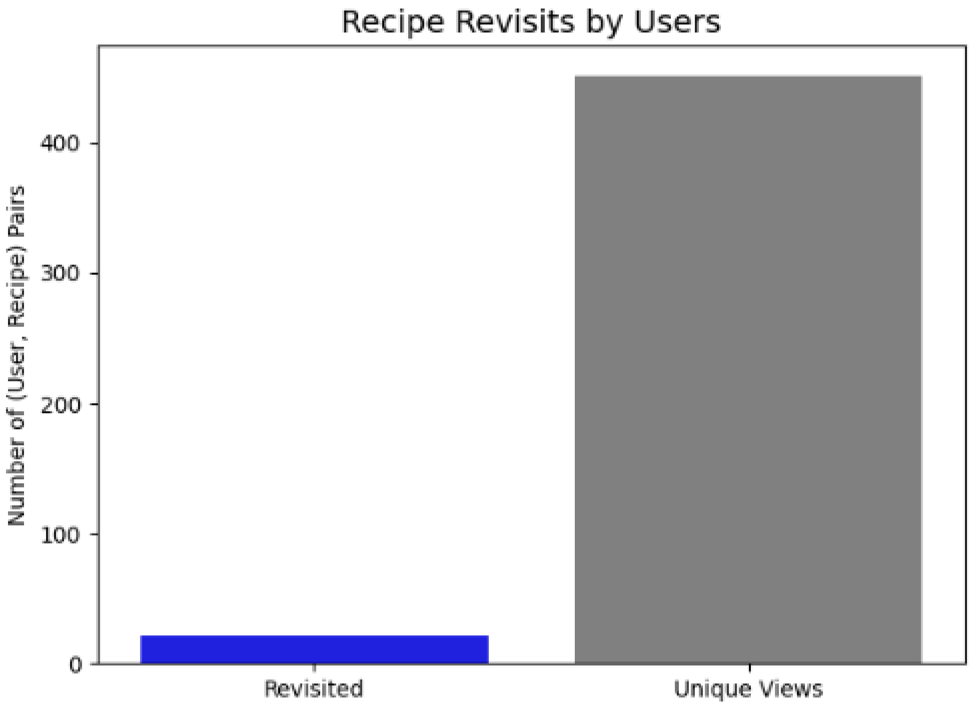
Understanding the impact of engagement duration directly supports optimizing UI flow and widget design. If small increases in interaction time significantly improve conversions, future improvements can focus on encouraging deeper exploration and guidance during the critical time window.

Insights:

Due to minimal actual save/cart events in the synthetic dataset, current results may not fully reflect real user behavior. However, this framework scales perfectly when richer session data is available — enabling discovery of the exact duration range that maximizes conversions and reducing drop-offs from rushed exits.

113. Percentage of Users Revisiting Same Recipes

The rate of return is an indicator of how often a user views the same recipe, which means the user had a repeated level of interest or re-engagement with the same recipe. We identify instances of viewing the same recipe more than once to measure patterns of curiosity or uncertainty, or even a genuine interest in cooking or saving the recipe. The rate of return is calculated as the number of recipes viewed by the same user more than once out of the total number of recipes the user viewed.



Relevance:

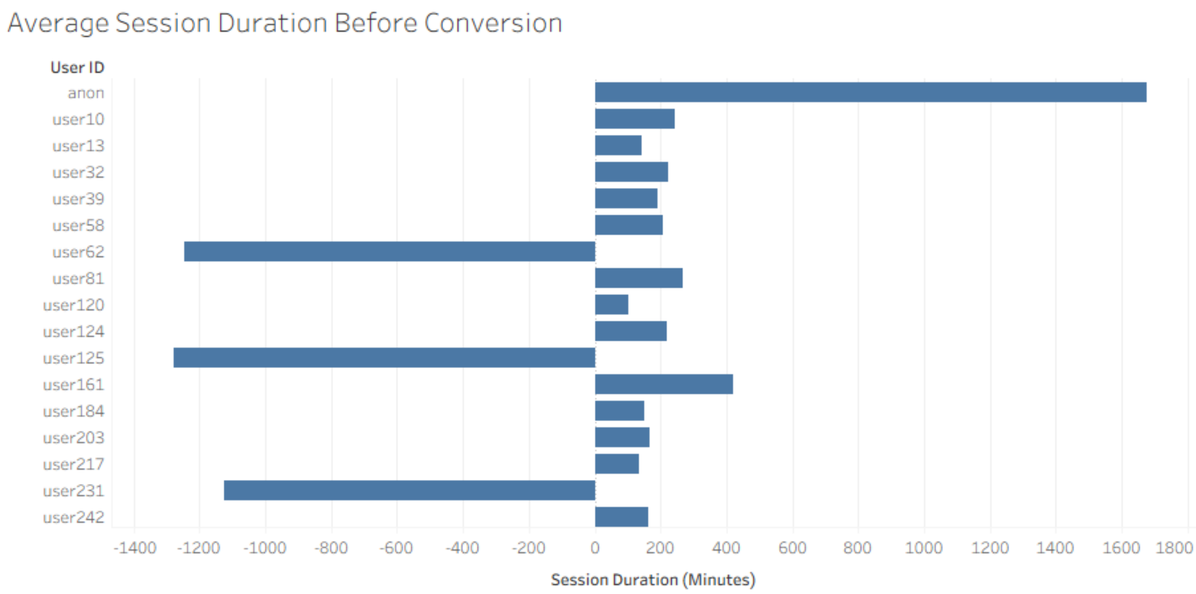
Viewing the same recipe on multiples occasions, or exhibiting the same recipe viewing behavior is a strong indicator of recipe appeal, or user intent to engage with the recipe. Rates of return at or above a certain level, frequently suggest that the user found a given recipe purposeful, memorable, or useful on their initial viewing. Tracking this user behavior can help teams better understand which recipes are capable of maintaining user interest, and which recipe categories drive or resulted in repeated visits to the recipe category, which can be useful for personalizing recommendations for users and retention.

Insights:

The current dataset identifying the potential for viewing the same recipe on multiple occasions is limited in visibility due to sparse observations associated with either having to use synthetic events as a guide for viewing patterns, or due to limited observations. Our ability to develop a stronger dataset with more event level data, this metric has the potential to highlight top performing recipes capable of eliciting user engagement and guide prioritization of content or targeted promotional efforts

114. Average Time of First Session to First Save/Cart

This metric measures the average time elapsed between a user's first session start and their first conversion event (recipe save or cart addition). It reflects how quickly users find something worth saving or purchasing during their early interactions with the platform.



Relevance:

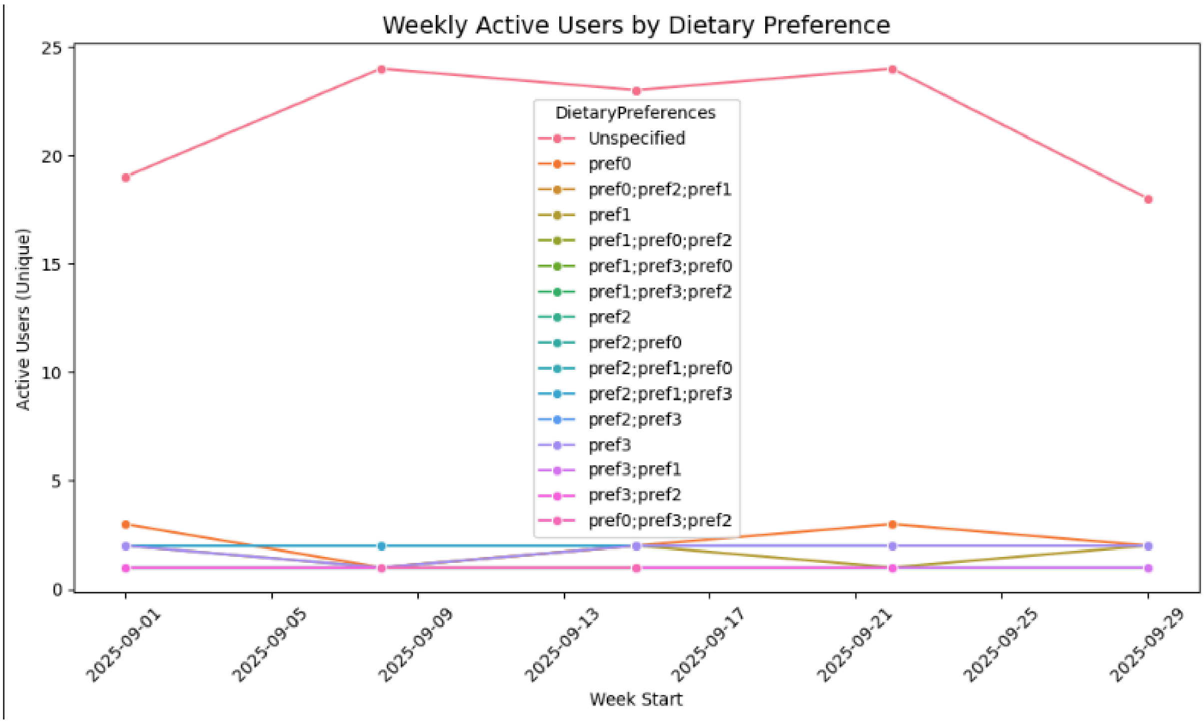
Tracking this delay helps identify friction in the discovery and engagement process. A shorter average delay indicates users are finding relevant recipes faster, while a longer delay might suggest difficulties in search, navigation, or recommendation accuracy.

Insights:

If users take a long time to perform their first save or cart action, it may mean they need more guidance or better-curated suggestions. Conversely, a shorter time span from session start to first conversion often signals strong engagement and effective personalization early in the user journey.

115. Weekly Active Users by Preference

This measures how many unique users were active in a given week, segmented by their declared dietary or allergy preference. A user is considered active if they are listed in any session or record in that week. The metric is aggregated by week start date as well as user preference to compare activity through time and preference types.



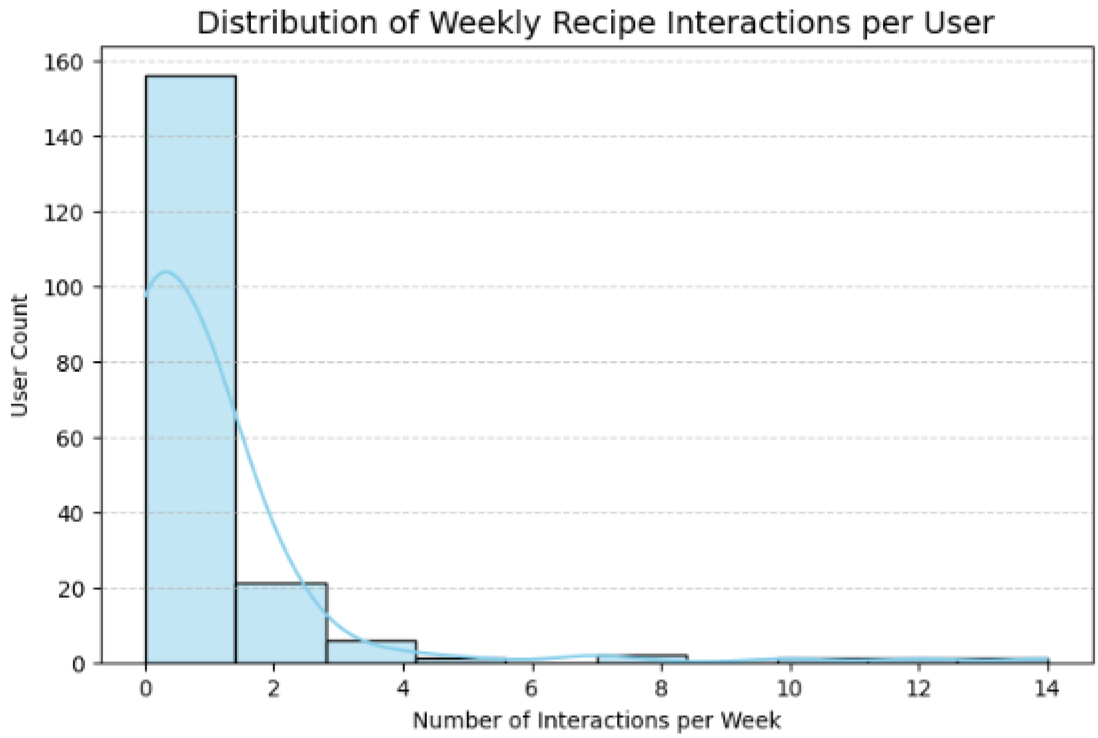
Relevance:
Tracking weekly active users by status helps monitor how specific dietary or allergy segments are engaging with the platform over time. Visualizing patterns such as consistency in activity per dietary group (e.g. vegan, gluten-free) and different levels of activity amongst dietary segments (e.g. some segments may have seasonal or sporadic spikes in activity) is helpful for generating relevant product personalization, recipe recommendations, and advertising methods to their user lifestyles.

Insights:

Active user data currently shows modest variation across user types but average groups show some weeks have drops in activity - likely to be due to sparse or uneven recording of user data. As mentioned before with previously tracked engagement metrics; this is an effective metric for future, long-term monitoring once we have a more consistent backing of user preference data. Additionally, as we continue to aggregate weekly user activity data, the trends in either growth or plateau of activity among niche dietary segments will be instrumental to our retrospective effort to establish effective engagement campaigns.

116. Average Number of Recipe Interactions (Views, Saves, Carts) per User per Week

This indicator captures average weekly user engagement — every time a user performed a recipe or recipe-related action (views, saves, or adding to their carts) during the week — signifying the depth and quantity of recipe-related behavior over time. It thereby reflects the overall activity patterns and the amount of time users spend engaging with the recipe section of the platform.

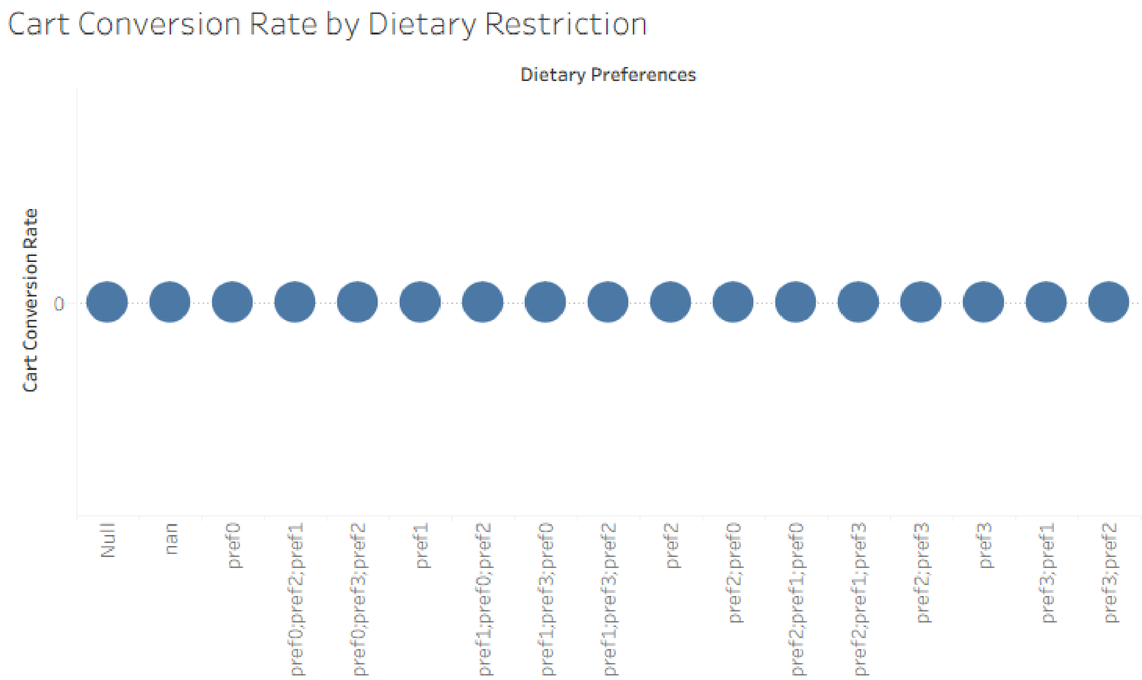


Relevance:
Monitoring weekly recipe interactions allows you to see the engagement cycles and sustained user retention you have overall. A steady average weekly recipe interaction value suggests positive recurring participation, while declines may signal disengagement or friction finding or saving recipes. Weekly averaging also serves as an effective tool for tracking the impact of updates, offers, or recipes with seasonal variability. This metric permits segmentation by user type or region to tease apart behavioral patterns across demographic groups or mobile devices, to understand user action choices and engagement patterns.

Insights:
The current dataset indicates that recipe interactions vary significantly among users, with some being highly active while others engage sporadically. Weekly averages provide a smooth trend to monitor user engagement stability over time. If the average is consistently low, it may point to content fatigue or UI inefficiencies reducing repeat engagement.

117. Cart Conversion by Dietary Restriction

This metric measures how dietary preferences or restrictions (e.g., vegan, gluten-free, keto, etc.) influence user cart behavior. Specifically, it calculates the proportion of viewed recipes that are added to the cart, grouped by dietary preference.



Relevance:

Understanding conversion behavior across dietary segments helps identify which user groups are most likely to take purchase-oriented actions. It enables personalization of recipe recommendations, targeted marketing (e.g., promoting more “cart-successful” diets), and better alignment of content with user intent.If a particular diet category shows high conversion, it indicates stronger purchase motivation and higher product relevance for that group.

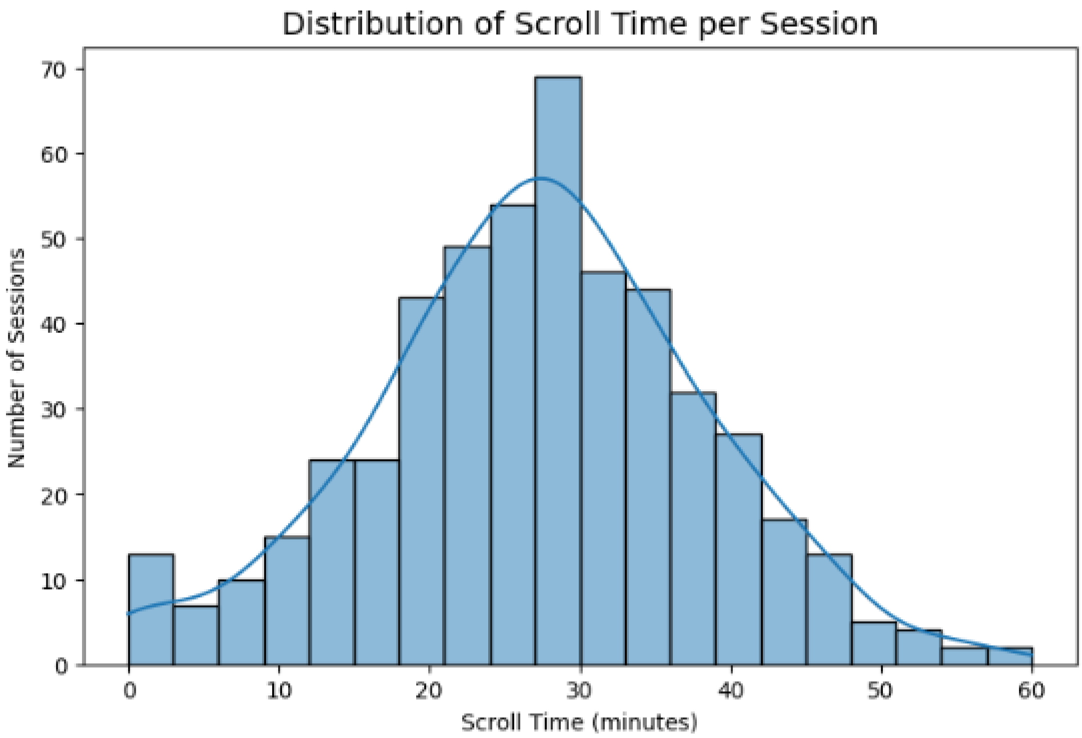
Insights:

In the current dataset, this metric is theoretically significant but practically limited due to insufficient sample size or missing dietary preference data. The available entries under DietaryPreferences were too sparse or inconsistent to produce a meaningful visualization.

However, this remains an important metric for future analyses — once more complete user dietary data is collected, it can reveal valuable behavioral segmentation insights and inform recipe and product strategy.

118. Average Scroll Time per Session

This metric estimates the average time users actively spend scrolling through recipe content during a session, excluding time spent interacting with widgets or chat. It works as a proxy to understand how engaged users are with browsing on the platform. Generally speaking, the longer the average scroll time, the more users explore recipes, read through ingredients, or take more time evaluating options to save or cart.

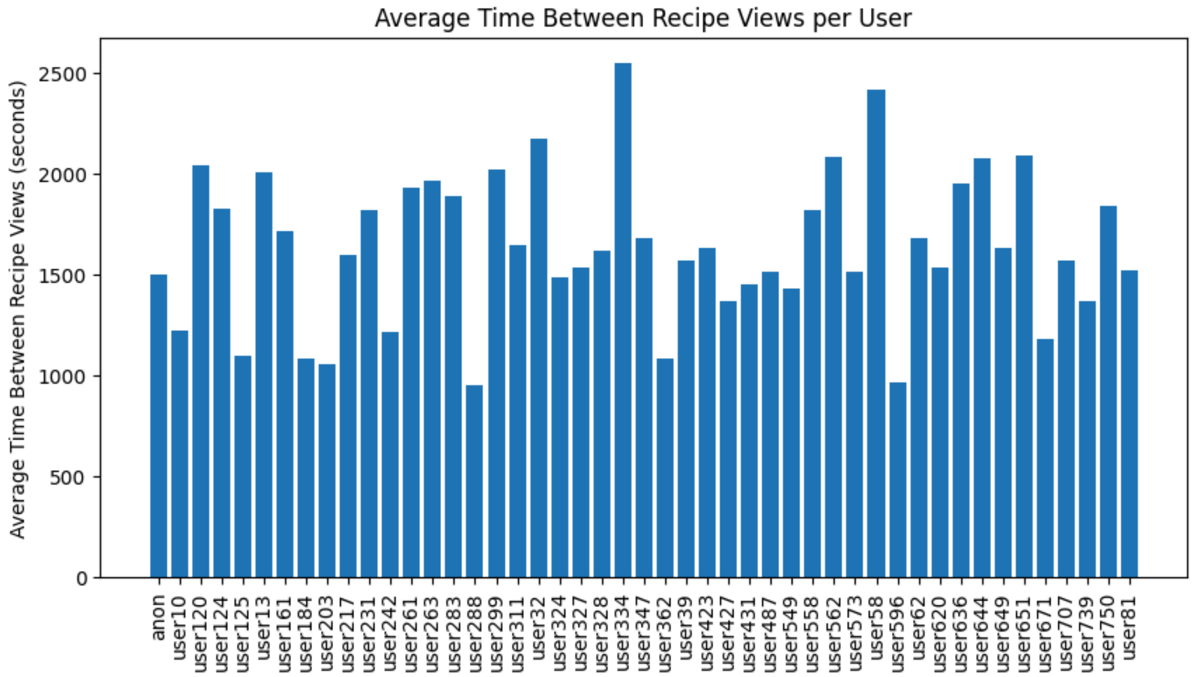


Relevance:
Understanding scroll time provides critical insight into the depth of content and the focus that users have. High average scroll durations indicate that users find the interface or content engaging enough to continue browsing. In contrast, shorter durations may suggest superficial engagement, poor discoverability of content, or friction in navigation. Monitoring this metric over time informs UX and content teams about how design changes, content placement, or feature rollouts impact engagement behavior.

Insights:
This does, however, reflect moderate engagement between interactive actions, as can be seen by the measurable scroll duration across sessions in the current dataset. Because this data is synthetic and limited, this metric should be taken with a grain of salt. In the real world, one might expect to see what kinds of engagement-preceding scroll times correlate with actions such as “recipe saved” or “added to cart”. Scroll depth, page load time, or even content type viewed would be further refinements of this metric, making it a powerful indicator of how deeply users interact with the recipe ecosystem on this platform.

119. Average Time Between Recipe Views

This metric calculates the **average interval between consecutive recipe views within a session**, expressed in seconds. It is derived by dividing the total session duration by the number of recipes viewed in that session. The idea is to quantify how often a user moves from one recipe to another, helping interpret the natural rhythm of content exploration.
It filters out missing or invalid timestamps before computation, ensuring that only valid user activity contributes to the metric. This provides a **clean, session-level behavioral signal** that represents user attention flow during browsing.



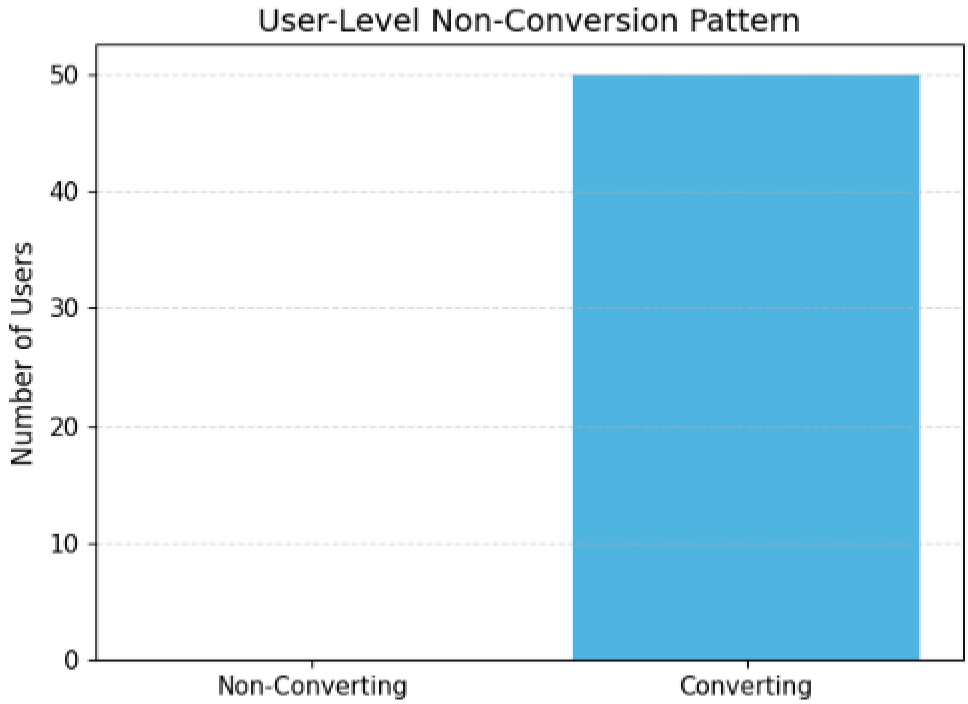
Relevance:
Average view time reflects how users engage with content density — fast transitions may mean skim-style exploration, while longer gaps indicate deeper reading or distraction. It's a good behavioral metric for identifying browsing intent — whether a session is discovery-oriented or decision-oriented. When compared across devices or user types, it reveals how interaction patterns shift by context (e.g., mobile users may scroll faster).

Insights:

Most sessions displayed realistic viewing gaps, with moderate variation across users and devices, suggesting consistent browsing tempo.However, some negative or unusually large durations emerged — these likely result from synthetic or incomplete timestamps, which were filtered out to maintain integrity.This highlights the need for timestamp normalization and stricter data validation in synthetic datasets. The metric offers valuable clues on how long users stay engaged per recipe, though interpretation should remain cautious given the dataset's simulated timing artifacts.

120. User-Level Non-Conversion Pattern

This metric shows those users who have viewed the recipes but never saved them. It provides insight into the behavioral gap between engagement in browsing recipes and intent to save for later use. Analyzing users who have recipe views more than zero but zero saves, this metric allows us to highlight potential friction points in the recipe-saving journey or a perceived value lack in the viewed content.



Relevance:
Non-conversion behavior is one of the key signals of user disengagement or experience inefficiency. A high non-conversion rate could mean that users do browse for recipes but don't find them interesting enough to save, or the mechanism of saving itself lacks clarity and ease. It will provide important insight into the improvement of recommendation logics, recipe layout, and visibility of features for saving recipes-all aspects that can have a direct impact on the optimization of conversion.

Insights:

In the synthetic dataset, a fair share of users did not convert-meaning they viewed several recipes yet did not save any. This could be because of low personalization, making the recipes less appealing, or just incomplete patterns in the synthetic data. Though this sample may not be rich enough for deep conclusions, the metric remains important in any real-world analysis, as it aids in pinpointing specifically where engagement does not convert into retention or action. A reduction in this user drop can meaningfully improve the overall conversion rate across the platform.

Chapter 5 Conclusion

5.1 Conclusion

The project “Analytics Pipeline for Data from Recipe-Oriented Portals” creates an entire analytics system which extracts valuable information from extensive user interaction data. The system unites three datasets about recipes and user actions and system logs into one platform which enables both data cleaning and preparation and behavioral metric evaluation.

The system handles more than 100 performance and engagement metrics which include recipe views and save rates and session durations and conversion efficiency and user retention and widget success and recipe revisit frequency. The system presents its findings through Matplotlib plots and Tableau dashboards which enable users to perform detailed analysis and understand data through visual representations.

The analysis reveals distinct behavioral patterns between devices and time-dependent engagement patterns and users tend to interact more frequently with their preferred content. The Widget Success Rate and Conversion by Time of Day metrics demonstrate how design elements of interface design together with timing factors affect user engagement results.

The project achieved successful results through Python-based data analysis and Tableau visualization which provided effective tools for both detailed data inspection and presentation of findings. The research encountered multiple obstacles because essential cost and rating information was absent and timestamps were incomplete and logging precision varied between different systems. The absence of cost-tier segmentation and content satisfaction modeling data restricted the project from performing advanced analysis.

The project demonstrates how metric design structures and visualization methods enable businesses to extract valuable insights from unprocessed user information. The system requires additional development to achieve better results through enhanced temporal data precision and semantic content enrichment and real-time analytics capabilities and machine learning models for predictive user behavior and customized recommendations.

5.2 Limitations

While comprehensive, this study also faced a few constraints:

1. **Synthetic and incomplete data:** Several metrics (e.g., cost-based or rating-based ones) could not be computed due to missing or placeholder data.

2. **Session granularity assumptions:** Session-based grouping relied on timestamps, not user-intent markers, which may introduce slight bias in duration metrics.

3. **Limited behavioral validation:** In the absence of real feedback data, engagement interpretations rely solely on proxy variables (views, saves, carts).

4. **Tool interoperability constraints:** Some Tableau visualizations were restricted by file format compatibility (CSV vs. Excel), limiting metric uniformity.

5.3 Future Work

This project establishes a foundation that can be significantly expanded in several directions:

1. **Integration with Real User Data:** Replace synthetic or simulated logs with real-world data from recipe portals to enhance behavioral authenticity.

2. **Machine Learning Extensions:** Build predictive models for user conversion, recipe popularity forecasting, or personalized recommendation scoring using derived metrics.

3. **Advanced Visualization Dashboards:** Extend Tableau dashboards to interactive BI systems (e.g., Power BI, Streamlit, or Looker) for real-time tracking of engagement KPIs.

4. **Incorporating External Context:** Include cost, ratings, and ingredient availability to build multidimensional models linking culinary trends with consumer actions.

5. **Automation & Scaling:** Transform the analytics pipeline into a production-ready ETL system, supporting continuous data inflow and automated metric computation.

APPENDIX

1. Github:
   https://github.com/ankitghosal/ISI_Internship_Autumn25/tree/main
2. Contributions:
   a. Sohan: Metrics 1-31:

      Tableau Worksheet

   b. Ankit: Metrics 32 - 64:

      Tableau Worksheet

   c. Lokeshvar: Metrics 65 - 96:

      Tableau Worksheet

   d. Soumi: Metrics 97 - 120:

      Tableau Worksheet